

# *Lectures on General Relativity*

*“To understand the universe we must understand how space and time bend”*

Saturday 19<sup>th</sup> September, 2026





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# LECTURES ON GENERAL RELATIVITY

**Miguel Jafert Serrano Mantilla**

*Professor: Fabio Duván Lora Clavijo*

*Universidad Industrial de Santander  
Sciences faculty  
Physics school  
Group of optics and signals treatment*

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# *Contents*

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>1</b> | <b>The History of General Relativity</b>                  | <b>3</b>  |
| 1.1      | Motion in the Ancient World . . . . .                     | 4         |
| 1.1.1    | The Aristotelian Cosmos . . . . .                         | 4         |
| 1.1.2    | Ptolemy and the Mathematics of the Heavens . . . . .      | 4         |
| 1.2      | The Arabic Custodians of Motion . . . . .                 | 5         |
| 1.3      | The Latin West Breaks with Aristotle . . . . .            | 6         |
| 1.3.1    | Buridan’s Impetus . . . . .                               | 6         |
| 1.3.2    | Copernicus and Kepler . . . . .                           | 6         |
| 1.3.3    | Galileo and the First Principle of Relativity . . . . .   | 7         |
| 1.4      | Newton and the Absolute Stage . . . . .                   | 8         |
| 1.5      | The Geometers Who Prepared the Ground . . . . .           | 8         |
| 1.5.1    | Gauss and the Curvature Hidden Inside a Surface . . . . . | 8         |
| 1.5.2    | Riemann’s Prophecy . . . . .                              | 9         |
| 1.6      | Einstein’s Long Road to General Relativity . . . . .      | 10        |
| 1.7      | Riemann, Hilbert and the Question of Theft . . . . .      | 11        |
| 1.8      | Confirmation: From the Trenches to the Eclipse . . . . .  | 12        |
| 1.9      | Beyond General Relativity . . . . .                       | 13        |
| <b>2</b> | <b>Differential Geometry</b>                              | <b>15</b> |
| 2.1      | Differential geometry . . . . .                           | 16        |
| 2.1.1    | Differential manifolds . . . . .                          | 16        |
| 2.1.1.1  | Smooth functions . . . . .                                | 19        |
| 2.1.1.2  | Curves and vectors . . . . .                              | 20        |
| 2.1.1.3  | Covectors and one-forms . . . . .                         | 26        |
| 2.1.1.4  | Tensors . . . . .                                         | 27        |
| 2.1.1.5  | Tensor fields . . . . .                                   | 28        |
| 2.1.1.6  | Commutator . . . . .                                      | 29        |
| 2.1.1.7  | Integral curves . . . . .                                 | 30        |
| 2.1.2    | The Lie’s derivative . . . . .                            | 31        |
| 2.1.2.1  | Pull back and push forward . . . . .                      | 31        |

|          |                                                                               |           |
|----------|-------------------------------------------------------------------------------|-----------|
| <b>3</b> | <b>The Theory of General Relativity</b>                                       | <b>33</b> |
| 3.1      | Introduction to general relativity . . . . .                                  | 34        |
| 3.1.1    | The incompatibility between the Newtonian gravity with the special relativity | 35        |
| 3.1.2    | Weak equivalence principle . . . . .                                          | 36        |
| 3.1.3    | Experimental evidence for a general relativity theory . . . . .               | 39        |
| 3.1.3.1  | Eötvös experiment . . . . .                                                   | 39        |
| 3.1.3.2  | Tidal forces . . . . .                                                        | 42        |
| 3.1.3.3  | Light deflection and gravitational red shift . . . . .                        | 42        |
| 3.1.4    | Einstein's equivalence principle and the space-time . . . . .                 | 45        |
| 3.2      | First steps on general relativity . . . . .                                   | 46        |
| 3.2.1    | Metric tensor . . . . .                                                       | 47        |
| 3.2.2    | Proper time and an introduction to paths . . . . .                            | 48        |
| 3.2.3    | Covariant derivative . . . . .                                                | 55        |
| 3.2.4    | Levi-Civita connection . . . . .                                              | 58        |
| 3.2.5    | Geodesics . . . . .                                                           | 60        |
| 3.2.5.1  | Affine transformations in Galilean relativity . . . . .                       | 61        |
| 3.2.5.2  | Affine transformations in special relativity . . . . .                        | 62        |
| 3.3      | Riemann's tensor . . . . .                                                    | 67        |
| 3.3.1    | Curvature of the Levi-Civita connection . . . . .                             | 67        |
| 3.3.1.1  | Einstein's equations . . . . .                                                | 74        |
| 3.3.2    | Properties related to the Riemann's tensor . . . . .                          | 75        |
| 3.3.3    | Geodesic deviation . . . . .                                                  | 78        |
| 3.4      | Gravitation and the Einstein's field equation . . . . .                       | 80        |
| 3.4.1    | Newtonian limit . . . . .                                                     | 81        |
| 3.4.2    | Einstein's field equations . . . . .                                          | 83        |
| 3.4.2.1  | Einstein-Hilbert action . . . . .                                             | 85        |
| 3.4.2.2  | Energy-momentum tensor . . . . .                                              | 89        |
| <b>4</b> | <b>Gravitational Waves</b>                                                    | <b>93</b> |
| 4.1      | Lorenz Gauge and gravitational waves . . . . .                                | 94        |
| 4.2      | Gravitational waves . . . . .                                                 | 97        |
| 4.2.1    | Traceless transverse gauge . . . . .                                          | 98        |
| 4.2.1.1  | Polarization and Stokes parameters for gravitational waves . . . . .          | 99        |
| 4.2.2    | Effects of the gravitational radiation on the matter . . . . .                | 102       |

# *List of Figures*

|     |                                                                                                                                                                                                                                                                                                                                                          |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1 | Schematic representation of the third postulate what it means is that in a differential manifold we can always go from one coordinate chard to another. . . . .                                                                                                                                                                                          | 17 |
| 2.2 | In the left there is the first chart $\phi_1$ which covers almost all the circumference but a point $p$ and to the right there is the other chard which covers almost all the circumference except a point $q$ diametrically opposite to $p$ . . . . .                                                                                                   | 18 |
| 2.3 | Each point on the sphere has a small vicinity which maps as a disc on $\mathbb{R}^2$ . . . . .                                                                                                                                                                                                                                                           | 19 |
| 2.4 | Representation of traditional map from $S^2$ to $\mathbb{R}^2$ we can see there are multi valued points which wont allow us to cover the manifold $S^2$ . . . . .                                                                                                                                                                                        | 20 |
| 2.5 | Graphical representation to find teh stereographic projection of $S^2$ to $\mathbb{R}^2$ . . . . .                                                                                                                                                                                                                                                       | 21 |
| 2.6 | Schematic representation of the map $f$ , this takes elements from the manifold and maps then to the rect of the reals $\mathbb{R}$ . . . . .                                                                                                                                                                                                            | 22 |
| 2.7 | The curve $\lambda$ maps from an open interval of the reals to the manifold, in the manifold this maps looks like, (remember this is just an scheme), a curve we see usually, from a point in that curve we can define a tangent vector $X_p$ . . . . .                                                                                                  | 23 |
| 2.8 | Two tangent vectors $X_p$ and $Y_p$ which both are tangent to $p$ but live in two different curves, $(\lambda_1, \lambda_2)$ but in the same vectorial space $T_p(M)$ . . . . .                                                                                                                                                                          | 24 |
| 3.1 | Graphical representation of the relative movement between the two test bodies. . . .                                                                                                                                                                                                                                                                     | 40 |
| 3.2 | In the left there are two particles which are in a free fall to the earth an the inertial reference frame (IRF) is to the side of the earth, he observes a diagonal trajectory of the bodies. In the right we set the IRF inside the box with the two bodies, and he observes that there is a strange force which makes the bodies get together. . . . . | 42 |
| 3.3 | In the left there is a light ray seeing from an inertial reference frame inside the box, and in the right there is the same ray but seeing from the earth. . . . .                                                                                                                                                                                       | 43 |
| 3.4 | Space-time diagram representation for the Doppler effect analized before. . . . .                                                                                                                                                                                                                                                                        | 45 |
| 3.5 | The light cone for the space-time structure, in the $x$ -axis there is $\mathbb{R}^3$ and in the $y$ axis there is the coordinate time $t$ , a space-like vector is a vector which lives outwards the cone, a light-like lives on the border of the cone and a time-like lives inside the cone. . . . .                                                  | 49 |
| 3.6 | Graphical representation of an integral time-like curve on a manifold $M$ . . . . .                                                                                                                                                                                                                                                                      | 50 |
| 3.7 | The real time-like trajectory a particle will follow is $\lambda(\tau)$ and the other two will be little different trajectories but with initial an final points fixed. . . . .                                                                                                                                                                          | 51 |

|      |                                                                                                                                                                                                                                                                                                                             |     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3.8  | Affine parameterization of the geodesics, here we illustrate the path (geodesic) a free particle will follow which is parameterized with a parameter $t$ but this is some application of the other parameter $u$ . . . . .                                                                                                  | 66  |
| 3.9  | Connection between $q$ and $p$ with a curve along a variation of $s$ with $X$ as tangent vector, in the same way $q$ and $r$ can be connected by a curve with $Y$ as tangent vector, $p$ and $u$ can be connected with a curve with $Y$ as tangent vector and same with $u$ and $r$ but with $X$ as tangent vector. . . . . | 70  |
| 3.10 | Representation of all the possible subsets of $S$ constant and $t$ constant, this is an analogous to what is done in the formulation $3 + 1$ where we separate the space-time and folds as a Cauchy space. . . . .                                                                                                          | 79  |
| 3.11 | Representation of the region (manifold $M$ ) of integration and its surrounding surface $\Sigma M$ . . . . .                                                                                                                                                                                                                | 88  |
| 4.1  | Caption . . . . .                                                                                                                                                                                                                                                                                                           | 103 |

## Abstract

These notes build the theory of general relativity from the ground up. After tracing how our understanding of gravity evolved from the ancient Greeks to Einstein's own revolution, the book develops the differential geometry needed to speak of curved space-time: manifolds, charts, tangent vectors and the metric tensor. With that language in hand, it turns to the theory itself, deriving the geodesic equation, the Riemann curvature tensor and the Einstein field equations that relate the curvature of space-time to the matter and energy within it. The final chapter linearises these field equations around a flat background to obtain the wave equation for gravitational waves, and studies their propagation and polarisation properties.

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# *Chapter 1*

## *The History of General Relativity*

*Every revolution in physics begins as a quiet correction to the certainties of the age before it.*

“If I have seen further it is by standing on the shoulders of Giants.”

– Isaac Newton

Before we bend space and curve time we must first understand why anyone would ever have thought to try. **The history of general relativity is not the history of a lone genius working in isolation, it is the history of nearly twenty five centuries of argument about a single, deceptively simple question:** why do bodies move the way they do? Every civilisation that has looked up at the night sky or dropped a stone from a height has had to answer this question in its own way, and the theory we are about to build in this book is, in a very real sense, the latest instalment of that argument rather than its final word. In this chapter we are going to walk through that argument from the Greeks to the physics of today, and along the way we will meet the geometers whose abstract mathematics quietly waited, sometimes for decades, for a physicist to notice that it was exactly what nature needed.

## 1.1 Motion in the Ancient World

### 1.1.1 The Aristotelian Cosmos

For the better part of two thousand years the Western understanding of motion belonged to Aristotle (384 to 322 B.C.), whose *Physics* divided all motion into two families. *Natural motion* was the motion a body underwent when left alone, and it depended entirely on what the body was made of: earth and water naturally fell towards the centre of the cosmos, while air and fire naturally rose away from it, and the heavenly bodies, made of a fifth incorruptible element he called *aether*, naturally moved in perfect eternal circles. *Violent motion*, by contrast, was any motion imposed on a body against its nature, and Aristotle's central postulate about it is one every student of mechanics eventually collides with: **a body moves only while something is in contact with it and pushing it**, so that the moment the pushing stops the motion must stop too (Aristotle, 1984). This is why a thrown spear troubled him so much. If the hand that threw it is no longer touching it, what keeps it moving through the air after it leaves the fingers? Aristotle's answer, the doctrine of *antiperistasis*, was that the air parted by the spear's tip rushes around to its rear and pushes it onwards, a mechanism that is elegant on paper and hopeless the moment one actually thinks about throwing something, and it would take the better part of two millennia and the combined efforts of Alexandria, Baghdad and Paris to finally dismantle it.

Underneath this physics of pushes and pulls lay a much grander architecture, the geocentric cosmos, in which a stationary Earth sat at the centre of a finite universe and everything else, the Moon, the Sun, the planets and the fixed stars, was carried around it on nested, concentric, crystalline spheres. It is worth pausing on how seriously this was meant: for Aristotle the centre of the cosmos was not merely a convenient coordinate origin, it was the natural resting place towards which heavy matter was drawn, so that geometry and physics were one and the same fact.

### 1.1.2 Ptolemy and the Mathematics of the Heavens

The trouble with perfect circles around a stationary Earth is that the planets simply refuse to move on them, drifting faster and slower across the year and occasionally, most embarrassingly

of all, reversing direction in the sky entirely, a phenomenon we now call retrograde motion and understand as nothing more than the view from a moving Earth overtaking a slower outer planet. Claudius Ptolemy, working in Alexandria around 150 A.D., inherited several centuries of Greek attempts to rescue the circles (Apollonius and Hipparchus among them) and assembled them into the *Almagest*, the most complete and mathematically serious astronomical text the ancient world produced (Ptolemy, 1998). Ptolemy kept the philosophical commitment to circular motion but smuggled in three devices to make the mathematics fit the sky: the *epicycle*, a small circle whose centre rides upon a larger circle called the *deferent*; the *eccentric*, which quietly moves the Earth slightly off the true centre of the deferent; and, most cunning of all, the *equant*, an empty point in space from which the planet's motion *appears* uniform even though it is uniform nowhere else. The equant in particular is a fascinating piece of intellectual dishonesty dressed up as rigour, because it abandons the very principle of uniform circular motion it was invented to save, and it is precisely this point, as we shall see in a moment, that the astronomers of Islam would spend centuries trying to repair.

## 1.2 The Arabic Custodians of Motion

When the intellectual centre of gravity of the ancient world shifted from Alexandria towards Baghdad, Damascus and later Maragha and Samarkand, it carried Aristotle and Ptolemy with it, but it did not simply preserve them, it interrogated them. The crucial bridge figure is not, strictly speaking, an Arab at all: John Philoponus, a Christian philosopher working in sixth century Alexandria, wrote a commentary on Aristotle's *Physics* in which he attacked antiperistasis directly, proposing instead that the thrower impresses onto the projectile itself a finite, internal quantity of motive power, an *incorporeal kinetic force*, which then dissipates gradually as the projectile travels, explaining why a spear eventually falls without needing the air to conspire behind it. Philoponus's Greek text was largely forgotten in Byzantium but survived and was studied intensely by scholars writing in Arabic, and it is through that transmission, not through the Latin West, that his idea re-entered the mainstream of physics.

The Persian polymath Ibn Sina, known in Latin as Avicenna (980 to 1037), sharpened Philoponus's idea considerably in his encyclopaedic *Kitab al-Shifa* (The Book of Healing). He argued that the impressed force, which he called *mayl* (inclination), is a genuine, quantifiable property of the moving body itself rather than of the medium around it, and, in a passage that reads almost eerily like a draft of Newton's first law, he claimed that in a void this *mayl* would never be consumed and the body would continue moving forever, since nothing in a true vacuum could exist to wear it down (Rashed, 1996). This is not yet inertia in the modern sense, because Ibn Sina still thought of *mayl* as a cause that must be continuously present to sustain motion rather than as the mere absence of a cause to stop it, but the intuition that unresisted motion is self-perpetuating had, for the first time, been stated clearly and defended on physical grounds rather than dismissed as an impossible thought experiment about an empty universe Aristotle himself had declared could not exist.

Astronomy saw an even more consequential development. Frustrated by the equant's abandonment of true uniform circular motion, the mathematicians of the Maragha observatory in thirteenth century

Persia, led by Nasir al-Din al-Tusi, invented a purely geometrical device, now called the *Tusi couple*, a small circle rolling inside a larger one, whose combined motion can reproduce a straight line oscillation using only uniformly rotating circles, with no equant required. A century later in Damascus, Ibn al-Shatir used exactly this device to rebuild the entire Ptolemaic planetary system without a single equant, producing models that were mathematically superior to Ptolemy's own (Saliba, 1987). What makes this genuinely remarkable, and worth pausing on for a student of physics rather than only of history, is that when Nicolaus Copernicus published *De Revolutionibus Orbium Coelestium* in 1543, the geometrical diagrams he used to eliminate the equant from his own, now heliocentric, system are mathematically identical to Ibn al-Shatir's, down to the labelling of points in some manuscripts. Historians still debate exactly how the Maragha school's mathematics reached Copernicus, whether through Byzantine Greek intermediaries in Italy or some now lost channel, but the technical debt is not seriously disputed, and it is a useful reminder that the Copernican revolution did not spring from nowhere: it stood, quite literally, on Islamic geometrical shoulders.

## 1.3 The Latin West Breaks with Aristotle

### 1.3.1 Buridan's Impetus

The idea of an impressed motive force crossed into Latin Europe through translations of Arabic and Greek texts flooding into the universities of the thirteenth century, and it found its sharpest medieval expression in Jean Buridan, rector of the University of Paris in the 1340s. Buridan proposed that the mover impresses upon the projectile a genuine, permanent quantity he called *impetus*, proportional to the amount of matter in the body and to its speed (a description that should already sound suspiciously familiar to anyone who has met the word *momentum*), and that this impetus would in principle persist forever were it not continuously eroded by air resistance and by the body's own natural tendency to fall (Clagett, 1959). Buridan went further than Ibn Sina in one crucial respect: he applied impetus theory to the heavens themselves, suggesting that God, at the Creation, might simply have impressed upon the celestial spheres an impetus that meets no resistance in the frictionless heavens and therefore never decays, a proposal that quietly detaches the eternal circular motion of the planets from any need for a continuously acting mover, Aristotle's prime mover included. His student Nicole Oresme pushed the mathematics further still, inventing a graphical method of plotting a body's speed against time as a geometrical figure and showing that the area under such a figure gives the distance travelled, a genuinely startling anticipation, two and a half centuries early, of the mean speed theorem that Galileo would later use to characterise uniformly accelerated fall.

### 1.3.2 Copernicus and Kepler

Copernicus's 1543 relocation of the Sun to the centre of the cosmos was, in itself, primarily a mathematical and aesthetic argument (he found the resulting system simpler and more harmonious), and it would take Johannes Kepler, working with the extraordinarily precise naked eye observations

that Tycho Brahe had spent a lifetime accumulating, to turn heliocentrism into a genuinely predictive physical theory. Between 1609 and 1619 Kepler announced three laws that any general relativity student should keep close at hand, because they are the observational bedrock that Newton, and much later Einstein, would both have to reproduce: planets move on ellipses with the Sun at one focus, they sweep out equal areas in equal times, and the square of their orbital period is proportional to the cube of their orbit's semi-major axis. Kepler arrived at the ellipse only after years of fighting his own preference for circles, famously calling his discovery a *cartload of dung* in a fit of frustration with the messy, imperfect shape nature had apparently chosen over the geometrical perfection he had hoped for, and it is a nice historical irony that this same ellipse would reappear, itself only an approximation, when Einstein finally explained in 1915 why Mercury's own elliptical orbit slowly, stubbornly, refuses to close.

### 1.3.3 Galileo and the First Principle of Relativity

It is with Galileo Galilei that impetus theory finally becomes something closer to modern inertia. In his *Two New Sciences* of 1638, written under house arrest and smuggled out of Italy for publication in the Protestant Netherlands, Galileo established through careful (and, for the tools available to him, remarkably precise) experiments with balls rolling down inclined planes that a freely falling body accelerates uniformly, covering distances in the ratio of the odd numbers one, three, five and so on in successive equal intervals of time, and he reasoned that a body moving on a perfectly smooth horizontal surface, meeting no resistance and no incline either to speed it up or slow it down, would continue moving forever at constant speed (Galilei, 1974). This is, for all practical purposes, Newton's first law stated half a century early.

Just as important for our story, though, is an idea Galileo had already set out six years earlier, in 1632, in the *Dialogue Concerning the Two Chief World Systems*. There he asks the reader to imagine themselves below decks on a smoothly sailing ship, with no portholes, watching fish swim in a bowl, butterflies fly and drops of water fall from a bottle, and he argues that not one of these phenomena would betray to the observer whether the ship is standing still in harbour or sailing at constant velocity across a perfectly calm sea (Galilei, 1953). **No mechanical experiment performed entirely inside a closed cabin can distinguish uniform motion from rest**, and this is the first clean statement of what we now call the *principle of relativity*, the very principle that, generalised first to electromagnetism by Einstein in 1905 and then to accelerated frames by the equivalence principle we shall meet in chapter 3, becomes the seed from which the entire theory of general relativity eventually grows. Tradition, almost certainly apocryphal but too delicious to leave out, has Galileo muttering *E pur si muove* ("and yet it moves") under his breath after being forced by the Roman Inquisition in 1633 to formally recant the Copernican system his own physics had done so much to defend.

## 1.4 Newton and the Absolute Stage

Isaac Newton's *Philosophiae Naturalis Principia Mathematica*, published in 1687, is the book that finally welded Galileo's terrestrial kinematics to Kepler's celestial geometry into a single mechanics. Newton's three laws of motion, and above all his law of universal gravitation, showed that the very same force that pulls an apple to the ground also holds the Moon in its orbit and drives the planets along Kepler's ellipses, a genuinely staggering unification for its time (Newton, 1999). But Newton needed a stage on which this mechanics could play out, and he built one deliberately: *absolute space*, an unmoving, featureless, infinite container existing independently of anything inside it, and *absolute time*, flowing uniformly and identically for every observer everywhere in the universe. He even offered an experimental argument for taking absolute space seriously, the famous rotating bucket: spin a bucket of water and its surface climbs the sides into a concave shape, and since this happens whether or not the bucket is rotating relative to any other nearby matter, Newton argued that the water must be rotating relative to something, namely space itself.

What is often forgotten by students who only encounter Newton through the tidy inverse square law is how deeply uncomfortable Newton himself was with one consequence of his own theory: gravity, as he had written it down, acts instantaneously at a distance across empty space, with no mediating mechanism whatsoever, and this troubled him far more than it troubles most introductory textbooks today. In a 1693 letter to the theologian Richard Bentley he wrote, with striking force, that

“that gravity should be innate, inherent and essential to matter, so that one body may act upon another at a distance through a vacuum, without the mediation of any thing else, by and through which their action and force may be conveyed from one to another, is to me so great an absurdity that I believe no man who has in philosophical matters a competent faculty of thinking can ever fall into it.” (Newton, 1958)

Newton had no candidate mechanism to offer in its place, and so, with characteristic caution, he famously declined to speculate publicly, writing in the General Scholium he added to later editions of the *Principia* that *hypotheses non fingo*, “I feign no hypotheses” (Newton, 1999). It would take more than two centuries, and an entirely new branch of geometry that did not yet exist in Newton's lifetime, before anyone could offer a mechanism at all: general relativity's answer, that gravity is not a force transmitted through space but the curvature of space-time itself, is in a very real sense the belated reply to Newton's own unease.

## 1.5 The Geometers Who Prepared the Ground

### 1.5.1 Gauss and the Curvature Hidden Inside a Surface

For gravity to become geometry, geometry first had to learn how to describe curvature without appealing to anything outside itself, and this is exactly what Carl Friedrich Gauss achieved in 1827

in his *Disquisitiones Generales circa Superficies Curvas* (General Investigations of Curved Surfaces) (Gauss, 1828). Gauss showed that the curvature of a surface, a quantity that seems at first to depend entirely on how that surface bends through the three dimensional space surrounding it, can in fact be computed using only measurements made *within* the surface itself, lengths and angles as measured by an ant living on it, with no reference whatsoever to the ambient space. He was so struck by this fact that he named it his *Theorema Egregium*, the “remarkable theorem”, and remarkable it is: it is the very first proof that curvature is an *intrinsic* property of a space, not something a space merely borrows from a higher dimensional stage on which it happens to sit. This is precisely the property that will let us, in chapter 2, speak of the curvature of space-time without ever having to ask what space-time is supposedly curving *into*.

It is often repeated, and worth mentioning with the appropriate degree of caution, that Gauss was not only a mathematician but also, in his day job, the director of a two decade long geodetic survey of the Kingdom of Hanover, and that he measured the angles of an enormous triangle formed by the peaks of the Brocken, the Hohenhagen and the Inselsberg, partly to test whether the angles of a very large physical triangle really do sum to exactly 180 degrees, as flat Euclidean geometry demands, or whether the physical space we live in might, just possibly, be very slightly curved. Whether Gauss genuinely intended this as a test of the geometry of physical space, or whether the story has been embellished by later admirers eager to cast him as a prophet of general relativity, remains disputed among historians, and no deviation from 180 degrees was in any case ever detected, the curvature of space-time being far too gentle at the scale of a mountain range for eighteenth century theodolites to have any hope of catching it. What is beyond dispute is that Gauss had, at the very least, begun to entertain the question of whether Euclidean geometry is a truth of pure reason or an empirical fact about our particular universe, and that question is the hinge on which this entire chapter, and indeed this entire book, turns.

### 1.5.2 Riemann’s Prophecy

Gauss’s intrinsic geometry described curved surfaces, objects with only two dimensions. The decisive generalisation, to spaces of any number of dimensions whatsoever, was the achievement of Gauss’s own student, Bernhard Riemann, and it arrived in a single lecture. In 1854, required by University of Göttingen custom to deliver a probationary lecture (a *Habilitationsvortrag*) before the assembled faculty in order to earn the right to teach, the shy and sickly Riemann chose, on Gauss’s own suggestion, the topic “Über die Hypothesen, welche der Geometrie zu Grunde liegen” (On the Hypotheses which lie at the Bases of Geometry). Gauss himself, by then an old man near the end of his life and famously difficult to impress, sat in the audience, and by every surviving account he was profoundly moved by what he heard.

In that lecture Riemann introduced what we would now call an  $n$ -dimensional manifold, a space that need only look locally like ordinary flat Euclidean space, and he showed how to equip such a space with a quantity, the *metric*, encoding the distance between any two infinitesimally close points, from which a notion of curvature at every point could then be built by pure calculation, generalising Gauss’s Theorema Egregium to any dimension and to spaces nobody could ever hope to visualise by

embedding them in something larger. This is the mathematical object we will meet formally as  $g_{\mu\nu}$  in chapter 2, and without it there simply is no general relativity to write down. But Riemann did something else that evening, something far stranger for a lecture ostensibly about pure mathematics, something translated from the German by the English mathematician William Kingdon Clifford two decades later and read ever since by physicists with a peculiar chill of recognition: he wondered aloud, in the lecture's closing paragraphs, whether the geometry of physical space might not be fixed and given after all, but might instead be determined by the matter distributed within it.

“we must seek the ground of its metric relations outside it, in binding forces which act upon it” (Riemann, 1873)

is how Clifford rendered Riemann's own conjecture, and Riemann explicitly named this a question for physics, not mathematics, to eventually answer, adding that the answer could only come from experience and observation of the actual physical world. Riemann died of tuberculosis in 1866, at only thirty nine years old, on the shores of Lake Maggiore in Italy, sixty years before anyone would show that he had, in that single lecture, more or less written down the mathematical vessel into which Albert Einstein would one day pour a physical theory of gravity.

## 1.6 Einstein's Long Road to General Relativity

Albert Einstein's 1905 theory of special relativity had already overthrown Newton's absolute time, welding space and time together into a single four dimensional structure and showing that simultaneity itself depends on the observer's motion. But special relativity, as its name honestly admits, is special: it applies only to unaccelerated, inertial observers, and it could offer no account whatsoever of gravity, since Newtonian gravity's instantaneous action at a distance is flatly incompatible with special relativity's insistence that no signal, gravitational or otherwise, can outrun the speed of light.

The idea that would eventually resolve this came to Einstein in 1907, while he was still working at the Swiss patent office in Bern, and he later called it, in a 1922 lecture in Kyoto, *der glücklichste Gedanke meines Lebens*, “the happiest thought of my life”: a person falling freely from a rooftop feels no gravity whatsoever, their own weight vanishing entirely for as long as the fall lasts. From this simple, almost childlike observation Einstein extracted his *equivalence principle*, the proposal that being at rest in a gravitational field is locally indistinguishable from accelerating through empty space with no gravity at all, a principle that, as we shall see explicitly in chapter 3, is nothing less than Galileo's shipboard relativity extended from uniform motion to acceleration itself, and it is the true conceptual seed of everything that follows in this book.

Turning that seed into a working theory took Einstein eight more gruelling years. He understood by around 1912, while teaching in Prague, that a theory built on the equivalence principle would need a gravitational field capable of bending the paths of light rays and of clocks running at different rates in different places, and that flat Minkowski space-time (the space-time of special relativity) simply could not accommodate this. What he needed was some kind of curved geometry, and he had no idea such a thing already existed in full generality, sitting in the mathematics literature

since 1854, until his old university classmate from Zurich, the mathematician Marcel Grossmann, came to his rescue. According to Einstein's own later recollection, and as reported by his biographer Abraham Pais, Einstein turned to Grossmann in something close to desperation with the words "*Grossmann, du musst mir helfen, sonst werd' ich verrückt!*" ("Grossmann, you must help me or I shall go crazy!") (Pais, 1982). Grossmann answered by introducing Einstein to precisely Riemann's geometry, together with the *absolute differential calculus* that the Italian mathematicians Gregorio Ricci-Curbastro and Tullio Levi-Civita had spent the intervening decades building on top of it, the very tensor calculus this book will spend the whole of chapter 2 constructing from scratch.

In 1913 Einstein and Grossmann published together a first attempt, the so called *Entwurf* (draft) theory, which used Riemann's curved metric but, out of a mixture of genuine physical worry and outright mathematical error, stopped short of demanding that the field equations hold in every coordinate system (Einstein and Grossmann, 1913). The *Entwurf* theory predicted the wrong value for the century old, already precisely measured anomaly in the precession of Mercury's perihelion, and Einstein knew it. The final months, in Berlin during the autumn of 1915, were by his own account among the most intense of his life: in a sequence of four papers delivered to the Prussian Academy of Sciences on four successive Thursdays in November, he corrected his errors in full public view, week by week, until on 25 November 1915 he presented the field equations essentially in the form we use today (Einstein, 1915), and applied them almost immediately to recover Mercury's precession exactly, a moment he later described as having given him heart palpitations severe enough to worry he might be physically unwell.

## 1.7 Riemann, Hilbert and the Question of Theft

It is sometimes claimed, in the looser corners of popular science writing, that Einstein "stole" general relativity from Riemann, and it is worth being precise about why this particular accusation does not survive contact with the historical record, even though a genuine and fascinating priority dispute does sit at the heart of this story, just with a different antagonist. Riemann died in 1866, forty nine years before Einstein wrote down a single field equation, and he never worked on gravity, never proposed a field equation, and never claimed to have solved the problem this chapter has been tracing since Aristotle: Riemann supplied a mathematical *language*, the geometry of curved manifolds, not a physical theory of gravitation, and Einstein always said so plainly and gratefully, both in his 1913 paper with Grossmann and later in his own *Autobiographical Notes* (Einstein, 1949). Crediting Riemann with the mathematics he actually built, generously and by name, is simply accurate history, not theft.

The real, and rather more dramatic, priority dispute concerns not Riemann but David Hilbert, the towering Göttingen mathematician who was, quite independently, hunting for the same field equations in the very same autumn of 1915. Einstein had visited Göttingen that summer and lectured on his still unfinished theory, and Hilbert, characteristically supremely confident (he is reported to have remarked, only half in jest, that "*every boy in the streets of Göttingen understands more about four dimensional geometry than Einstein*"), so thoroughly had David Hilbert's own city absorbed the mathematics Einstein was still struggling to apply), set to work on the problem from a different

direction entirely, a variational principle built from the very Einstein-Hilbert action we shall derive in chapter 3. Hilbert submitted a paper to the Göttingen Academy titled “Die Grundlagen der Physik” on 20 November 1915, five days before Einstein’s own final, decisive paper of 25 November (Hilbert, 1915), and for decades afterwards a persistent rumour circulated that Einstein had somehow seen and silently borrowed from Hilbert’s unpublished proofs, or conversely that Hilbert had quietly lifted the final covariant equations from Einstein’s own Göttingen lecture.

The matter was only properly settled in 1997, when the historians of science Leo Corry, Jürgen Renn and John Stachel tracked down the actual surviving galley proofs of Hilbert’s paper, printer’s proofs dated December 1915, sitting quietly in the Göttingen archive (Corry et al., 1997). Those proofs, it turned out, did *not* yet contain the fully generally covariant field equations that appeared in Hilbert’s paper when it was finally published, months later, in 1916: Hilbert had revised his own manuscript after that publication date, almost certainly after seeing Einstein’s completed November papers, to bring it into line with Einstein’s result. **Einstein reached the correct, generally covariant field equations first, and independently**, while Hilbert arrived at essentially the same equations by a beautiful and genuinely independent variational route, close behind him and very possibly influenced, in the final polish of his own paper, by Einstein’s success. Hilbert himself, to his considerable credit, never once publicly disputed Einstein’s priority and always referred to the result as “Einstein’s equations”, and the two men remained on entirely cordial, mutually respectful terms for the rest of their lives. If there is a lesson here for a student just beginning this subject, it is that the discovery of general relativity was not a single flash of individual genius but the convergence of an entire community, Gauss’s intrinsic curvature, Riemann’s manifolds, Ricci and Levi-Civita’s calculus, Grossmann’s tutoring, and Hilbert’s own parallel and generous rivalry, all funnelled through Einstein’s singular physical insight that gravity simply *is* the shape of space-time.

## 1.8 Confirmation: From the Trenches to the Eclipse

Extraordinary claims of this kind demand to be tested, and their testing produced two of the most romantic episodes in the whole history of physics. The first came almost immediately. Karl Schwarzschild, a distinguished German astrophysicist who had volunteered for military service and was at the time calculating artillery trajectories on the Russian front, read Einstein’s November 1915 papers and, within a matter of weeks, working between his military duties, found the first exact, non approximate solution to Einstein’s notoriously difficult field equations, the solution describing the curved space-time around a single spherical, non rotating mass, which we will study closely in chapter 3 and which today bears his name (Schwarzschild, 1916). Schwarzschild mailed his solution to Einstein directly from the front; Einstein, visibly delighted, presented it to the Prussian Academy on Schwarzschild’s behalf in January 1916. Schwarzschild had, tragically, already contracted a rare and painful autoimmune skin disease at the front, and he died only a few months later, in May 1916, never learning that his solution would one day predict the existence of black holes.

The second episode made Einstein a global celebrity almost overnight. General relativity predicted that starlight passing close to the Sun should be deflected by the Sun’s gravity by twice the amount Newtonian gravity alone would predict, an effect only observable during a total solar eclipse, when the

Sun’s own overwhelming brightness is briefly blotted out and stars that ordinarily hide directly behind it become visible. On 29 May 1919, two expeditions organised chiefly by the British astronomers Arthur Eddington and Frank Dyson, one to the island of Príncipe off the coast of West Africa and one to Sobral in northern Brazil, photographed the stars around the eclipsed Sun and found, after months of careful measurement, a deflection matching Einstein’s relativistic prediction and decisively ruling out the smaller Newtonian value (Dyson et al., 1920). When the results were announced at a joint meeting of the Royal Society and the Royal Astronomical Society in London that November, the news spread with astonishing speed: the *New York Times* greeted it on 10 November 1919 with the headline “Lights All Askew in the Heavens”, and within weeks Einstein, until then a name known chiefly to physicists, had become the most famous scientist on Earth.

## 1.9 Beyond General Relativity

The theory did not stop working once it was confirmed, and it has continued, ever since, to reveal consequences even its own author found hard to accept. In 1917 Einstein had inserted an extra term, the *cosmological constant*, into his own field equations purely to force them to describe a static, unchanging universe, since a static cosmos was simply what everyone, Einstein included, assumed the real universe to be. The Russian mathematician Alexander Friedmann in 1922 (Friedmann, 1922) and, independently, the Belgian priest and astronomer Georges Lemaître in 1927 (Lemaître, 1927), showed that Einstein’s own unmodified equations naturally describe an expanding, evolving universe instead, a result Einstein initially, and famously, resisted. It took Edwin Hubble’s 1929 observation that distant galaxies are indeed systematically receding from us, and at a rate proportional to their distance, exactly as Friedmann and Lemaître had predicted, to settle the matter (Hubble, 1929), and Einstein is widely (if perhaps not entirely reliably) reported to have called his own cosmological constant the biggest blunder of his career.

General relativity’s most extreme prediction took even longer to be taken seriously. In 1939 J. Robert Oppenheimer and his student Hartland Snyder showed mathematically that a sufficiently massive star, having exhausted its nuclear fuel, must collapse without limit under its own gravity, its surface eventually crossing a point of no return beyond which not even light can climb back out (Oppenheimer and Snyder, 1939), but the paper was largely ignored for nearly three decades, overshadowed by the Second World War and by a lingering, widely shared conviction among physicists, Einstein among them, that nature would surely find some way to forbid so pathological an outcome. It was the physicist John Archibald Wheeler who, in a 1967 lecture, popularised a short, vivid name for these collapsed objects, *black holes*, a term that stuck instantly where more cumbersome earlier phrases such as “gravitationally completely collapsed object” never had (Wheeler and Ford, 1998), and it was Wheeler, too, who gave us the aphorism already quoted on this chapter’s own title page: spacetime tells matter how to move, and matter tells spacetime how to curve.

Einstein’s 1916 prediction of gravitational waves, tiny ripples of curvature radiating outward from accelerating masses, waited even longer for its confirmation, precisely one century, until on 14 September 2015 the two detectors of the Laser Interferometer Gravitational Wave Observatory (LIGO) simultaneously registered the unmistakable signature of two black holes, thirty six and

twenty nine times the mass of the Sun, spiralling into one another and merging over a billion years ago, a discovery announced to the world in February 2016 and one we shall return to properly in chapter 4 (Abbott, B. P. et al., 2016).

And there the story remains genuinely, actively unfinished. General relativity describes gravity magnificently at the scale of planets, stars and galaxies, yet it still refuses to speak the same language as quantum mechanics, the theory that governs the very small, and no one yet possesses a fully working theory of *quantum gravity* capable of describing, for instance, what truly happens at the very centre of a black hole or in the first unimaginably brief instant of the universe's own existence. String theory, loop quantum gravity and a handful of other, still more speculative approaches are today's competing attempts to write the next chapter, each borrowing, in one way or another, from the very same geometrical inheritance that runs through this one: Gauss's intrinsic curvature, Riemann's manifolds, and Einstein's insight that geometry itself can be physics. It is worth carrying this long procession of names in mind as we turn, in chapter 2, to build the mathematics properly from first principles, because every symbol we are about to write down, the metric, the connection, the curvature tensor, was earned by someone, often across a lifetime and occasionally at real personal cost, and general relativity is best understood not as a single flash of individual genius but as the latest, and very probably not the last, chapter in humanity's long and stubborn argument about why things fall.

## *Chapter 2*

### *Differential Geometry*

*Before space-time could bend, mathematics first  
had to learn how a surface curves upon itself.*

“Geometry is not true, it is advantageous.”

– Henri Poincaré

## 2.1 Differential geometry

### 2.1.1 Differential manifolds

This is one of the most attractive theories in physics and maths because it allowed physicist to arrive one of the most important theories of modern physics, the general relativity, a geometrical theory of the space and time bounded as one entity called *space-time* all the implications of our reality this theory allowed us to see beyond our planet, allowing us to travel beyond this earth and dreaming of really strange systems, galaxies o universes, here we're going to study briefly the foundations of the differential geometry and the differentiable manifolds.

If we want to make a correct study of the differential geometry we should begin from the sets theory, then topology, manifolds and then arrive to differential manifolds.

**Definition 1** *A  $n$ -dimensional differential manifold is a set  $M$  with a collection of subsets  $\mathcal{O}_\alpha$  such that:*

1.  $\bigcup_{i=1}^n \mathcal{O}_i = M$ . This means that the finite union of all the subsets covers the manifold  $M$ .
2.  $\forall \alpha \exists \phi_\alpha \mid \phi_\alpha : \mathcal{O}_\alpha \rightarrow U_\alpha \ \& \ U_\alpha \subset \mathbb{R}^n$ . For every subset  $\mathcal{O}_\alpha$  of the manifold  $M$  exists a bijective map which goes from the subset to an open subset of  $\mathbb{R}^n$ .
3. if  $\mathcal{O}_\alpha$  and  $\mathcal{O}_\beta$  intersect such that  $\mathcal{O}_\alpha \cap \mathcal{O}_\beta \neq \emptyset$ , then  $\phi_\beta \circ \phi_\alpha^{-1}$  maps from  $\phi_\alpha(\mathcal{O}_\alpha \cap \mathcal{O}_\beta) \subset U_\alpha \subset \mathbb{R}^n$  to  $\phi_\beta(\mathcal{O}_\alpha \cap \mathcal{O}_\beta) \subset U_\beta \subset \mathbb{R}^n$ . This map must be smooth (infinitely differentiable.)

Here we must say some comments or observations:

- The maps  $\phi_\alpha$  are usually called coordinate charts or coordinate systems, we will later see this is directly related to the strong equivalence principle.
- $\{\phi_\alpha\}$  is called an Atlas.
- A point  $p$  represented in a coordinate chard  $\phi_\alpha$  is represented as  $\phi_\alpha(p) = (x_\alpha^1(p), \dots, x_\alpha^n)$  and the components  $x_\alpha^i$  are called the coordinates of  $p$ .

Here we will use often the word *smooth* this refers to the differentiability of the geometrical object, we say something is smooth if it is of class  $C^\infty$  this means it is infinitely differentiable, usually (specially in physics) such objects are hard to find, so we work on objects of class  $C^k$  or  $k$ -times differentiable. Lets see some examples:

1.  $\mathbb{R}^n$  by its own represents a manifold whose atlas is a chart  $\phi : (x^1, \dots, x^n) \rightarrow (x^1, \dots, x^n)$ . This is a manifold of dimension  $n$ .

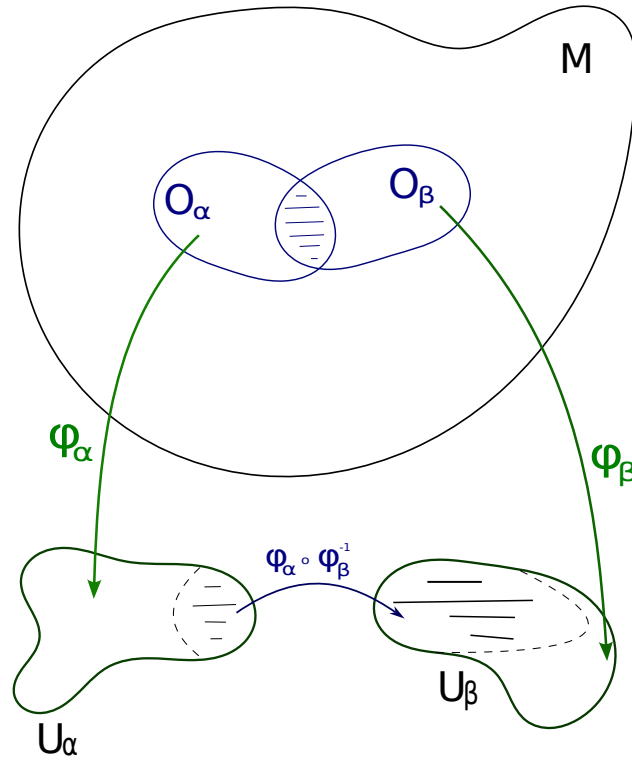


Figure 2.1: Schematic representation of the third postulate what it means is that in a differential manifold we can always go from one coordinate chart to another.

2.  $S^1$  also called the unitary circle;  $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ . It is a one dimension manifold because we only need one parameters to describe it and also at same time it is a subset of  $\mathbb{R}^2$  given by  $(\cos\theta, \sin\theta)$ , where  $\theta \in \mathbb{R}$ , lets see how to cover this manifold

We cannot set a closed interval lets say  $0 \leq \theta \leq 2\pi$  because then the map  $\phi = \theta$  would not be bijective because at  $\theta = 0$  and  $\theta = 2\pi$  it would have the same value, because of that we choose two coordinate charts  $\phi_1 = S^1 - \{p\}$   $\phi_1(p) = \theta_1$  and the point we remove is  $p = (1, 0)$  and the second map  $\phi_2 = S^1 - \{q\}$   $\phi_2(q) = \theta_2$  where the point we had removed is  $q = (-1, 0)$ .

From here we can see what is the Atlas for this manifold it is

$$A = \{\phi_1, \phi_2\}, \quad (2.1)$$

so now, how using this both coordinate charts can we built on  $S^1$ , how are related  $\theta_1$  and  $\theta_2$  from the third postulate is easy to see that

$$\theta_2 = \phi_2 \circ \phi_1^{-1} = \theta_1 - 2\pi. \quad (2.2)$$

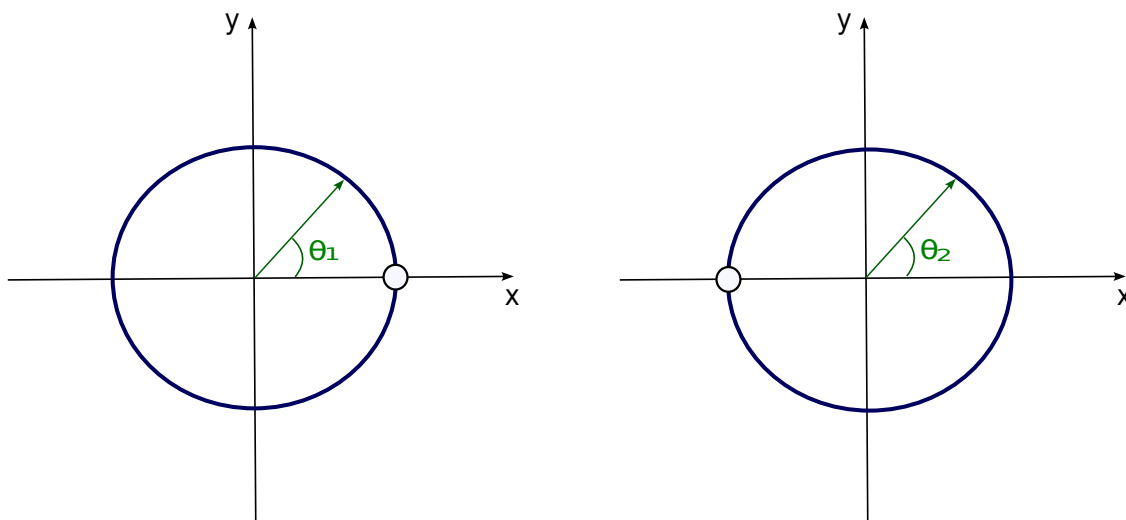


Figure 2.2: In the left there is the first chart  $\phi_1$  which covers almost all the circumference but a point  $p$  and to the right there is the other chart which covers almost all the circumference except a point  $q$  diametrically opposite to  $p$ .

This is also easy to interpret geometrically, this means to get  $\theta_2$  from the chart  $\phi_1$  we only have to do a complete turn of  $\theta_1$  (this is because we had chosen two point diametrically opposite.)

3.  $S^2$ . This is an sphere of unitary radius  $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ . We can choose any point to have an analogous solution to  $S^1$  because any point on the sphere has a enough small vicinity which is a one to one onto a disc of  $\mathbb{R}^2$ , see figure 2.3,

This also illustrates that the map certainly does not preserve neither length nor angles. Consider the spherical coordinates we always use in our courses of calculus, an example of map with  $\theta = x$  and  $\phi = y$ , therefore the sphere seems to be mapped into the rectangle

$$0 \leq x \leq \pi, \quad 0 \leq y \leq 2\pi.$$

The map breaks on the pole, when  $\theta = 0$ . When a point is mapped to the line  $x = 0$ ,  $0 \leq y \leq 2\pi$ , see figure 2.4, then in the pole there is no even a map.

The second problem rise at the points whose polar coordinate is  $\phi = 0$  they are mapped to both places  $y = 0$  and  $y = 2\pi$  therefore there is no map.

If instead of having a closed interval we take an open interval as  $0 < x < \pi$ ,  $0 < y < 2\pi$  using two coordinate charts, the other one map can be taken in the line  $\phi = 0$  in the equator, then  $\phi$  goes from  $\phi = \pi/2$  to  $\phi = 3\pi/2$ .

There is another way to choose the maps for  $S^2$ , called the stereographic projection of the manifold on  $\mathbb{R}^2$ .

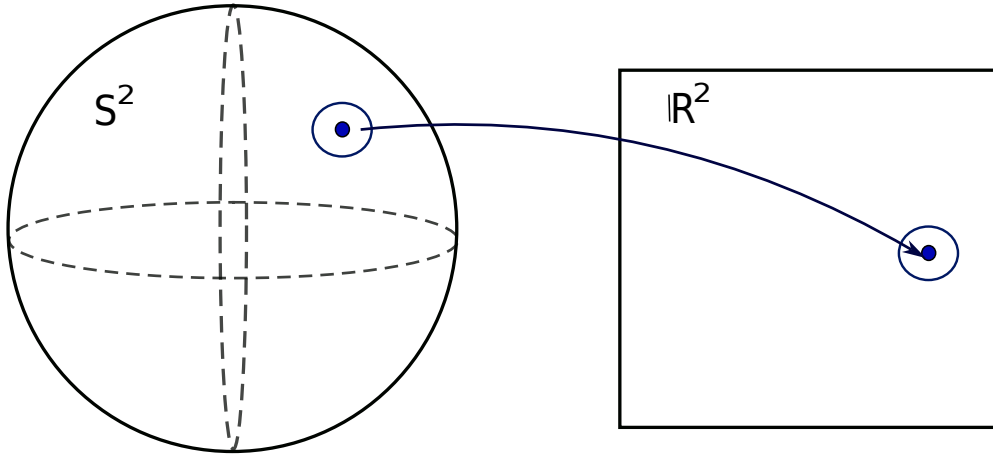


Figure 2.3: Each point on the sphere has a small vicinity which maps as a disc on  $\mathbb{R}^2$ .

The sphere is tangent to the plane at a point  $N$  diametrically opposite to the tangent point. From a point  $P$  on the sphere and  $Q$  in  $\mathbb{R}^2$  this defines the map:  $P$  maps to  $Q$ , the coordinates of  $P$  in  $S^2$  are the coordinates of  $Q$  in  $\mathbb{R}^2$ . It is a one to one map except on  $N$  in the vicinity of  $N$  there is a map which tends to infinity there can be chosen another coordinate chart.

**Definition 2** Two Atlases are compatible if their union is another Atlas. The union of all the compatible atlases is called the **complete atlas**. It is an atlas that it is not in any other atlas.

### 2.1.1.1 Smooth functions

If  $\phi : \mathcal{O} \rightarrow U$  is a coordinate chart and  $f : M \rightarrow \mathbb{R}$  then  $f \circ \phi^{-1}$  is a map from  $U \subset \mathbb{R}^n$  to  $\mathbb{R}$ . Because of that every scalar field implicitly goes from the manifold.

**Definition 3** A function  $f : M \rightarrow \mathbb{R}$  is smooth if and only if, for all coordinate chart  $\phi$  can be define a map  $F \equiv f \circ \phi^{-1}$

$$F : U \rightarrow \mathbb{R} \tag{2.3}$$

is a smooth function.

In general relativity, a function  $f : M \rightarrow \mathbb{R}$  is sometimes called scalar field.

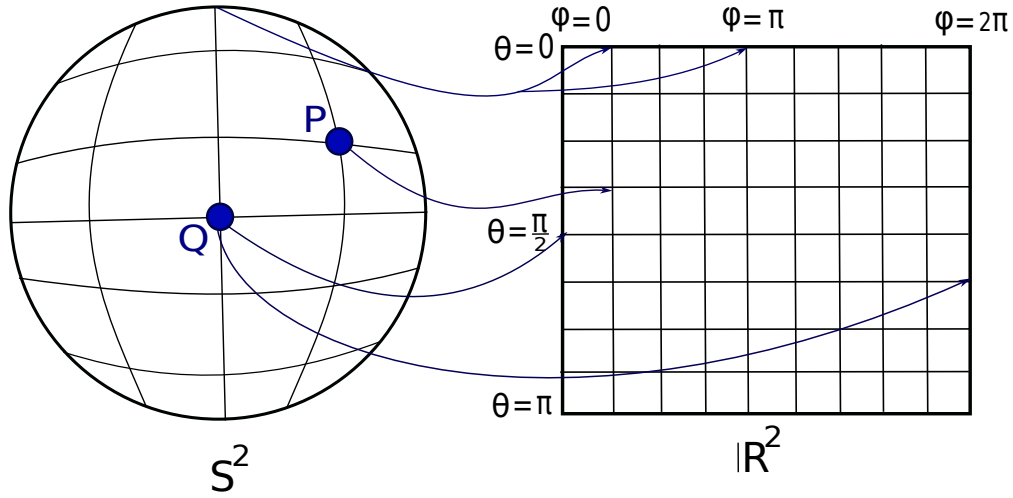


Figure 2.4: Representation of traditional map from  $S^2$  to  $\mathbb{R}^2$  we can see there are multi valued points which won't allow us to cover the manifold  $S^2$ .

#### 2.1.1.2 Curves and vectors

**Definition 4** A smooth curve on a differentiable manifold  $M$  is a smooth function

$$\lambda : I \rightarrow M \quad (2.4)$$

where  $I$  is an open interval in  $\mathbb{R}$ .

This means that  $\phi_\alpha \circ \lambda$  is a map from  $I$  to  $\mathbb{R}^n$  for all the coordinate charts  $\phi_\alpha$  this is represented in the figure 2.7.

The elements of  $I$  can be thought as the parameters that parameterize the curve. Let

$$f : M \rightarrow \mathbb{R} \ \& \ \lambda : I \rightarrow M, \quad (2.5)$$

then we can define the following map

$$f \circ \lambda : I \rightarrow \mathbb{R}. \quad (2.6)$$

This is in fact, the form of a parameterized function  $f(\lambda(t))$  and we talk about its derivative respect to the parameter  $t$ :

$$\frac{d}{dt} [(f \circ \lambda)(t)] = \frac{d}{dt} f(\lambda(t)) = \frac{\partial f}{\partial \lambda} \frac{d\lambda}{dt}. \quad (2.7)$$

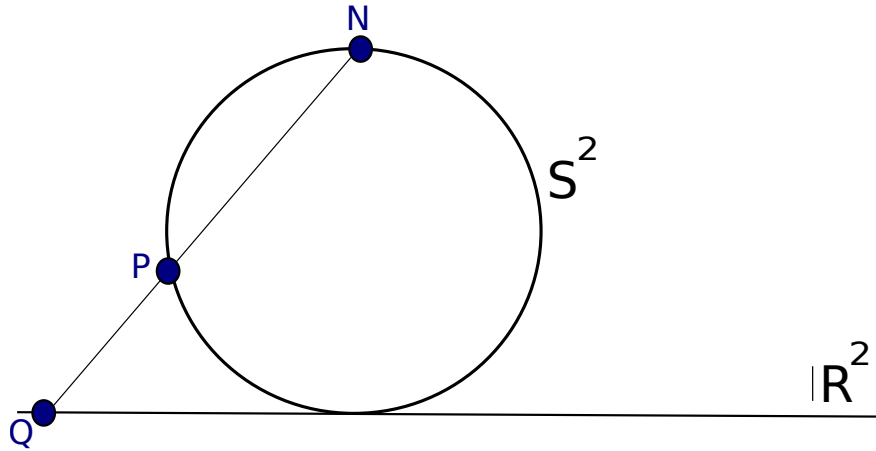


Figure 2.5: Graphical representation to find the stereographic projection of  $S^2$  to  $\mathbb{R}^2$ .

**Definition 5** Let  $\lambda : I \rightarrow M$  a smooth curve with  $\lambda(0) = p$  represents the point  $p$  on the manifold, see figure 2.7. The vector tangent to  $\lambda$  at  $p$  is a linear map  $X_p$  from  $M$  to  $\mathbb{R}$  define by:

$$X_p(f) = \left\{ \frac{d}{dt} [f(\lambda(t))] \right\}_{t=0}. \quad (2.8)$$

This tangent vector satisfies some really important algebraic properties

1. The tangent vector is a linear map:

$$X_p(f + g) = X_p(f) + X_p(g), \quad (2.9)$$

$$X_p(\alpha f) = \alpha X_p(f). \quad (2.10)$$

2. The tangent vector satisfies the Leibniz's rule:

$$X_p(fg) = gX_p(f) + fX_p(g). \quad (2.11)$$

Lets consider a coordinated chart defined in the vicinity of  $P$

$$\phi(x^1, \dots, x^n), \quad (2.12)$$

and we're going to consider the following function  $F = f \circ \phi^{-1}$  there because  $\phi$  is a bijective map we can write

$$f \circ \lambda = f \circ \phi^{-1} \circ \phi \circ \lambda, \quad (2.13)$$

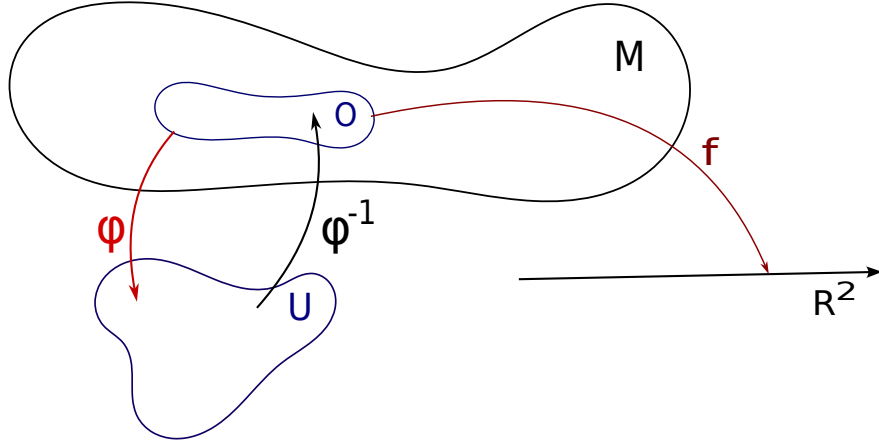


Figure 2.6: Schematic representation of the map  $f$ , this takes elements from the manifold and maps them to the rect of the reals  $\mathbb{R}$ .

or

$$f \circ \lambda = F \circ \phi \circ \lambda. \quad (2.14)$$

The tangent vector, using the chain rule, is given by

$$X_p(f) = \left\{ \frac{d}{dt} [f(\lambda(t))] \right\}_{t=0} = \left( \frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left( \frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0}, \quad (2.15)$$

this means we, in general, can write every tangent vector as a linear combination of some components in product with some basis elements.

**Theorem 1** *The set of all tangent vectors to  $p$  form an  $n$ -dimensional vectorial space, called the tangent space  $T_p(M)$*

**Demonstration:**

Let  $\lambda_1$  and  $\lambda_2$  two curves such that  $\lambda_1(0) = \lambda_2(0) = p$  and let  $X_p$  and  $Y_p$  their tangent vectors, respectively we must define the multiplication by a constant and the addition of two vectors. Let  $\phi(x^1, \dots, x^n)$  a coordinate chart defined in a vicinity of  $p$ . Consider the following curve

$$\nu(t) = \phi^{-1} \{ \alpha [\phi(\lambda_1(t)) - \phi(p)] + \beta [\phi(\lambda_2(t)) - \phi(p)] + \phi(p) \}, \quad (2.16)$$

we can see that at  $t = 0$  the curve is  $\nu(0) = p$ . Let  $Z_p$  the tangent vector to this curve on  $p$ , then

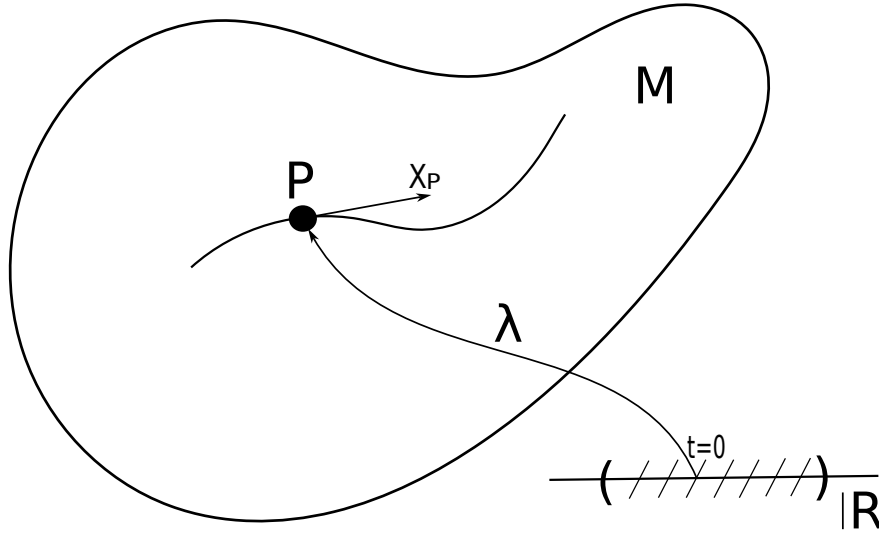


Figure 2.7: The curve  $\lambda$  maps from an open interval of the reals to the manifold, in the manifold this maps looks like, (remember this is just an scheme), a curve we see usually, from a point in that curve we can define a tangent vector  $X_p$ .

$$Z_p(f) = \left( \frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left\{ \frac{d}{dt} [\alpha(x^\mu(\lambda_1(t)) - x^\mu(p)) + \beta(x^\mu(\lambda_2(t)) - x^\mu(p)) + x^\mu(p)] \right\}_{t=0}, \quad (2.17)$$

$$Z_p(f) = \left( \frac{\partial F(x)}{\partial x^\mu} \right)_{\phi(p)} \left[ \alpha \left( \frac{d}{dt} x^\mu(\lambda_1(t)) \right)_{t=0} + \beta \left( \frac{d}{dt} x^\mu(\lambda_2(t)) \right)_{t=0} \right], \quad (2.18)$$

$$Z_p(f) = \alpha X_p(f) + \beta Y_p(f), \quad (2.19)$$

$$Z_p(f) = (\alpha X_p + \beta Y_p)(f). \quad (2.20)$$

Since this is true for any smooth function  $f$  we have  $Z_p(f) = \alpha X_p(f) + \beta Y_p(f)$ . Therefore  $Z_p$  is tangent to the curve  $\nu$  at  $p$ . Then is deduced the set of tangent vectors to  $p$  form a vectorial space.

Now, we gotta prove it is  $n$ -dimensional, to do this lets define a basis. Let  $1 \leq \mu \leq n$ . Consider the curve  $\lambda_\mu$  along  $p$  defined by:

$$\lambda_\mu(t) = \phi^{-1} (x^1(p), \dots, x^{\mu-1}(p), x^\mu(p) + t, x^{\mu+1}(p), \dots, x^n(p)). \quad (2.21)$$

The tangent vector to this curve on  $p$  is denoted by  $\left( \frac{\partial}{\partial x^\mu} \right)_p$ . To see this lets see that:

$$\left( \frac{\partial}{\partial x^\mu} \right)_p (f) = \left( \frac{\partial F}{\partial x^\mu} \right)_{\phi(p)} \quad (2.22)$$

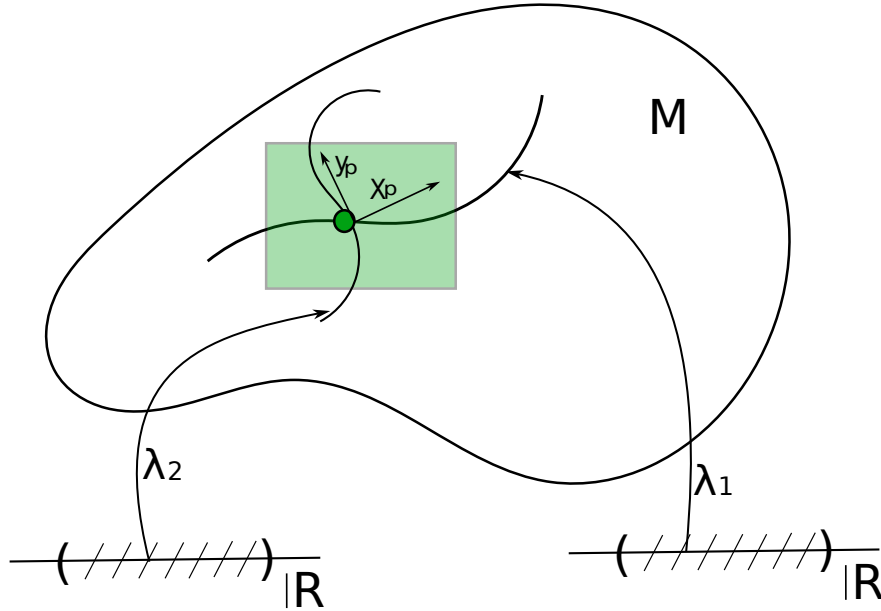


Figure 2.8: Two tangent vectors  $X_p$  and  $Y_p$  which both are tangent to  $p$  but live in two different curves,  $(\lambda_1, \lambda_2)$  but in the same vectorial space  $T_p(M)$ .

this is easy to see because we had define that  $F = f \circ \phi^{-1}$  but it is evaluated at  $\phi(p)$  so the global effect of the coordinated chart is balance and it became evaluated at  $p$  instead of  $\phi(p)$ . Lets see the  $n$  tangent vectors  $\left(\frac{\partial}{\partial x^\mu}\right)_p$  are linearly independent, if the vectors are tangent then the following relation is fulfilled

$$\sum_{\mu=1}^n a_\mu \left(\frac{\partial}{\partial x^\mu}\right)_p = 0 \rightarrow a_1 = a_2 = \dots = a_n = 0. \quad (2.23)$$

$$\begin{aligned} \sum_{\mu=1}^n a_\mu \left(\frac{\partial}{\partial x^\mu}\right)_p &= 0, \\ \left(\frac{\partial}{\partial x^\nu}\right) \sum_{\mu=1}^n a_\mu \left(\frac{\partial}{\partial x^\mu}\right)_p &= \sum_{\mu=1}^n a_\mu \left(\frac{\partial^2}{\partial x^\nu \partial x^\mu}\right)_p \frac{\partial x^\mu}{\partial x^\nu} = 0, \\ &= \sum_{\mu=1}^n a_\mu \left(\frac{\partial^2}{\partial x^\nu \partial x^\mu}\right)_p \delta_\nu^\mu = a_\nu \left(\frac{\partial^2}{\partial (x^\nu)^2}\right)_p = 0 \end{aligned}$$

and because we're working of smooth functions such second derivative must exist and no always be zero because otherwise we would be working on a trivial manifold, the empty set  $\emptyset$  thus using the elemental theory of algebra

$$a_\nu \left( \frac{\partial^2}{\partial (x^\nu)^2} \right)_p = 0 \rightarrow a_\nu = 0 \quad \forall \nu \in [1, n]. \quad (2.24)$$

Finally, once proven they're linearly independent we then have

$$X_p(f) = \left( \frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0} \left( \frac{\partial}{\partial x^\mu} \right)_p (f) \quad (2.25)$$

this is true for every  $f$  therefore:

$$X_p = \left( \frac{dx^\mu(\lambda(t))}{dt} \right)_{t=0} \left( \frac{\partial}{\partial x^\mu} \right)_p. \quad (2.26)$$

$X_p$  can be written as a linear combination of  $n$  tangent vectors  $\left( \frac{\partial}{\partial x^\mu} \right)_p$ . This  $n$  vectors then form a base for  $T_p(M)$ , which establish the tangent space is  $n$ -dimensional.

The base  $\left\{ \left( \frac{\partial}{\partial x^\mu} \right)_p : \mu = 1, \dots, n \right\}$  is dependent of the coordinate chart: it is necessary to choose a coordinate chart  $\phi$  defined in a vicinity of  $p$ . Choosing a different chart it would give to  $T_p(M)$ . A basis chosen like this is called a coordinate basis.

We often want to represent the something in one coordinate system or in another, so we must know how to transform the components of the vector from one representation to another. Let  $\phi = (x^1, \dots, x^n)$  and  $\bar{\phi} = (\bar{x}^1, \dots, \bar{x}^{\bar{n}})$  two coordinate charts thus

$$\left( \frac{\partial}{\partial x^\mu} \right)_p (f) = \left( \frac{\partial}{\partial x^\mu} (f \circ \phi^{-1}) \right)_{\phi(p)} = \left\{ \frac{\partial}{\partial x^\mu} [(f \circ \bar{\phi}^{-1}) \circ (\bar{\phi} \circ \phi^{-1})] \right\}_{\phi(p)}, \quad (2.27)$$

$$\bar{F} = f \circ \bar{\phi}^{-1} \quad : \quad \bar{F} = \bar{F}(\bar{X}). \quad (2.28)$$

$$\begin{aligned} \left( \frac{\partial}{\partial x^\mu} \right)_p (f) &= \left( \frac{\partial}{\partial x^\mu} [\bar{F}(\bar{x}(x))] \right)_{\phi(p)} \\ &= \left( \frac{\partial x^{\bar{\nu}}}{\partial x^\mu} \right)_{\phi(p)} \left( \frac{\partial \bar{F}(\bar{x})}{\partial x^{\bar{\nu}}} \right)_{\bar{\phi}(p)} \\ &= \left( \frac{\partial x^{\bar{\nu}}}{\partial x^\mu} \right)_{\phi(p)} \left( \frac{\partial}{\partial x^{\bar{\nu}}} \right)_p (f). \end{aligned}$$

In this way we have found the transformation rule for the components of a vector from one coordinate system to another which is given by

$$\left( \frac{\partial}{\partial x^\mu} \right)_p = \left( \frac{\partial x^{\bar{\nu}}}{\partial x^\mu} \right)_{\phi(p)} \left( \frac{\partial}{\partial x^{\bar{\nu}}} \right)_p, \quad (2.29)$$

every vector can be seen then as

$$X_p = x^\nu \left( \frac{\partial}{\partial x^\nu} \right)_p = x^\nu \left( \frac{\partial x^{\bar{\mu}}}{\partial x^\nu} \right) \left( \frac{\partial}{\partial x^{\bar{\mu}}} \right)_p = x^{\bar{\mu}} \left( \frac{\partial}{\partial x^{\bar{\mu}}} \right)_p. \quad (2.30)$$

### 2.1.1.3 Covectors and one-forms

**Definition 6** Let  $V$  a real vectorial space. The dual space  $V^*$  of  $V$  is a vectorial space which maps linearly from  $V$  to the reals

$$\tilde{P} : V \rightarrow \mathbb{R}. \quad (2.31)$$

**Lemma:** If  $V$  is  $n$ -dimensional, then  $V^*$  too. If  $\{\mathbf{e}_\mu : \mu = 1, \dots, N\}$  is a basis of  $V$  then  $V^*$  has a base  $\{f^\mu : \mu = 1, \dots, N\}$  (dual basis), define by  $f^\mu(\mathbf{e}_\nu) = \delta^\mu_\nu$ . If  $X = x^\alpha \mathbf{e}_\alpha$  then:

$$f^\alpha(x) = f^\alpha(x^\beta \mathbf{e}_\beta) = x^\beta f^\alpha(\mathbf{e}_\beta) = x^\alpha. \quad (2.32)$$

As  $V$  and  $V^*$  have the same dimension, we can then say they're isomorphic, therefore there exists an isomorphism.

**Theorem 2** If  $V$  is dimensionally finite, then  $(V^*)^*$  is naturally isomorphic to  $V$ .

$$\phi : V \rightarrow (V^*)^* \equiv \phi(x)(\omega) = \omega(x) \quad \forall \omega \in V^*. \quad (2.33)$$

**Definition 7** The dual space of  $T_p(M)$  is called the cotangent space in  $p$  and is represented as  $T_p^*(M)$ .

**Definition 8** Let  $f : M \rightarrow \mathbb{R}$  a smooth function, a covector  $(df)_p$  is given by  $(df)_p(x) = x(f)$  for any  $x \in T_p(M)$ . It is also called the gradient of  $f$  at the point  $p$ . This covector or one-form satisfies the following properties

1. If  $\eta \in V^*$  then

$$\eta(\mathbf{e}_\mu) = \eta_\nu f^\nu(\mathbf{e}_\mu) = \eta_\mu. \quad (2.34)$$

2. If  $X \in T_p(M)$  then:

$$\eta(x) = x^\alpha \eta_\alpha \in \mathbb{R}. \quad (2.35)$$

A one-form is a really special geometrical object, because we constantly use it in physics but usually is late in our studies when we start to think about it, for example, if vectors are represented as directed arrows, how does it look like a covector or one-form? To see this let's consider the interaction between the vector  $\vec{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$  and the one-form  $\tilde{q} = [x, y]$  thus

$$\tilde{q}(\vec{A}) = x + 2y. \quad (2.36)$$

The equation 2.36 represents a linear rect such that their normal vector has components  $[1, 2]^T$  this means that meanwhile vectors are directed arrows, one-form represent orthogonal planes which folds the vector, the number of planes that folds the vector is the norm of the vector.

What is the most common operation that gets an scalar and returns orthogonal vectors to the scalar field? The gradient, in fact, the gradient is the most common example of an one-form, they represent the orthogonal vector to the equi-potential surfaces, that is the reason why we denote the uno-form in a point  $p$  as  $(df)_p$ , this also allow us to realize that electric field in a Coloumbian Gauge is not a vector it is an uno-form and the vector we always work with is the magnetic field.

#### 2.1.1.4 Tensors

A tensor of kind  $\binom{r}{s}$  in  $p$  is a multilinear map

$$T : T_p^*(M) \times \dots \times T_p^*(M) \times T_p(M) \times \dots \times T_p(M) \rightarrow \mathbb{R} \quad (2.37)$$

which maps  $r$  uno-forms and  $s$  vectors into the reals, we can also represent it as

$$T = T(\tilde{q}_1, \dots, \tilde{q}_r, \vec{A}_1, \dots, \vec{A}_s) \rightarrow \mathbb{R}, \quad (2.38)$$

and we can calculate the components of the tensor as

$$T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r} = T(f^{\mu_1}, \dots, f^{\mu_r}, \mathbf{e}_{\nu_1}, \dots, \mathbf{e}_{\nu_s}). \quad (2.39)$$

**Definition 9** If  $S$  is a tensor of type  $\binom{p}{q}$  and  $T$  is a tensor of type  $\binom{r}{s}$  then the tensorial product between  $S$  and  $T$ , denoted by  $S \otimes T$  and it is a tensor of type  $\binom{p+r}{q+s}$

$$S \otimes T = (\omega_1, \dots, \omega_p, \eta_1, \dots, \eta_r, x_1, \dots, x_q, y_1, \dots, y_s) = S(\omega_1, \dots, \omega_p, \eta_1, \dots, \eta_r) T(x_1, \dots, x_q, y_1, \dots, y_s). \quad (2.40)$$

### 2.1.1.5 Tensor fields

**Definition 10** A vectorial field is a map  $X$  which maps any point  $p \in M$  to a vector  $X_p$  in  $p$

$$X(f) : M \rightarrow \mathbb{R} \ \& \ X(f) : p \rightarrow X_p(f). \quad (2.41)$$

These vectorial fields talk about how a physical quantity could change from a point to another inside a manifold, usually we see a vectorial field as an arrows fields but that is in the  $\mathbb{R}^n$  manifold, this last definition is more general and those abstract arrows can be interpreted as the changing rate of our physical quantities, and this is smooth if  $f$  is smooth.

**Definition 11** A covectorial field is a map  $\omega$  from a point  $p \in M$  to a covector  $\omega_p$  in  $p$

$$\omega(\vec{X}) : M \rightarrow \mathbb{R} \ \& \ \omega(\vec{X}) : p \rightarrow \omega_p(\vec{X}_p). \quad (2.42)$$

This is analogous to the vectorial field but in this case there is no analogous to an arrows space, this would be the manifold foliations and we must remember that the smoothness of a vectorial or covectorial fields, depends on the smoothness of the components and the functions  $f$ .

**Definition 12** A  $\binom{r}{s}$  tensor field is a map  $T$ , which maps from any point  $P \in M$  to an  $\binom{r}{s}$  tensor  $T_p$  in  $p$

$$T(\eta_1, \dots, \eta_r, \vec{X}_1, \dots, \vec{X}_s) : M \rightarrow \mathbb{R}, \quad (2.43)$$

$$P \rightarrow T_p \left( (\eta_1)_p, \dots, (\eta_r)_p, (\vec{X}_1)_p, \dots, (\vec{X}_s)_p \right). \quad (2.44)$$

The tensor fields can be thought as the generalization of the vector fields and covector fields, this tensor field is the hearth of many physical theories, not only the general or special relativity, in solid state physics, in particle physics, even in the crystals optics, it is very important because, as we will see later, with this tensor we can define the notion of events distance <sup>1</sup> and also will allow us to define physical quantities that do not depend on the coordinate chart such quantities as the Maxwell's tensor, which tell us how much electromagnetic energy density is in our system, the Faraday's electromagnetic tensor which tell us how are related the electric and magnetic fields and the conservations laws in a dynamical system, the pressure tensor, which tell us about the pressure fields in every direction of a fluid, even in a magnetohydrodynamical fluid.

Tensors had been the hearth of the 20th century physics but now there are more geometrical objects, such as spinors, such are define ad fractional rank tensors or even the twistors of Penrose which contain both tensors and spinors.

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<sup>1</sup>this is also related to a topology in the manifold  $M$

## 2.1.1.6 Commutator

We're going to analyze two vectorial fields  $\vec{X}$ ,  $\vec{Y}$  and we're going to take a smooth function  $f$ , and also let's define another smooth function  $\vec{Y}(f)$  so is then  $\vec{X}(\vec{Y}(f))$  a vector field? To answer this we first must see this new function have to satisfy the linearity and Leibniz rule

$$\vec{X}(\vec{Y}(fg)) = \vec{X}(f\vec{Y}(g) + g\vec{Y}(f)) = f\vec{X}(\vec{Y}(g)) + g\vec{X}(\vec{Y}(f)) = \vec{X}(f)\vec{Y}(g) + \vec{X}(g)\vec{Y}(f).$$

we then can appreciate this new function  $\vec{X}$  is not a vector field it does not fulfill the Leibniz rule, but we can define another map  $\vec{Z}(f)$  as:

$$\vec{Z}(f) = \vec{X}(\vec{Y}(f)) - \vec{Y}(\vec{X}(f)), \quad (2.45)$$

which is a vectorial field.

**Definition 13** The commutator of two vector field  $\vec{X}$  and  $\vec{Y}$  is a vector field defined as

$$[\vec{X}, \vec{Y}](f) = \vec{X}(\vec{Y}(f)) - \vec{Y}(\vec{X}(f)) \quad \forall f \in C^\infty. \quad (2.46)$$

We can expand the definition of commutator to see how does it operates in a given coordinated chart  $\phi_1 = (x^1, \dots, x^n)$

$$\begin{aligned} [\vec{X}, \vec{Y}](f) &= \vec{X}\left(Y^\alpha \frac{\partial F}{\partial x^\alpha}\right) - \vec{Y}\left(X^\beta \frac{\partial F}{\partial x^\beta}\right) \\ &= X^\beta \frac{\partial}{\partial x^\beta} \left(Y^\alpha \frac{\partial F}{\partial x^\alpha}\right) - Y^\alpha \frac{\partial}{\partial x^\alpha} \left(X^\beta \frac{\partial F}{\partial x^\beta}\right) \\ &= \left(X^\beta \frac{\partial}{\partial x^\beta} Y^\alpha \frac{\partial F}{\partial x^\alpha} - Y^\alpha \frac{\partial X^\beta}{\partial x^\alpha} \frac{\partial F}{\partial x^\beta}\right) \\ [\vec{X}, \vec{Y}](f) &= \left(X^\alpha \frac{\partial Y^\beta}{\partial X^\alpha} - Y^\alpha \frac{\partial X^\beta}{\partial Y^\alpha}\right) \frac{\partial F}{\partial x^\beta}, \end{aligned}$$

$$[\vec{X}, \vec{Y}](f) = [\vec{X}, \vec{Y}]^\beta \left(\frac{\partial}{\partial x^\beta}(f)\right)_p, \quad (2.47)$$

$$[\vec{X}, \vec{Y}]^\beta = \left(X^\alpha \frac{\partial Y^\beta}{\partial X^\alpha} - Y^\alpha \frac{\partial X^\beta}{\partial Y^\alpha}\right). \quad (2.48)$$

The commutator satisfies some four important relations:

1.  $[\vec{X}, \vec{Y}] = -[\vec{Y}, \vec{X}]$  ; the commutator is antisymmetric.
2.  $[\vec{X}, \vec{Y} + \vec{Z}] = [\vec{X}, \vec{Y}] + [\vec{X}, \vec{Z}]$ ; the commutator is a linear map.
3.  $[\vec{X}, f\vec{Y}] = f[\vec{X}, \vec{Y}] + \vec{X}(f)\vec{Y}$ ; the commutator is closed under the product with any smooth function  $f$ .
4.  $[\vec{X}, [\vec{Y}, \vec{Z}]] + [\vec{Y}, [\vec{Z}, \vec{X}]] + [\vec{Z}, [\vec{X}, \vec{Y}]] = 0$  ; the Jacobi identity.

We will later see this commutator has a lot of physical information because we can set the Riemann tensor, which tell us how is the geometry of the system, in term of the commutator and it allows it to be independent from a coordinate chart also we will see it later will allow us to see if some basis is a coordinate basis or not.

### 2.1.1.7 Integral curves

In the common language the term curve, vector field are used often but in fact, if we look carefully, they usually are separated but they are not, if we turn back to our elemental calculus classes we can remember in  $\mathbb{R}^3$  a curve is defined as a vector whose components are parameterized functions, in we look carefully it is exactly what we've been developing to this point, a curve is a map from a interval  $I$  parameterized with a parameter  $t$  to the manifold  $\mathbb{R}^3$  which is represented as a vector  $\lambda = [X(t), Y(t), Z(t)]^T$ .

It is time to formalize all we've studied in our calculus classes using as a mental picture the speed of a flux, which (no only in special relativity) is usually written as  $\vec{U}(\vec{X}) \in \mathbb{R}^3$  a vector field. The behavior of that speed field is characterized by an initial position  $\vec{X}_0$  such that

$$\frac{d\vec{X}}{dt} = \vec{U}(\vec{X}(t)) ; \vec{X}(0) = \vec{X}_0. \quad (2.49)$$

The solution of the equation 2.49 is what we will call *integral curve of  $\vec{U}$  across  $\vec{X}_0$* .

**Definition 14** Let  $\vec{X}$  a vector field over a manifold  $M$  and  $p \in M$ . An integral curve of  $\vec{X}$  across  $p$  is a curve whose tangent in every point is  $\vec{X}$

Let  $\lambda$  an integral curve of  $\vec{X}$  with  $\lambda(0) = p$  and we're going to choose a coordinate chart  $\phi^\mu = (x^1, \dots, x^n)$  therefore

$$\frac{X^\alpha}{dt}(\lambda(t)) = x^\alpha = x^\alpha(\lambda(t)), \quad (2.50)$$

$$X^\alpha(0) = X_p^\alpha. \quad (2.51)$$

The solution to this equation is unique if and only if the initial conditions are set otherwise we would have a dynamical system which can be autonomous or not what will allow us to talk about chaos.

### 2.1.2 The Lie's derivative

There is a truly important concept in general relativity and differential geometry the derivative, but there is no one derivative, there are many kinds and all have got their meaning and applications, there is the ordinary derivative, the partial derivative, the gradient, the divergence, the rotational, the Jacobian, the covariant derivative, the fractional derivative here we are go developpe one special derivative that allow us to pull or push elements from a manifold to another manifold and find out the notion of derivative along a curve and the relation between manifolds which, in general, are different, this is the pull back and push forwards operations, with those we can developpe the *Lie's derivative* a kind of primitive and primordial derivative, the scalar, partial and even the covariant derivative are particular cases of this derivative.

#### 2.1.2.1 Pull back and push forward

Let  $M$  and  $N$  two different manifolds, and  $f$  a smooth function that maps from  $N$  to  $\mathbb{R}$  and let  $\phi$  a bijection which maps from  $M$  to  $N$  Therefore there exists an inverse map  $\phi^{-1} : N \rightarrow M$  such that  $\phi \circ \phi^{-1} = I_N$  where  $I_N$  is the identity function on the manifold  $N$ , and  $\phi^{-1} \circ \phi = I_M$  where  $I_M$  is the identity function on the manifold  $M$ .



## *Chapter 3*

### *The Theory of General Relativity*

*Gravity is not a force reaching across space, it is  
the shape of space itself.*

“Spacetime tells matter how to move; matter tells spacetime how to curve.”

– John Archibald Wheeler

### 3.1 Introduction to general relativity

At the heart of Albert Einstein's general theory of relativity lies a profound shift in our understanding of gravity and space-time. In this section, we'll delve into the qualitative aspects of general relativity and draw comparisons with its predecessor, special relativity.

One of the central tenets of general relativity is the idea that massive objects, like planets and stars, bend the fabric of space-time around them. This bending, often depicted as a curved surface, is what we perceive as gravity. Imagine placing a heavy ball on a stretched rubber sheet. The sheet curves around the ball, and smaller objects nearby naturally roll toward it. Similarly, massive objects create curves in space-time, influencing the motion of other objects in their vicinity.

This insight was a monumental departure from the classical concept of gravity as a force of attraction between masses. Rather than envisioning a force acting at a distance, general relativity describes gravity as a consequence of the geometry of space-time itself.

Special relativity, introduced by Einstein in 1905, focuses on the relationship between space and time, particularly in situations involving relative motion. It famously established the constancy of the speed of light, which led to phenomena like time dilation and length contraction. Special relativity was designed to address situations where velocities approach the speed of light and encompassed the realm of "inertial frames" – frames that move at a constant velocity relative to one another.

General relativity extends this framework to non-inertial frames – those undergoing acceleration or influenced by gravitational fields. It unifies gravity with the fabric of space-time, encompassing all motion, including accelerated motion, within its purview. While special relativity deals with flat space-time and relative motion, general relativity introduces the dynamic curvature of space-time itself as a response to mass and energy.

One of the remarkable consequences of general relativity is its application to the cosmological scale. The curvature of space-time influences the trajectory of light, causing gravitational lensing – the bending of light around massive objects. This effect was confirmed during the 1919 solar eclipse observations, where stars near the Sun appeared slightly shifted due to the bending of their light around the Sun's gravitational field.

General relativity also provides the foundation for understanding black holes, where the gravitational curvature becomes so intense that not even light can escape – leading to the concept of an event horizon. These celestial phenomena have since become crucial tools for studying the universe's most extreme environments and for testing the boundaries of our understanding of physics.

In summary, general relativity profoundly transformed our understanding of gravity by revealing it as the result of the curvature of space-time caused by mass and energy. This departure from the classical Newtonian view revolutionized our comprehension of the cosmos, offering a unified description of gravity and motion on all scales.

While special relativity opened the door to the intertwining of space and time and addressed relative motion in inertial frames, general relativity extended this framework to include accelerated motion and the gravitational influence of massive objects. Together, these theories form the foundation of modern physics, guiding our exploration of the universe's inner workings and shaping our technological advancements.

### 3.1.1 The incompatibility between the Newtonian gravity with the special relativity

The Newton's universal law of gravity was one of the biggest hits in the history of physics, this is because he could explain almost all the gravitational phenomena in the planet (or any other) and in the solar system, with this theory Edmund Halley could also predict the trajectory of a far asteroid, the well known Halley's asteroid. The Newton's theory failed with one special planet in the solar system, Mercury, no one could explain the retrograde movement of this planet using this theory, it was Albert Einstein with his theory of general relativity who could solve this problem and generalize to high gravitational fields and also discovered that his theory of special relativity is incompatible with the Newton's theory.

Lets show that. The SR (special relativity from now on) has preferred observers <sup>1</sup> this are known as *inertial observers* when we associate then with a set of four coordinates  $(t, x, y, z)$ , this becomes an inertial reference frame. The different inertial frames are all related by Lorentz transformations, with this we can enunciate the *principle of special relativity*.

**Principle 1 *Principle of special relativity:*** *The laws of physics must take the same form in every inertial frame.*

This principle talks about the mathematical forms of the laws of physics this means that no matter how you transform the reference frame, the coordinate system, the mathematical form of the laws must be the same, there is no physical interest in a mathematical formulation fro a physical law which has a different form for every inertial frame.

Lets analyze the Newton's laws to see like this the compatibility between this and the SR

$$\nabla^2\phi = 4\pi G\rho \quad (3.1)$$

where  $\phi$  is the gravitational potential,  $\rho$  is the mass density, and  $G$  is the gravitational or Cavendish's constant. Due to the Lorentz transformations the spatial and temporal coordinates mix each other it is obvious that this equations won't be longer a Poisson equation under a Lorentz

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<sup>1</sup>here we must clarify that a preferred observer is not a coordinate card, an observer is related to world trajectories, therefore the observers are related more to a geometrical objects.

transformation, this implies that *the Newton laws are incompatible with the principle of special relativity*. So now let's analyze the behaviour of the solutions of the equation 3.1, these have the form,

$$\phi(t, \vec{x}) = -G \int \frac{\rho(t, \vec{x}')}{|\vec{x} - \vec{x}'|} d^3x', \quad (3.2)$$

solutions of this form also violate the principle of SR, because solutions of this form have instantaneous reactions which also violate the principles because *two simultaneous events (spatially separated) in an inertial reference frame, won't be simultaneous in all the other reference frames*.

It is worth to wonder when are compatible the SR with the Newtonian gravity, we will prove that this compatibility is given under low velocities, when we talk about low velocities we must say respect to what, from now on in the context of the relativity theory usually is respect to the speed of light  $C$ , so when  $v \ll C$  like in the solar system. All the planets in our solar system travel, in a relatively good approximation, in circular trajectories so we can use the Newton's law and the gravitation law to relate the potential with the speed, the gravitational potential is

$$\phi = -\frac{MG}{|\vec{r}|} \quad (3.3)$$

where  $M$  is the sun's mass, and  $\vec{r}$  is the positions vector from the sun to the planet, if the movement is perfectly circular there is a centripetal acceleration, thus

$$\frac{GM}{|\vec{r}|^2} = \frac{v^2}{|\vec{r}|} \rightarrow \frac{v^2}{C^2} = \frac{|\phi|}{C^2}. \quad (3.4)$$

Here we can see the direct relation there is between the gravitational potential with the relativistic effects in gravitational fields, intense gravitational fields imply relativistic effects, this also implies, yeh again, *the Newton theory is incompatible with relativity*. When  $|\phi|/C^2 \ll 1$  we have that the Newton's law are compatible. There is a certain dimensionless constant that tell us if an object is enough to produce relativistic effects, this is called the compactness  $c$  defined by:

$$c = \frac{GM}{C^2 r}. \quad (3.5)$$

This constant for the sun has a values of the order of  $\approx 10^{-6}$  and for a Neutron star of  $\approx 0.5$ , thus in the vicinity of a Neutron star the relativistic effects will be really important in contrast with our sun.

### 3.1.2 Weak equivalence principle

In the Newtonian theory there is a difference between two concepts, the *inertial mass* and the *gravitational mass*,  $m_I, m_g$  respectively, the distinctions between these two is really important,

because in the common language, they're the same, but we will see that the fact they're numerically the same implies a principle and they are not the same concepts, physically imply two different things.

- The inertial mass is related with the second law of Newton, this mass is an inertia factor which tell us how hard would be to move an object and appears mathematically as

$$\vec{F} = m_I \vec{a}. \quad (3.6)$$

- The gravitational mass is a factor which tells us how an object interacts with a gravitational field it is analogous to charge in the electromagnetism and appears mathematically as

$$\vec{F} = m_g \vec{g}. \quad (3.7)$$

To see how they are related, let's consider a particle falling freely in a gravitational field, thus

$$\vec{a} = \frac{\vec{F}}{m_I} = \frac{m_g}{m_I} \vec{g}. \quad (3.8)$$

It was Galileo, with his so famous experiments in the Piza tower, who postulated *the relation between the gravitational and inertial mass  $m_g/m_I$  are the same for all the materials*. Without any loss of generality, we're saying that in magnitude the gravitational mass and the inertial mass are equal. Later Eötvös proved experimentally that

$$\frac{m_g}{m_I} - 1 \approx \mathcal{O}(10^{-12}). \quad (3.9)$$

This separates gravity from all the other three forces, because all the other forces interact with matter depending on their compositions but this not, the main objective nowadays is to try to violate the relation 3.9 and we can say that *the equality between  $m_I$  and  $m_g$  for all the bodies is a form of the weak equivalence principle*.

The relation 3.9, this principle, in fact has a really heavy implication mathematically, because we start from a movement equation like,

$$m_I \ddot{\vec{x}} = m_g \vec{g}[\vec{x}(t), t], \quad (3.10)$$

but using 3.9 we then get

$$\ddot{\vec{x}} = \vec{g}[\vec{x}(t), t]. \quad (3.11)$$

Solutions of this equations are determined uniquely by the initial position and speed of the particle. This implies that **two different particle with the same position and speed will have the same trajectories**. Without this relation this wont be any longer true, there is the strenght and beauty of this principle. Lets now formulate this principle in this other, but physically equal, form:

**Principle 2 Weak equivalence principle:** *The trajectory of a test body falling freely depends only on its initial position and speed and it is independent of its chemical composition.*

We must remember that this is enunciate to test bodies, this kind of test bodies have got auto-gravitational effects negligible respect to the field of the sources and this is not extended, it is almost a particle respect to the field size, this to avoid effects like torques. Lets analyze the mechanical behavior of a reference frame that fulfill the transformations of Galileo's group.

$$t' = t, \quad (3.12)$$

$$\vec{X}' = \vec{X} - \vec{Y}(t), \quad (3.13)$$

$\vec{Y}(t)$  is related with the origin of the new reference frame. And we will set  $\ddot{\vec{Y}}(t) = \vec{a}$  so the acceleration of the new reference frame is given by the Galileo's relativity theory by

$$\ddot{\vec{X}}' = \vec{g} - \vec{a} = \vec{g}'. \quad (3.14)$$

Here we can see the acceleration of the new reference frame can be interpreted as a measure of the gravity acceleration from this reference frame, lets analyze some special cases of the equation 3.14.

1.  $\vec{g} = 0$  which implies that  $\vec{g}' = -\vec{a}$  we can see that *the uniform acceleration is indistinguishable from an uniform gravitational acceleration*.
2.  $\vec{g} = cte$  &  $\vec{a} = \vec{g}$ . For this kind of situations we can then find an inertial reference frame associated to the free fall, in this reference frame the gravitational effects are zero. This is equivalent to be next to a falling object like a paratrooper.

If we haven't got uniformity, then we will see that *each coordinate chard of the inertial reference frame can be such that this is associated to a geometrical manifold, such manifolds will allow us to define neighborhoods which are locally Minkowskian*.

### 3.1.3 Experimental evidence for a general relativity theory

Many physics students feel attraction to the theory of general relativity and gravitation because the mathematical structure behind this theory which by its own is really abstract but with this theory it takes a beautiful physical interpretation but many people forget how this theory was born.

Is the experimentation, the experiences we live everyday what allow physicist do physics, because of that there are some experiments which cannot be explained with the special relativity theory of which are the first steps in the search of a general theory of relativity, here we're going to check some of those experiments and analyze their physical information.

#### 3.1.3.1 Eötvös experiment

The Hungarian Baron Loránd Eötvös was a mathematician, engineer and physicist who made an amazing experiment to test the equivalence principle, here we're going to analyze his experiment. Lets suppose a pair of objects to which a force is applied

$$\vec{F}^{(1)} = m_I^{(1)} \vec{a}^{(1)} = m_g^{(1)} \vec{g}, \quad (3.15)$$

$$\vec{F}^{(2)} = m_I^{(2)} \vec{a}^{(2)} = m_g^{(2)} \vec{g}. \quad (3.16)$$

He, in his experiment, postulate that the acceleration of the two objects aren't the same

$$\vec{a}^{(2)} = \frac{m_g^{(2)}}{m_I^{(2)}} \vec{g} \neq \frac{m_g^{(1)}}{m_I^{(1)}} \vec{g} = \vec{a}^{(1)}. \quad (3.17)$$

We're going to have on count the gravitational force

$$\vec{F}_g = -GM_O M_G \frac{\vec{r}}{r^3}, \quad (3.18)$$

and the centrifugal force due to the relative rotational motion between the two forces with angular frequency  $\vec{\omega}$ , see figure 3.1.

$$\vec{F}_\omega = m_I \omega^2 \vec{r}_{axis}, \quad (3.19)$$

$$\vec{r}_{axis} = \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega}, \quad (3.20)$$

$$\vec{F}_\omega = m_I [\vec{\omega} \times (\vec{\omega} \times \vec{r})]. \quad (3.21)$$

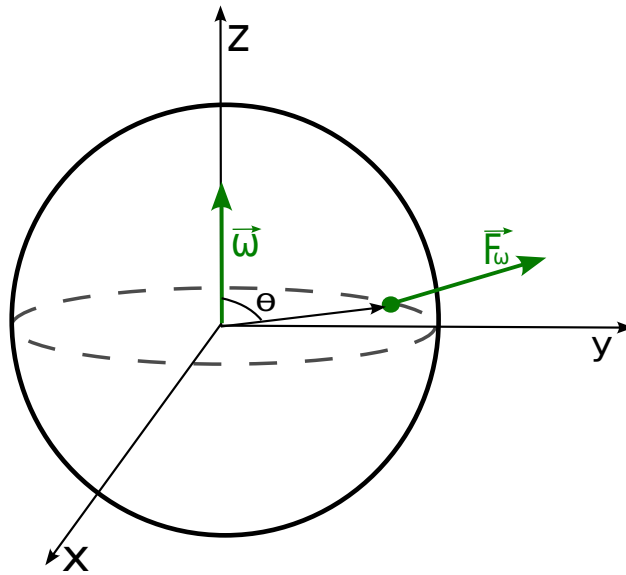


Figure 3.1: Graphical representation of the relative movement between the two test bodies.

Then the forces balance is

$$\vec{F}^{(i)} = \vec{F}_g^{(i)} + \vec{F}_{\omega}^{(i)}.$$

And lets define a parameter  $\alpha$  which relates the strength relation between the gravitational and centrifugal force as

$$\alpha = \frac{|\vec{F}^{(1)} \times \vec{F}^{(2)}|}{|\vec{F}^{(1)}||\vec{F}^{(2)}|}, \quad (3.22)$$

if  $\alpha = 0$  we'll have that the forces are parallel. In magnitude in the earth the gravitational force is stronger than the centrifugal because

$$\frac{GM_O}{|\vec{r}|} / \frac{1}{\omega^2 |\vec{r}|^2} \approx 300.$$

Lets calculate the parameter  $\alpha$  so, first we're going to calculate the numerator

$$\begin{aligned}
\vec{F}^{(1)} \times \vec{F}^{(2)} &= \left( -GM_O m_g^{(1)} \frac{\vec{r}}{r^3} + m_I^{(1)} [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \right) \times \left( -GM_O m_g^{(2)} \frac{\vec{r}}{r^3} + m_I^{(2)} [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \right) \\
&= - \left( \frac{GM_O m_g^{(1)} m_I^{(2)}}{r^3} \right) (\vec{r} \times \vec{\omega}) \times (\vec{\omega} \times \vec{r}) - \left( \frac{GM_O m_g^{(2)} m_I^{(1)}}{r^3} \right) [\vec{\omega} \times (\vec{\omega} \times \vec{r})] \times \vec{r} \\
&= - \frac{GM_O \omega^2}{r^3} \left\{ m_g^{(1)} m_I^{(2)} \vec{r} \times \left[ \vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] - m_g^{(2)} m_I^{(1)} \left[ \vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \times \vec{r} \right\} \\
&= - \frac{GM_O}{r^3} (\vec{r} \times \vec{\omega}) (\vec{\omega} \cdot \vec{r}) [m_g^{(1)} m_I^{(2)} - m_g^{(2)} m_I^{(1)}],
\end{aligned}$$

$$|\vec{F}^{(1)} \times \vec{F}^{(2)}| = \frac{GM_O}{r^3} |\vec{r} \times \vec{\omega}| (\vec{\omega} \cdot \vec{r}) [m_g^{(1)} m_I^{(2)} - m_g^{(2)} m_I^{(1)}]. \quad (3.23)$$

Now, for the denominator we must calculate each norm, because both have the same form we'll do for one and only change their respective indices

$$\begin{aligned}
|\vec{F}^{(i)}| &= \left\{ \frac{(GM_O m_g^{(i)})^2}{r^4} + (m_I^{(i)} \omega^2)^2 \left[ \vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \cdot \left[ \vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] - 2GM_O m_g^{(i)} m_I^{(i)} \frac{\omega^2}{r^3} \vec{r} \cdot \left[ \vec{r} - \frac{(\vec{\omega} \cdot \vec{r}) \vec{\omega}}{\omega^2} \right] \right\}^{1/2} \\
&= \left\{ \frac{(GM_O m_g^{(i)})^2}{r^4} + (m_I^{(i)} \omega^2)^2 \left[ r^2 - \frac{2(\vec{\omega} \cdot \vec{r})^2}{\omega^2} + \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] - 2GM_O m_g^{(i)} m_I^{(i)} \frac{\omega^2}{r^3} \left[ r^2 - \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] \right\}^{1/2} \\
&= \frac{GM_O m_g^{(i)}}{r^2} \left\{ 1 + \left[ \frac{m_I^{(i)} \omega^2}{GM_O m_g^{(i)}} \right] \left[ r^2 - \frac{(\vec{\omega} \cdot \vec{r})^2}{\omega^2} \right] \left[ \frac{m_I^{(i)} r^2 \omega^2}{GM_O m_g^{(i)}} - \frac{2}{r} \right] \right\}^{1/2}.
\end{aligned}$$

Taking a first order approximation, or what is the same, saying the biggest contribution to the norm of each force is due to the gravitational interaction we then have:

$$|\vec{F}^{(i)}| \approx \frac{GM_O m_g^{(i)}}{r^2}, \quad (3.24)$$

then the parameter  $\alpha$  becomes

$$\alpha \approx \left[ \frac{m_I^{(1)}}{m_g^{(1)}} - \frac{m_I^{(2)}}{m_g^{(2)}} \right] \frac{|\vec{r} \times \vec{\omega}| (\vec{\omega} \cdot \vec{r})}{GM_O}, \quad (3.25)$$

$$\alpha \approx \left[ \frac{m_I^{(1)}}{m_g^{(1)}} - \frac{m_I^{(2)}}{m_g^{(2)}} \right] \frac{\cos \theta \sin \theta}{300}. \quad (3.26)$$

With the result of the equation 3.26 we can see that this experiment can not be done for some who is in the equator nor the poles because  $\theta = 0$  &  $\theta = \pi/2$  respectively, fortunately Eötvös was in Hungary a country beyond the equator and he found that  $\alpha$  was zero in a part for  $10^9$ , this means it is zero until the ninth significant number, from there on there are errors, and mathematically from 3.26 we can see that if it is zero this means by the elementary theorem of algebra that

$$\frac{m_I^{(1)}}{m_g^{(1)}} = \frac{m_I^{(2)}}{m_g^{(2)}}. \quad (3.27)$$

### 3.1.3.2 Tidal forces

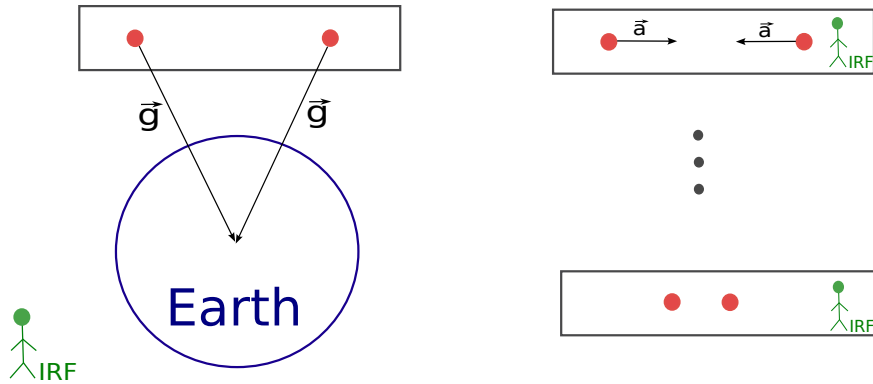


Figure 3.2: In the left there are two particles which are in a free fall to the earth and the inertial reference frame (IRF) is to the side of the earth, he observes a diagonal trajectory of the bodies. In the right we set the IRF inside the box with the two bodies, and he observes that there is a strange force which makes the bodies get together.

In the figure 3.2 we can see an scheme of the effect of the tidal forces on two particles inside a box. This box is in a free fall to the earth, an observer with an inertial reference frame next to the earth will observe a kind of diagonal trajectory for both particles to the center of the earth but if we place our inertial reference frame inside the box we will see that both particles will join together why? In the Newtonian mechanics we would say that there is a fictitious force <sup>2</sup> that makes the objects attract each other but in fact, there is no imaginary force, there is a non homogeneous gravitational field, so *the tidal forces are an effect of the non homogeneity of the gravitational field, these are physical and important forces.*

### 3.1.3.3 Light deflection and gravitational red shift

In the figure 3.1 we can see that even the light is affected by the gravity, in the Newtonian theory this was something impossible to imagine!, we need then a theory to describe this effect.

<sup>2</sup>Newton never used the term fictitious force, that term was introduced by the Mach's formalism for the mechanics.

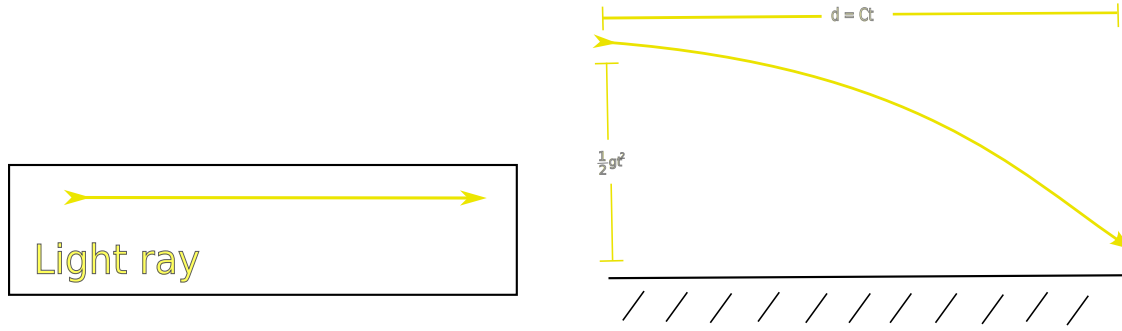


Figure 3.3: In the left there is a light ray seeing from an inertial reference frame inside the box, and in the right there is the same ray but seeing from the earth.

There are some other effects of the gravity on the light, like the color shift, this is a gravitational Doppler effect, so, lets suppose two observers Alice (A) and Bob (B), Alice is in a free fall at a distance  $h$  from Bob and sends a light pulse, then Bob gets it and in the moment he gets it Alice sends another beam what is the relation between the Alice and bob interval times?. To solve this we will suppose the gravity is uniform and we will set a reference frame such that  $\vec{g} = \vec{a}$  so for Alice

$$Z_A = h + \frac{1}{2}gt^2 \quad (3.28)$$

$$Z_B = \frac{1}{2}gt^2. \quad (3.29)$$

Alice sends the first signal at a time  $t_1$  then the position of the beam  $Z$  is

$$Z = Z_A(t_1) - C(t - t_1) = h + \frac{1}{2}gt_1^2 - C(t - t_1), \quad (3.30)$$

it arrives to Bob at  $t = T_1$  then

$$h + \frac{1}{2}gt_1^2 - C(T_1 - t_1) = \frac{1}{2}gT_1^2. \quad (3.31)$$

Alice sends the second signal at  $t = t_1 + \Delta\tau_A$  then

$$Z = Z_A(t_1 + \Delta\tau_A) - C(t - t_1 - \Delta\tau_A). \quad (3.32)$$

And it lands on Bob at  $t = T_1 + \Delta\tau_B$  thus

$$h + \frac{1}{2}g(t_1 + \Delta\tau_A)^2 - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1 + \Delta\tau_B)^2 \quad (3.33)$$

Lest expand 3.33 then

$$h + \frac{1}{2}g(t_1 + \Delta\tau_A)^2 - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1 + \Delta\tau_B)^2, \quad (3.34)$$

$$h + \frac{1}{2}g(t_1^2 + 2t_1\Delta\tau_A + \Delta\tau_A^2) - C(T_1 + \Delta\tau_B - t_1 - \Delta\tau_A) = \frac{1}{2}g(T_1^2 + 2T_1\Delta\tau_B + \Delta\tau_B^2). \quad (3.35)$$

Using the equation 3.31 in the right hand side of the last equation and simplifying the  $h$  and  $\frac{1}{2}gt_1^2$  then we have

$$\begin{aligned} gt_1\Delta\tau_A + \frac{1}{2}g\Delta\tau_A^2 - C(\Delta\tau_B - \Delta\tau_A) &= gT_1\Delta\tau_B + \frac{1}{2}g\Delta\tau_B^2, \\ \Delta\tau_A(gt_1 + C) + \frac{1}{2}g\Delta\tau_A^2 &= \Delta\tau_B(gT_1 + C) + \frac{1}{2}g\Delta\tau_B^2. \end{aligned}$$

Dividing all by  $C$  and remembering that in Earth  $g\Delta\tau_i \ll C$  then we have

$$\Delta\tau_A \left(1 + \frac{gt_1}{C}\right) = \Delta\tau_B \left(1 + \frac{gT_1}{C}\right) \quad (3.36)$$

$$\Delta\tau_B = \left(1 + \frac{gt_1}{C}\right) \left(1 + \frac{gT_1}{C}\right)^{-1} \Delta\tau_A, \quad (3.37)$$

now using the geometric series  $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$  to first order because, in the Earth  $gt_i \leq C$  thus

$$\Delta\tau_B \approx \left[1 + \frac{gt_1}{C} - \frac{gT_1}{C} - g^2 \frac{t_1 T_1}{C^2}\right] \Delta\tau_A \approx \left[1 - \frac{g}{C}(T_1 - t_1)\right] \Delta\tau_A. \quad (3.38)$$

The times seems to run slower in  $B$  than in  $A$  we can conclude then that *the time runs slower where the gravitational field is stronger*. Here we can also see two things, the first one is that the compactness appears again because

$$\Delta\tau_B \approx \left[1 - \frac{gh}{C^2}\right] \Delta\tau_A = \left[1 + \frac{\phi_B - \phi_A}{C^2}\right] \Delta\tau_A, \quad (3.39)$$

then we see the compactness appears and tells us how fast will run the time of an observer. The other thing we can see is a blue shift because using the relation between time and frequency  $\Delta\tau_i = \frac{\lambda_i}{C}$  we then have

$$\lambda_B \approx \left(1 - \frac{gh}{C^2}\right) \lambda_A \quad (3.40)$$

and because the second term in the parenthesis is positive the wavelength of *Bob* will be each time shorter, show there will be a blue shift, or what is the same a Doppler effect when the ambulance (Alice with her laser) comes closer to Bob.

### 3.1.4 Einstein's equivalence principle and the space-time

Einstein introduced the space-time structure to his theory, this to make a coupled theory of space and time, in the Newtonian theory the space is absolute and the time is absolute but not together, this was the Einstein's contribution to physics, neither time nor space are absolute separately but joint they are, such a structure has a special geometry, called a pseudo-Reinmannian geometry, this because in these geometries there are negative and positive or even null vector modulus, their physical interpretation will be discussed later. A space-time diagram is a representation of the physical events, for example for the blue shift problem the diagram is represented in the figure 3.4.

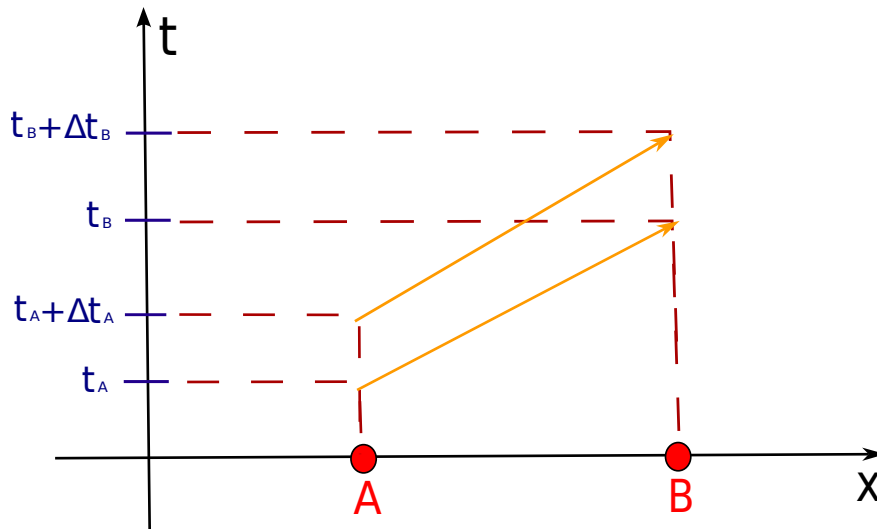


Figure 3.4: Space-time diagram representation for the Doppler effect analyzed before.

Lets study this, in a Minkowskian structure we have got the following relation

$$C^3 d\tau^2 = C^2 dt^2 - dx^2 - dy^2 - dz^2, \quad (3.41)$$

but lets see what happens if we modify it a bit like this

$$C^2 d\tau^2 = \left(1 + \frac{2\phi}{C^2}\right) C^2 dt^2 - \left(1 - \frac{2\phi}{C^2}\right) (dx^2 + dy^2 + dz^2), \quad (3.42)$$

as we can see for Alice and Bob there are no physical displacement, only temporal, so their proper times are, using the equation 3.42 are

$$\Delta\tau_A^2 = \left(1 + \frac{2\phi_A}{C^2}\right) \Delta t^2, \quad (3.43)$$

$$\Delta\tau_B^2 = \left(1 + \frac{2\phi_B}{C^2}\right) \Delta t^2. \quad (3.44)$$

Now, taking the square root of both equations and using Taylor expansion to first order we then have

$$\Delta\tau_A \approx \left(1 + \frac{\phi_A}{C^2}\right) \Delta t, \quad (3.45)$$

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B}{C^2}\right) \Delta t. \quad (3.46)$$

It is easy to see that the relation between their proper times is

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B}{C^2}\right) \left(1 + \frac{\phi_A}{C^2}\right)^{-1} \Delta\tau_A \quad (3.47)$$

Using now the geometrical series because in the earth the compactness is really low  $\phi/C^2 \ll 1$  finally we then have

$$\Delta\tau_B \approx \left(1 + \frac{\phi_B - \phi_A}{C^2}\right) \Delta\tau_A. \quad (3.48)$$

This is a really strong result because changing the plane geometric structure of the Minkowskian space-time we will return the result we had obtain before with the light blue shift, this is a straight proof that we need a more geometrical an general theory of the relativity.

## 3.2 First steps on general relativity

From now on we will move to a more physical territory using all the abstract tools developed by the mathematicians, first we're going to learn how to crawl, then to walk and at the end how to run, because the general relativity is in fact a theory of *the electrodynamics of bodies in movement* the heart of a movement, from a physical point of view, is the notion of position and even more important, the notion of distance, this in mathematics, in a topological space can be defined using a topology on the topological space, here in the same way we're going to define a metric on the manifold, what we will call a metric tensor.

### 3.2.1 Metric tensor

Lets see how to make the notion of distance over  $M$ , for example, lets look at  $\mathbb{R}^3$ . Let  $\vec{X}(t)$  a curve on  $\mathbb{R}^3$  and an open interval  $a < t < b$  so the arc length  $L$  is given by

$$L = \int_a^b ds = \int_a^b \left[ \frac{d\vec{X}}{dt} \cdot \frac{d\vec{X}}{dt} \right]^{1/2} dt, \quad (3.49)$$

where  $\vec{X}$  is the tangent vector to the curve, we see here rises the notion of inner product of these tangent vectors does it mean that always an inner product give us a notion of distance? The fast answer is yes, but the long answer is no, the notion of inner product is more related to the notion of a topology of the vector space, in our case of the tangent space to a point  $p$  on the manifold  $M$ .

The notion of distance is a kind of combinations between the notion of topology or inner product with the physical structure of the problem, there are many maps or inner products that satisfies the axioms of a topology or the inner product is the physical meaning what rises the notion of distance, therefore to find such a distance lets define the metric tensor  $\mathbb{G}$

$$\mathbb{G} : T_p(M) \times T_p(M) \rightarrow \mathbb{R}, \quad (3.50)$$

and in our notation the application of the metric tensor  $\mathbb{G}$  to two vectors is

$$\mathbb{G}(\vec{A}, \vec{B}) = \vec{A} \cdot \vec{B}. \quad (3.51)$$

**Definition 15** A metric tensor in  $p \in M$  is a tensor  $\mathbb{G}$  of type  $\binom{0}{2}$ , known also as bilinear form, such that satisfies the following properties:

1.  $\mathbb{G}(\vec{X}, \vec{Y}) = \mathbb{G}(\vec{Y}, \vec{X}) \forall \vec{X}, \vec{Y} \in T_p(M)$ .
2.  $\mathbb{G}(\vec{X}, \vec{Y}) = 0 \forall \vec{Y} \in T_p \Leftrightarrow \vec{X} = \vec{0}$  this means the metric tensor is not degenerated.

In differential geometry we're always interested in a metric such that it is Riemannian this means *the metric has a signature mostly positive*, this means the metric tensor is positive-defined. In general relativity we're interested in Lorentizan metrics, they are those whose signature has the form  $- + + \dots$  here the metric is not necessarily positive-defined.

**Definition 16** A Riemannian manifold es a pair  $(M, \mathbb{G})$  where  $M$  is a differentiable manifold and  $\mathbb{G}$  is the metric tensor field, which can be Riemannian or not.

*In general relativity such a pair is known as space-time.*

In this way we can calculate a length curve  $L$  as

$$L = \int_a^b \left[ \mathbb{G}(\vec{X}, \vec{X}) \right]_{\lambda(t)}^{1/2}, \quad (3.52)$$

and given a coordinate chart we can write the metric tensor as

$$\mathbb{G} = g_{\alpha\beta} dx^\alpha \otimes dx^\beta. \quad (3.53)$$

**Definition 17** *The metric  $\mathbb{G}$  is not degenerated, thus this must be invertible. The inverse metric is a symmetric tensor field of type  $\binom{2}{0}$  denoted by  $\mathbb{G}^{-1}$  such that*

$$g^{ac} g_{cb} = \delta_b^a. \quad (3.54)$$

**Definition 18** *The Lorentzian signature is defined such that for any point  $p$  in the manifold we can choose an orthonormal basis  $\{\mathbf{e}_\alpha\}$  such that*

$$\mathbb{G}(\mathbf{e}_\alpha, \mathbf{e}_\beta) = g_{\alpha\beta} \equiv \text{Diagonal}[-1, 1, 1, 1, \dots]. \quad (3.55)$$

This basis is not unique and we must set a relation between basis

$$\begin{aligned} \mathbf{e}_{\bar{\alpha}} &= \Lambda_{\bar{\alpha}}^\beta \mathbf{e}_\beta \rightarrow g_{\bar{\alpha}\bar{\beta}} = \mathbb{G}(\mathbf{e}_{\bar{\alpha}}, \mathbf{e}_{\bar{\beta}}) = \mathbb{G}(\Lambda_{\bar{\alpha}}^\mu \mathbf{e}_\mu, \Lambda_{\bar{\beta}}^\nu \mathbf{e}_\nu) \\ &= \Lambda_{\bar{\alpha}}^\mu \Lambda_{\bar{\beta}}^\nu \mathbb{G}(\mathbf{e}_\mu, \mathbf{e}_\nu) = \Lambda_{\bar{\alpha}}^\mu \Lambda_{\bar{\beta}}^\nu g_{\mu\nu}. \end{aligned}$$

All the basis are related by Lorentz transformations.

### 3.2.2 Proper time and an introduction to paths

Due to the nature of the space-time manifold  $(\mathbb{R}^4, \mathbb{G})$  the vectors in this space have a particular behavior.

**Definition 19** *On a Lorentzian manifold  $(M, \mathbb{G})$  a non zero vector  $\vec{X} \in T_p(M)$  is*

- A time-like vector  $\mathbb{G}(\vec{X}, \vec{X}) < 0$ .
- A null or light-like vector  $\mathbb{G}(\vec{X}, \vec{X}) = 0$ .
- A space-like vector  $\mathbb{G}(\vec{X}, \vec{X}) > 0$ .

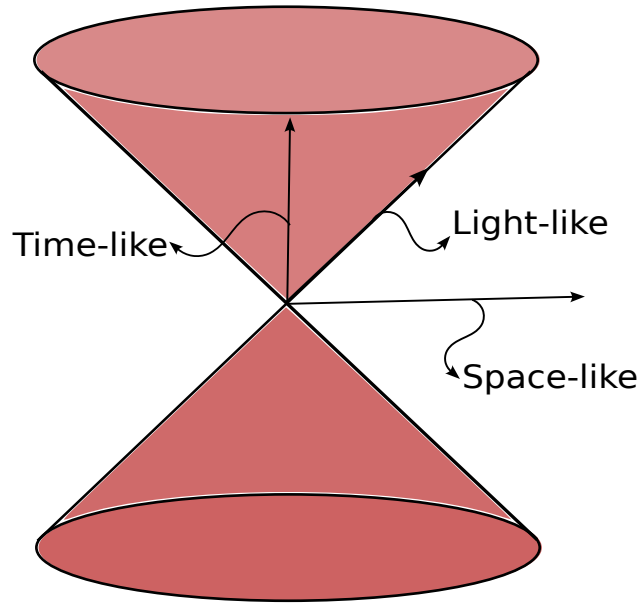


Figure 3.5: The light cone for the space-time structure, in the  $x$ -axis there is  $\mathbb{R}^3$  and in the  $y$  axis there is the coordinate time  $t$ , a space-like vector is a vector which lives outwards the cone, a light-like lives on the border of the cone and a time-like lives inside the cone.

In the figure 3.5 there is a graphical representations of these three kinds of vectors, if we adopt a mostly positive signature, and from the postulates of the special relativity we can see the vectors with physical meaning are the time and light like vectors, because we can reach and not even get over the light speed so everything will be inside the cone and the objects that travel at the speed of light will be on the cone border, a space-link thus wont have any physical meaning.

We will see we can associate an observer with a time-like vector, therefore we'll focus on the properties of these vector, moreover on the characteristics of their tangent space  $T_p(M)$ , first we must see that if  $\{\mathbf{e}_\mu\}$  is basis of orthonormal vectors in  $p$ , then the metric tensor has components

$$\mathbb{G}(\mathbf{e}_\alpha, \mathbf{e}_\mu) = \eta_{\alpha\beta} \quad (3.56)$$

such that  $T_p(M)$  has exactly the same structure that the Minkowski space-time.

The time-like trajectory in a Lorentizan manifold  $(M, \mathbb{G})$  is given by an integral curve such that  $\vec{X}$  is always time-like or in a more formal way  $\mathbb{G}(\vec{X}, \vec{X}) < 0$  see figure 3.6.

Lets see the behavior of this curve. Let  $\lambda(v)$  a smooth time-like curve such that  $\lambda(0) = p$  and we can find an expression for a tangent vector  $\vec{X}^a$ , which is time-like, therefore we can define a proper time

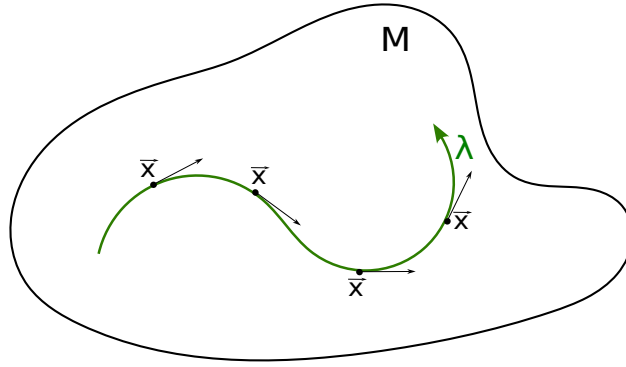


Figure 3.6: Graphical representation of an integral time-like curve on a manifold  $M$ .

$$\frac{d\tau}{dv} = -\sqrt{-(g_{ab}\vec{x}^a\vec{x}^b)|_{\lambda(v)}} \quad \& \quad \tau(0) = 0. \quad (3.57)$$

We must remember  $ds^2 = d\vec{X} \cdot d\vec{X} = g_{\alpha\beta}dx^\alpha dx^\beta$  this is given a coordinate basis in a coordinate chart  $\phi = \{x^0, x^1, x^2, x^3\} \in \mathbb{R}^4$  thus

$$\frac{dx^2}{dv^2} = g_{\alpha\beta} \frac{dx^\alpha}{dv} \frac{dx^\beta}{dv}, \quad (3.58)$$

using a Minkowski space-time we have

$$d\tau^2 = -g_{\mu\nu}dx^\mu dx^\nu = -ds^2, \quad (3.59)$$

this allows us to see that  $\vec{X} = X^\alpha \frac{d}{dx^\alpha}$  therefore  $X^\alpha = \frac{dx^\alpha}{dv}$  we can then integrate and find an expression for the proper time

$$\tau = \int_0^{v_q} \left[ - \left( g_{\mu\nu} \frac{dx^\mu}{dv} \frac{dx^\nu}{dv} \right) \right]_{\lambda(v)}^{1/2} du. \quad (3.60)$$

We will see later that the time-like vectors can be parameterized from a proper time like 3.60, for the null-like ones we gotta chose an special curve, this is because the equation 3.60 ins related to geodesics. We're going to suppose  $v = \tau$ , the parameter is the proper time, then developing the equation 3.60

$$\begin{aligned} \tau &= \int_0^\tau \left[ - \left( g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) \right]^{1/2} d\tau, \\ \tau &= \int_0^\tau [-U^\mu U_\mu]^{1/2} d\tau. \end{aligned}$$

For the expression 3.60 become the proper time, we gotta impose that the tangent quadri-vector has norm  $-1$  for any coordinate chart.

In the study of mechanics there is an important question, what is the path a test particle will follow in certain conditions? And why, from an infinity family of trajectories, did the particle choose its trajectory? To answer this a mathematical theory was developed, what we in physics known as the variational principles, let  $M$  a manifold and  $\lambda(\tau)$  a time-like integral curve on  $M$  and  $\lambda(\tau')$  alternative integral curves, see figure 3.7, thus the proper time is given by

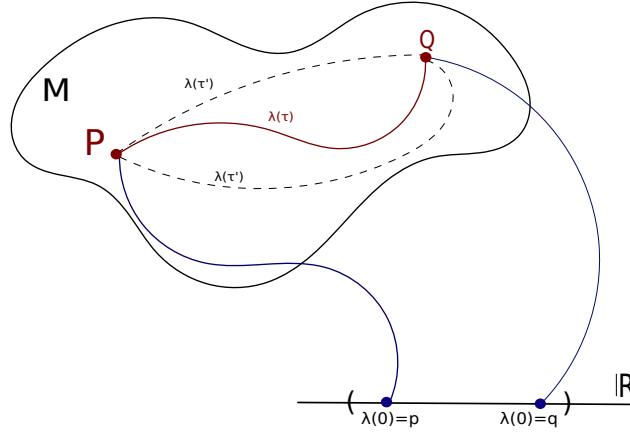


Figure 3.7: The real time-like trajectory a particle will follow is  $\lambda(\tau)$  and the other two will be little different trajectories but with initial and final points fixed.

$$\tau[\lambda(v)] = \int_0^1 G(x(\lambda(v)), \dot{x}(\lambda(v))) du, \quad (3.61)$$

$$G = [-g_{\mu\nu}(x(\lambda(v))) \dot{x}^\mu(\lambda(v)) \dot{x}^\nu(\lambda(v))]^{1/2}, \quad (3.62)$$

Using the equations 3.61 and 3.62 we want to formulate the following problem, how must be the magnitude of the metric  $G$  such that the proper time is stationary this means, it is minimum, or maximum or a seat value. Lets find time-like curve  $\lambda(v)$  to extreme the integral 3.61.

Let us denote the correct chart as  $x(\lambda)$  and the incorrect one as

$$X(\lambda) = x(\lambda) + \eta(\lambda), \quad (3.63)$$

where  $\eta(\lambda)$  is the difference between the real and the wrong path, and because the initial and end point are fixed, thus

$$\eta(\lambda(0)) = \eta(\lambda(1)) = 0. \quad (3.64)$$

The integral  $\tau[\lambda]$  taken along the wrong curve  $X(\lambda)$  must be larger than that along the right curve  $x(\lambda)$ , no matter how close the former is to the latter. To express this requirement, we will introduce a parameter  $\alpha$  and redefine  $X(\lambda)$  to be

$$X(\lambda) = x(\lambda) + \alpha\eta(\lambda), \quad (3.65)$$

The integral  $S$  taken along the curve  $X(\lambda)$  now depends on the parameter  $\alpha$ , so let's call it  $\tau(\alpha)$ . The right curve  $x(\lambda)$  is obtained from 3.65 by setting  $\alpha = 0$ . Thus the requirement that  $\tau$  is minimum for the right curve  $x(\lambda)$  implies that  $\tau(\alpha)$  is stationary at  $\alpha = 0$ . With this result, we have converted our problem to the traditional problem from elementary calculus of making sure that an ordinary function [namely  $\tau(\alpha)$ ] has a minimum at  $\alpha$  specified point  $\alpha = 0$ . To ensure this, we must just check that the derivative  $\frac{d\tau}{d\alpha}$  is zero when  $\alpha = 0$ .

The integral  $\tau(\alpha)$  is

$$\tau(\alpha) = \int_0^1 G(x(\lambda), \dot{x}(\lambda), \lambda) dv = \int_0^1 G(x + \alpha\eta, \dot{x} + \alpha\dot{\eta}) dv. \quad (3.66)$$

To differentiate 3.66 with respect to  $\alpha$ , we note that  $\alpha$  appears in the integrand  $G$ , so we need to evaluate  $\frac{\partial G}{\partial \alpha}$ . Since  $\alpha$  appears in two of the arguments of  $G$ , this gives two terms, namely (using the chain rule)

$$\frac{\partial G(x + \alpha\eta, \dot{x} + \alpha\dot{\eta}, \lambda)}{\partial \alpha} = \eta \frac{\partial G}{\partial x} + \dot{\eta} \frac{\partial G}{\partial \dot{x}}, \quad (3.67)$$

and for  $\frac{d\tau}{d\alpha}$  which has to be zero

$$\frac{d\tau}{d\alpha} = \int_0^1 \frac{\partial G}{\partial \alpha} dv = \int_0^1 \left( \eta \frac{\partial G}{\partial x^\mu} + \dot{\eta} \frac{\partial G}{\partial \dot{x}^\mu} \right) dv = 0. \quad (3.68)$$

We need to rewrite the second term on the right using integration by parts'

$$\int_0^1 \dot{\eta}(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} = \left[ \eta(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} \right]_0^1 - \int_0^1 \eta(\lambda) \frac{d}{d\lambda} \left( \frac{\partial G}{\partial \dot{x}^\mu} \right) du, \quad (3.69)$$

Because  $\eta$  is zero in the initial and end point is zero, the second term goes to zero, thus

$$\int_0^1 \dot{\eta}(\lambda) \frac{\partial G}{\partial \dot{x}^\mu} du = - \int_0^1 \eta(\lambda) \frac{d}{d\lambda} \left( \frac{\partial G}{\partial \dot{x}^\mu} \right) du. \quad (3.70)$$

Inserting this equation into 3.68 we find that

$$\int_0^1 \eta(\lambda) \left( \frac{d}{dv} \left( \frac{\partial G}{\partial \dot{x}^\mu} \right) - \frac{\partial G}{\partial x^\mu} \right) du = 0, \quad (3.71)$$

thus we have arrived into the Euler- Lagrange equation for  $G$ :

$$\frac{d}{dv} \left( \frac{\partial G}{\partial \dot{x}^\mu} \right) - \frac{\partial G}{\partial x^\mu} = 0. \quad (3.72)$$

Computing the partial derivatives we will have

$$\frac{\partial G}{\partial \dot{x}^\mu} = -\frac{1}{G} g_{\mu\nu} \dot{x}^\nu, \quad (3.73)$$

$$\frac{\partial G}{\partial x^\mu} = -\frac{1}{2G} \partial_\mu g_{\nu D} \dot{x}^\nu \dot{x}^D. \quad (3.74)$$

Now, before solving the Euler-Lagrange equation for  $G$  lets give a deeper look into the relation between the curve, the proper time and  $G$

$$G = \frac{d\tau}{d\lambda} \rightarrow \frac{d}{d\lambda} = \frac{d\tau}{d\lambda} \frac{d}{d\tau} = G \frac{d}{d\tau}. \quad (3.75)$$

Using this equation, the Euler lagrange equation thus we will have the following development. Multiplying the equation 3.72 by  $G$  we will have

$$G \frac{d}{d\tau} \left( \frac{\partial G}{\partial \dot{x}^\mu} \right) - G \frac{\partial G}{\partial x^\mu} = 0,$$

using the chain rule, therefore

$$\begin{aligned} \frac{d}{d\tau} \left[ \frac{\partial}{\partial \dot{x}^\mu} \left( \frac{1}{2} G^2 \right) \right] - \frac{\partial}{\partial x^\mu} \left( \frac{1}{2} G^2 \right) &= 0, \\ \frac{1}{2} G^2 &= \frac{1}{2} g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = \mathcal{L}. \end{aligned}$$

$\mathcal{L}$  is in fact the Lagrangian of the system, thus the equation 3.72 becomes

$$\frac{d}{d\tau} \left( \frac{\partial}{\partial \dot{x}^\mu} \mathcal{L} \right) - \frac{\partial}{\partial x^\mu} \mathcal{L} = 0. \quad (3.76)$$

Now, we gotta calculate the partial derivatives

$$\frac{\partial}{\partial x^\mu} \mathcal{L} = \frac{1}{2} \frac{\partial}{\partial x^\mu} (g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta) = \frac{1}{2} (\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta), \quad (3.77)$$

$$\frac{\partial}{\partial \dot{x}^\mu} \mathcal{L} = \frac{1}{2} \frac{\partial}{\partial \dot{x}^\mu} (g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta) = \frac{1}{2} (g_{\alpha\beta} \delta^\alpha_\mu \dot{x}^\beta + g_{\alpha\beta} \dot{x}^\alpha \delta^\beta_\mu) = \frac{1}{2} (g_{\mu\beta} \dot{x}^\beta + g_{\alpha\mu} \dot{x}^\alpha). \quad (3.78)$$

Using both equation in the Euler-Lagrange equation then

$$\begin{aligned} \frac{d}{d\tau} \left[ \frac{1}{2} (\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta) \right] &= \frac{1}{2} (g_{\mu\beta} \ddot{x}^\beta + \partial_l g_{\mu\beta} \dot{x}^\beta \dot{x}^\alpha + g_{\alpha\mu} \ddot{x}^\alpha + \partial_l g_{\alpha\mu} \dot{x}^\alpha \dot{x}^l), \\ \frac{1}{2} (g_{\mu\beta} \ddot{x}^\beta + \partial_l g_{\mu\beta} \dot{x}^\beta \dot{x}^\alpha + g_{\alpha\mu} \ddot{x}^\alpha + \partial_l g_{\alpha\mu} \dot{x}^\alpha \dot{x}^l) &- \frac{1}{2} (\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta), \end{aligned}$$

in the first two terms we can use the dummies indices  $\alpha = \beta$  and in the third one  $l = \alpha$ , in the fourth  $\beta = l$  thus

$$g_{\alpha\mu} \ddot{x}^\alpha + \frac{1}{2} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0. \quad (3.79)$$

Multiplying the equation 3.79 by  $g^{\mu\lambda}$  we will have

$$\delta^\lambda_\alpha \ddot{x}^\alpha + \frac{1}{2} g^{\mu\lambda} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0, \quad (3.80)$$

$$\ddot{x}^\lambda + \frac{1}{2} g^{\mu\lambda} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0. \quad (3.81)$$

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma^\lambda_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (3.82)$$

$$\Gamma^\lambda_{\alpha\beta} = \frac{1}{2} g^{\lambda\mu} (\partial_\alpha g_{\beta\mu} + \partial_\beta g_{\alpha\mu} - \partial_\mu g_{\alpha\beta}). \quad (3.83)$$

The equation 3.82 it is not more than the differential equation for the geodesics in the space-time, we will proof this in few for now, we must make some definitions.  $\Gamma^\mu_{\alpha\beta}$  are known as the Cristoffel symbols, they need a kind of connection, that is because we have derivatives along the curve, but a curve at any point can have a different tangent space, so each tangent vector is different and we need something to connect those spaces, here that connection is the Levi-Civita connection, which one we will develop later. A particular case is when the Cristoffel symbols is zero, we will obtain that the quadri-acceleration is zero, this is the domain of the spetial relativity so we can see physically that *the presence of no Cristoffel symbols mean a zero quadri-acceleration, or a flat space-time, a Minkowski space, therefore the presence of accelerations or the presence of a non zero Cristoffel symbols means the presence of a curved space-time.*

### 3.2.3 Covariant derivative

In the Euler-Lagrange equation for the metric determinant  $G$  we used derivatives, but here the derivatives are kind of special, why? To answer this, first let's examine a vectorial field  $V^a$  and let's see if its gradient is or not a tensor

$$T^\alpha_\beta = \partial_\beta V^\alpha. \quad (3.84)$$

To see how does  $T^\alpha_\beta$  transforms let's suppose two coordinate charts  $\phi = \{x^0, x^1, x^2, x^3\}$  and  $\bar{\phi} = \{\bar{x}^0, \bar{x}^1, \bar{x}^2, \bar{x}^3\}$ , thus

$$T^{\bar{\alpha}}_{\bar{\beta}} = \partial_{\bar{\beta}} V^{\bar{\alpha}} = \partial_{\bar{\beta}} \left( \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} V^\mu \right) = \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} \frac{\partial}{\partial x^\nu} \left( \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} V^\mu \right) = \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} \left( \frac{\partial}{\partial x^\nu} V^\mu \right) + \left( \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} V^\mu \right) \frac{\partial}{\partial x^\nu} \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu},$$

$$T^{\bar{\alpha}}_{\bar{\beta}} \neq \frac{\partial x^{\bar{\alpha}}}{\partial x^\mu} \frac{\partial x^\nu}{\partial x^{\bar{\beta}}} T^\mu_\nu.$$

Thus  $T^\alpha_\beta$  as we had defined does not transform as a tensor, therefore we must look carefully on how do we take derivatives. The problem is in the fact that  $T^\alpha_\beta$  as we had defined is only the derivatives of the components, but how about the derivatives of the basis vectors? They are also changing along the manifold, thus their derivatives must have been taken on count, to solve this we must introduce the notion of the *covariant derivative*.

**Definition 20** A covariant derivative  $\vec{\nabla}$  on a manifold  $M$  is a map from two vector fields  $X, Y$  to a vector field with the following properties.

Let  $X, Y, Z$  three smooth vector fields and  $f, g$  two smooth functions, the covariant derivative satisfies the following properties:

1.  $\vec{\nabla}_{fX+gY} Z = f \vec{\nabla}_X Z + g \vec{\nabla}_Y Z.$
2.  $\vec{\nabla}_X (Y + Z) = \vec{\nabla}_X Y + \vec{\nabla}_X Z.$
3.  $\vec{\nabla}_X (fY) = f \vec{\nabla}_X Y + (\vec{\nabla}_X f) Y.$

From these properties we can see that

$$\vec{\nabla} Y : X \rightarrow \vec{\nabla}_X Y \quad (3.85)$$

is a linear map from  $T_p(M)$  to itself, and

$$\vec{\nabla} f : X \rightarrow \vec{\nabla}_X f = x(f) \quad (3.86)$$

which is the definition of gradient. Lets see an example, let  $X = x^\mu \mathbf{e}_\mu$ ,  $Y = y^\mu \mathbf{e}_\mu$ , thus lets calculate the covariant derivative

$$\begin{aligned} \vec{\nabla}_X Y &= \vec{\nabla}_X (y^\mu \mathbf{e}_\mu) = X(y^\mu) \mathbf{e}_\mu + y^\mu \vec{\nabla}_X \mathbf{e}_\mu \\ &= x^\nu \mathbf{e}_\nu (y^\mu) \mathbf{e}_\mu + y^\mu \vec{\nabla}_{x^\nu \mathbf{e}_\nu} \mathbf{e}_\mu = x^\nu \mathbf{e}_\nu (y^\mu) \mathbf{e}_\mu + y^\mu x^\nu \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu, \\ \vec{\nabla}_X Y &= x^\nu \left[ \mathbf{e}_\nu (y^\mu) \mathbf{e}_\mu + y^\mu \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu \right] = x^\nu \left[ \mathbf{e}_\nu (y^\mu) \mathbf{e}_\mu + y^\rho \Gamma_{\nu\rho}^\mu \mathbf{e}_\mu \right] \end{aligned} \quad (3.87)$$

where

$$\Gamma_{\nu\mu}^\rho \mathbf{e}_\rho = \vec{\nabla}_{\mathbf{e}_\nu} \mathbf{e}_\mu. \quad (3.88)$$

From the equation 3.87 we see then the covariant derivative is a map from a tangent space to itself, and whose components are given by

$$\left[ \vec{\nabla}_X Y \right]^\mu = x^\nu \left[ \mathbf{e}_\nu (y^\mu) + \Gamma_{\nu\rho}^\mu y^\rho \right], \quad (3.89)$$

and if we choose a coordinate chart, the components of the covariant derivative will be denoted by  $y^\mu_{;\nu}$  are

$$y^\mu_{;\nu} = \partial_\nu y^\mu + \Gamma_{\nu\rho}^\mu y^\rho. \quad (3.90)$$

Here we see the importance of the derivatives of the basis vectors and their direct relation with the Cristoffel symbols, therefore we gotta do a definition

**Definition 21** In a basis  $\{\mathbf{e}_\mu\}$  the components of the connections are given by  $\Gamma_{\rho\nu}^\mu \mathbf{e}_\mu$  called the Cristoffel symbols and are defined by

$$\vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\nu = \Gamma_{\rho\nu}^\mu \mathbf{e}_\mu. \quad (3.91)$$

Now, lets generalize the covariant derivative to tensor fields. If  $T$  is a tensor field  $\binom{r}{s}$  then  $\vec{\nabla} T$  is a tensor field of kind  $\binom{r}{s+1}$ . To show this, lets start with the one-forms:

$$\begin{aligned} \left( \vec{\nabla}_X \eta \right) (Y) &= \vec{\nabla}_X (\eta(Y)) - \eta \vec{\nabla}_X Y = \vec{\nabla}_X (\eta_\mu y^\mu) - \eta_\mu \left( \vec{\nabla}_X Y \right)^\mu, \\ &= x(\eta_\mu) y^\mu + \eta_\mu X(y^\mu) - \eta_\mu (x^\nu \mathbf{e}_\nu (y^\mu) + \Gamma_{\nu\rho}^\mu y^\rho x^\nu), \end{aligned}$$

$$\left(\vec{\nabla}_X \eta\right)(Y) = [X(\eta_\mu) - \Gamma_{\mu\nu}^\rho X^\nu \eta_\rho] y^\mu. \quad (3.92)$$

Therefore  $\vec{\nabla}_X \eta$  is a co-vector field, and the application  $\vec{\nabla} \eta$  is a tensor field of rank  $\binom{0}{2}$  such that

$$\left(\vec{\nabla}_X \eta\right)_\mu = X(\eta_\mu) - \Gamma_{\mu\nu}^\rho x^\nu = x^\nu \mathbf{e}_\nu(\eta_\mu) - \Gamma_{\mu\nu}^\rho x^\nu \eta_\rho, \quad (3.93)$$

and the covariant derivative of the one-form, denoted by  $\eta_{\mu;\nu}$  is given by

$$\eta_{\mu;\nu} = \mathbf{e}_\nu(\eta_\mu) - \Gamma_{\mu\nu}^\rho \eta_\rho. \quad (3.94)$$

Now, lets generalize all this development for a tensor field  $T$  of rank  $\binom{r}{s}$ , so lets take the covariant derivative  $\vec{\nabla} T$

$$\begin{aligned} \vec{\nabla}_\rho T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} &= \partial_\rho T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} + \Gamma_{\sigma\rho}^{\mu_1} T^{\sigma \mu_2 \dots \mu_r}_{\nu_1 \dots \nu_s} + \dots + \Gamma_{\sigma\rho}^{\mu_r} T^{\mu_1 \dots \sigma}_{\nu_1 \dots \nu_s} \\ &\quad - \Gamma_{\nu_1 \rho}^\sigma T^{\mu_1 \dots \mu_r}_{\sigma \nu_2 \dots \nu_s} - \dots - \Gamma_{\nu_s \rho}^\sigma T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \sigma}, \end{aligned}$$

where the positive sings represents the Cristoffel symbols for the covariant derivatives of the vectors that compose the tensor and the negative ones are for the derivatives of the one-form that compose the tensor, and we must note that *the covariant derivative is taking a vector and carry it along another one.*

The second Newton's law, the uncoupled Maxwell laws, the Navier-Stokes equation, almost all the equations in physics are written in terms of second derivatives, therefore we gotta study the properties of second covariant derivatives, the usual partial derivative has the following property

$$\partial_\nu \partial_\mu f = \partial_\mu \partial_\nu f, \quad (3.95)$$

and let  $f$  a smooth functions, lets calculate its covariant derivatives

$$\vec{\nabla}_\nu \vec{\nabla}_\mu f = \partial_\nu \partial_\mu f - \Gamma_{\mu\nu}^\rho \partial_\rho f, \quad (3.96)$$

$$\vec{\nabla}_\mu \vec{\nabla}_\nu f = \partial_\mu \partial_\nu f - \Gamma_{\nu\mu}^\rho \partial_\rho f. \quad (3.97)$$

From these equation we can calculate the anti-symmetric part of these second covariant derivatives given by

$$\vec{\nabla}_{[\nu} \vec{\nabla}_{\mu]} f = -\Gamma_{[\mu\nu]}^\rho \partial_\rho f. \quad (3.98)$$

Thus if we want the second derivative to commute the Cristoffel symbol must be symmetric in the lower indices, but in general for a vector or a tensor this is not generally true.

**Definition 22** A connection  $\vec{\nabla}$  is free of torsion, if  $\vec{\nabla}_A \vec{\nabla}_B f = \vec{\nabla}_B \vec{\nabla}_A f$  for any smooth function  $f$  which is equivalent to  $\Gamma_{[\mu\nu]}^\rho = 0$ .

To have a first step on the understating of the torsion, for example think on a particle in a circular motion, we know there is a curvature factor, but the particle keeps in the circular motion, if there is any force which changes the angular momentum therefore a torque appears, such a system has torsion.

**Theorem 3** For any two free of torsion conditions, if  $X, Y$  are two vector fields, then

$$[X, Y] = \vec{\nabla}_X Y - \vec{\nabla}_Y X. \quad (3.99)$$

Let  $\phi = \{x^0, x^1, x^2, x^3\}$  a coordinate chart, thus

$$\begin{aligned} x^\nu \vec{\nabla}_\nu y^\mu - y^\nu \vec{\nabla}_\nu x^\mu &= x^\nu \partial_\nu y^\mu + \Gamma_{\sigma\nu}^\mu x^\nu y^\sigma - y^\nu \partial_\nu x^\mu - \Gamma_{\nu\sigma}^\mu y^\nu x^\sigma \\ &= (x^\nu \partial_\nu y^\mu - y^\nu \partial_\nu x^\mu) + \Gamma_{\sigma\nu}^\mu x^\nu y^\sigma - \Gamma_{\nu\sigma}^\mu y^\nu x^\sigma + (\Gamma_{\sigma\nu}^\mu x^\sigma y^\nu - \Gamma_{\sigma\nu}^\mu x^\sigma y^\nu) \\ &= [X, Y]^\mu + 2\Gamma_{[\sigma\nu]}^\mu x^\nu y^\sigma, \end{aligned}$$

now, if there is no torsion, therefore the Cristoffel symbol commutes in the lower indices, thus

$$x^\nu \vec{\nabla}_\nu y^\mu - y^\nu \vec{\nabla}_\nu x^\mu = [X, Y]^\mu. \quad (3.100)$$

We can also define a torsion tensor as follows

$$T(X, Y) = \vec{\nabla}_X Y - \vec{\nabla}_Y X - [X, Y]. \quad (3.101)$$

Even if the theory is free of torsion, the second derivatives don't commute for a tensor field. The torsion is analogous to the rotation plane of the particle, *we can have curvature but not torsion*.

### 3.2.4 Levi-Civita connection

**Theorem 4** Let  $M$  a manifold with a metric  $\mathbb{G}$  therefore there is a Riemannian manifold  $(M, \mathbb{G})$ , there exist a free of torsion connections  $\vec{\nabla}$  such that the metric is covariantly constant  $\vec{\nabla} \mathbb{G} = 0$  (this is known as the Levi-Civita connection of metric connection).

**Demonstration:** Let  $X, Y, Z$  three vector fields, thus using the Leibniz rule and assuming the metric is covariantly constant, then

$$X (\mathbb{G}(Y, Z)) = \vec{\nabla}_X [\mathbb{G}(Y, Z)] = \mathbb{G} (\vec{\nabla}_X Y, Z) + \mathbb{G} (Y, \vec{\nabla}_X Z), \quad (3.102)$$

$$Y (\mathbb{G}(Z, X)) = \mathbb{G} (\vec{\nabla}_Y Z, X) + \mathbb{G} (Z, \vec{\nabla}_Y X), \quad (3.103)$$

$$Z (\mathbb{G}(Y, Z)) = \mathbb{G} (\vec{\nabla}_Z X, Y) + \mathbb{G} (X, \vec{\nabla}_Z Y), \quad (3.104)$$

$$H = X (\mathbb{G}(Y, Z)) = \vec{\nabla}_X [\mathbb{G}(Y, Z)] + Y (\mathbb{G}(Z, X)) - Z (\mathbb{G}(Y, Z)) \quad (3.105)$$

and if we assume the theory if free of torsion, then

$$H = 2\mathbb{G} (\vec{\nabla}_X Y, Z) - \mathbb{G} ([X, Y], Z) - \mathbb{G} ([Z, X], Y) + \mathbb{G} ([Y, Z], X), \quad (3.106)$$

thus

$$\mathbb{G} (\vec{\nabla}_X Y, Z) = \frac{1}{2} [X (\mathbb{G}(Y, Z)) + Y (\mathbb{G}(Z, X)) - Z (\mathbb{G}(X, Z))] \quad (3.107)$$

$$+ \mathbb{G} ([X, Y], Z) + \mathbb{G} ([Z, X], Y) - \mathbb{G} ([Y, Z], X), \quad (3.108)$$

$$\mathbb{G} (\vec{\nabla}_{fX} Y, Z) = f\mathbb{G} (\vec{\nabla}_X Y, Z) = \mathbb{G} (f\vec{\nabla}_X Y, Z), \quad (3.109)$$

To finish the prove we must prove that  $\mathbb{G}$  is in fact a metric tensor, therefore just rest to prove it is not degenerate, so lets do the following change of variable  $X \rightarrow Xf$

$$\begin{aligned} \mathbb{G} (\vec{\nabla}_{Xf} Y, Z) &= \frac{1}{2} [fX (\mathbb{G}(Y, Z)) + Y (\mathbb{G}(Z, Xf)) - Z (\mathbb{G}(Xf, Y))] \\ &\quad + \mathbb{G} ([Xf, Y], Z) + \mathbb{G} ([Z, Xf], Y) - \mathbb{G} ([Y, Z], xf) \\ &= \frac{1}{2} [fX (\mathbb{G}(Y, Z)) + fY (\mathbb{G}(Z, X)) - fZ (\mathbb{G}(X, Y))] \\ &\quad + f\mathbb{G} ([X, Y], Z) + f\mathbb{G} ([Z, X], Y) - f\mathbb{G} ([Y, Z], x) = \mathbb{G} (f\vec{\nabla}_X Y, Z), \end{aligned}$$

$$\mathbb{G} (\vec{\nabla}_{fX} Y - f\vec{\nabla}_X Y, Z) = 0. \quad (3.110)$$

Lets calculate the components of the Levi-Civita connection respect to a coordinate basis

$$[\mathbf{e}_\mu, \mathbf{e}_\nu] = \Gamma_{\mu\nu}^\sigma \mathbf{e}_\sigma - \Gamma_{\nu\mu}^\sigma \mathbf{e}_\sigma = 0. \quad (3.111)$$

The equation 3.111 is valid when the theory is free of torsion, thus, lets calculate the following application

$$\mathbb{G}(\vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\nu, \mathbf{e}_\sigma) = \frac{1}{2} [\mathbf{e}_\rho(g_{\nu\sigma}) + \mathbf{e}_\nu(g_{\sigma\rho}) - \mathbf{e}_\sigma(g_{\rho\sigma})] = \frac{1}{2} [\partial_\rho(g_{\nu\sigma}) + \partial_\nu(g_{\sigma\rho}) - \partial_\sigma(g_{\rho\sigma})],$$

$$\mathbb{G}(\Gamma_{\rho\nu}^\tau \mathbf{e}_\tau, \mathbf{e}_\sigma) = \Gamma_{\rho\nu}^\tau g_{\tau\sigma}, \quad (3.112)$$

$$\Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}). \quad (3.113)$$

### 3.2.5 Geodesics

When there is only gravitational interaction the trajectories a test particle will follow are given by the geodesics of the space-time but, first of all, what is a geodesic? A **geodesic** is the shortest path between two points on a manifold. In this order of ideas lets recover the equation we obtained the expression 3.82

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu (X(\tau)) \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0, \quad (3.114)$$

there  $\frac{dx^\mu}{d\tau} = X^\mu$  the components of  $X$ . If we want to talk about geodesics and tangent vectors we must have something to relate different tangent vectors from one point to another on the manifold, we must make an isomorphism to be able to connect the tangent spaces, therefore lets work on the equation 3.82,

$$\frac{d^2 x^\mu}{d\tau^2} = \frac{d}{d\tau} X^\mu (X(\tau)) = \frac{dx^\nu}{d\tau} \frac{\partial x^\mu}{\partial x^\nu} = X^\nu \partial_\nu X^\mu, \quad (3.115)$$

$$X^\nu (\partial_\nu X^\mu + \Gamma_{\nu\rho}^\mu X^\rho) = 0, \quad (3.116)$$

$$X^\nu \vec{\nabla}_\nu X^\mu = 0. \quad (3.117)$$

Therefore to find the geodesic and a relation between the tangent spaces, we must take  $X^\mu$  and project it along the connections over itself.

**Definition 23** Let  $M$  a manifold with connection  $\vec{\nabla}$ , an affinely parameterized geodesic is an integral curve of a vector field  $X$  that satisfies:

$$\vec{\nabla}_X X = 0. \quad (3.118)$$

In geometry, an affine transformation or affine map (also called affinity) between two affine spaces (in particular, two vector spaces) consists of a linear transformation followed by a translation:

$$X \rightarrow \mathbb{A}X + b, \quad (3.119)$$

In the finite-dimensional case, every affine transformation can be represented by a matrix  $\mathbb{A}$  and a vector  $b$  of the vector space. Geometrically, an affine transformation in a Euclidean space is a transformation that preserves:

- The collinearity (and coplanarity) relationships between points, that is, points that fell on the same line (or on the same plane) before the transformation, are preserved after an affine transformation.
- The ratios between distances along a line, that is, for three different aligned points  $P_1, P_2, P_3$  the reasons  $|P_2P_1| / |P_3P_2|$  before and after the transformation are the same.

An affine transformation  $f : \mathbb{A} \rightarrow \mathbb{A}'$  between two affine spaces  $(\mathbb{A}, E, \delta)$  and  $(\mathbb{A}', E', \delta')$  is an application between two points that acts linearly on the vectors that join them. Formally,  $f$  is an affine transformation if there exists a linear map  $g$  such that  $\forall p, q \in \mathbb{A}$

$$f(p)\vec{f}(q) = g(\vec{pq}). \quad (3.120)$$

Before going on, let's remember the usefulness of this kind of affine transformations.

### 3.2.5.1 Affine transformations in Galilean relativity

In the classical mechanics there is a really important group that rules the inertial frames and the mechanics, the well known as the Galilean group.

**Definition 24** *The **Galilean group** is a group  $G = \langle \mathbb{T}, \circ \rangle$  composed by the Galilean transformations under the compositions bilinear operation.*

The Galilean transformations are the following three

1. The displacement of the origin of the reference frame

$$T_d(\vec{r}, t) \rightarrow (\vec{r} + \vec{r}', t + t'). \quad (3.121)$$

2. The rotations of the coordinate axis of the reference frame

$$T_r(\vec{r}, t) \rightarrow (\mathbb{R}\vec{r}, t) : \mathbb{R} \in SU(3). \quad (3.122)$$

3. Motion at uniform velocity  $\vec{V}$ 

$$T_M(\vec{r}, t) \rightarrow (\vec{r} + \vec{V}t, t). \quad (3.123)$$

It is easy to prove that the position of a particle can be determined using these three transformations, in the following form

$$(\vec{r}', t') = (T_d \circ T_r) \vec{r} + \vec{r}t, \quad (3.124)$$

we see it is naturally an affine transformation all the mechanics we studied in the school or in the first year of our degrees is written in terms of affine transformations between inertial reference frames which move one each other at constant velocity.

**3.2.5.2 Affine transformations in special relativity**

In the same way we're gonna deduce the affine transform for Minkowskian space-time, let  $\vec{x}'$  a vector which obeys an affine transformation as:

$$\vec{x}' = \mathcal{L}(\vec{x}) = \Lambda \vec{x} + \vec{a} \quad : \quad \vec{x}, \vec{a} \in \mathbb{R}^4, \quad \Lambda \in G_L,$$

Where  $G_L$  is the Lorentz's group, but before doing that, lets discover the linear transform case:

$$\vec{x}' = \Lambda \vec{x} \quad (3.125)$$

The transform  $\Lambda$  must give invariant the light cone so let  $\vec{v}, \vec{w} \in \mathbb{R}^4$ :

$$\langle \Lambda \vec{v}, \Lambda \vec{w} \rangle = \langle \vec{v}, \vec{w} \rangle \rightarrow (\Lambda \vec{v})^T \eta (\Lambda \vec{w}) = \vec{v}^T \eta \vec{w}$$

$$\vec{v}^T \Lambda^T \eta \Lambda \vec{w} = \vec{v}^T \eta \vec{w}$$

$$\eta = \Lambda^T \eta \Lambda \quad (3.126)$$

The equation 3.126 is a similarity transform, is easily to see that  $\Lambda$  only could have two values for its determinant  $|\Lambda| = \pm 1$ .

Demonstration:

$$|\eta| = |\Lambda^T \eta \Lambda| = |\Lambda^T| * |\eta| * |\Lambda|, \quad (3.127)$$

We can see in 3.126 that  $|\eta| = -1$  and using the properties  $|A| = |A^T|$  we can rewrite the equation 3.127 as:

$$1 = |\Lambda|^2 \rightarrow |\Lambda| = \pm 1.$$

This is only possible in three cases:

1. Temporal or parity inversions:

$$\Lambda = \text{diag}[\mp 1, \pm 1, \pm 1, \pm 1]$$

Respectively

2. Unitary transformations:

$$\Lambda = \begin{bmatrix} 1 & \vec{O}^T \\ \vec{O} & R \end{bmatrix} : R \in SO(3)$$

3. Speed boosts, for example a x boost:

$$\Lambda = \begin{bmatrix} \gamma & -v_x \gamma & 0 & 0 \\ -v_x \gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} : \gamma = \frac{1}{\sqrt{1 - v_x^2}} \quad C = 1 \quad (3.128)$$

If we are out from the natural coordinates  $\gamma$  changes to  $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{C^2}}}$

As the Galilean group the Lorentz's group is made by 6 elements, 3 rotations and 3 speed boosts, but in general:

$$\mathcal{L}(\vec{x}) = \Lambda \vec{x} + \vec{a} : \vec{x}, \vec{a} \in \mathbb{R}^4$$

That general form is more general than the last one, we obtain it from the Lorentz's group  $G_L$  and 4 translations, we call it the *Poincaré's group*  $G_{\mathcal{P}}$ .

Let  $\mathcal{O} \wedge \bar{\mathcal{O}}$  inertial observers such that there is a relative movement between them, thus:  $\bar{\mathcal{O}} \rightarrow v : v < 1$  relative to  $\mathcal{O} \wedge v > 1$  in the  $x$  direction, thus the coordinates transformed are:

$$\begin{aligned} \bar{t} &= At + Bx \\ \bar{x} &= Ct + Dx \\ \bar{y} &= y \\ \bar{z} &= z \end{aligned}$$

Now, because the light cone is invariant we've got the following equation:

$$\begin{aligned} -t^2 + x^2 + y^2 + z^2 &= -\bar{t}^2 + \bar{x}^2 + \bar{y}^2 + \bar{z}^2 \\ -t^2 + x^2 &= -(A^2 - C^2)t^2 + (D^2 - B^2)x^2 + 2(CD - AB)tx \end{aligned}$$

$$A^2 - C^2 = 1$$

$$D^2 - B^2 = 1$$

$$AB = CD$$

This is a inconsistent system of equations, we've only 3 equations and fourth variables! But there're some special function that satisfies that equations, look at this equation:  $\cosh^2(\phi) - \sinh^2(\phi) = 1$  THAT'S IT! There is our solution, thus:

$$\begin{aligned} A &= D = \cosh(\phi) \\ C &= B = -\sinh(\phi) \end{aligned}$$

Thus our Lorentz's transformation becomes:

$$\begin{bmatrix} \bar{t} \\ \bar{x} \end{bmatrix} = \begin{bmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \quad (3.129)$$

That is a beautiful solution! Looks like a rotation! But it is not a rotation, we don't even know what in the hell is  $\phi$  so lets find out what is  $\phi$ .

Let  $\bar{x} = 0$  thus  $0 = -t\sinh(\phi) + vt\cosh(\phi)$  solving for  $v$  we've got:

$$v = \tanh(\phi) \rightarrow \tanh^2(\phi) + \operatorname{sech}^2(\phi) = 1 \rightarrow \operatorname{sech}(\phi) = \sqrt{1 - v^2}$$

$$\gamma = \cosh(\phi) = \frac{1}{\sqrt{1 - v^2}} \quad (3.130)$$

$$\sinh(\phi) = \frac{v}{\sqrt{1 - v^2}} = \gamma v \quad (3.131)$$

With all of this, we can get the final form of our Lorentz's transformation:

$$\begin{aligned}
\bar{t} &= \gamma(t - vx) \\
\bar{x} &= \gamma(x - vt) \\
\bar{y} &= y \\
\bar{z} &= z
\end{aligned}$$

If we are not in natural coordinates, the transformation look like this:

$$\begin{aligned}
\bar{t} &= \gamma\left(t - \frac{v}{c^2}x\right) \\
\bar{x} &= \gamma(x - vt) \\
\bar{y} &= y \\
\bar{z} &= z
\end{aligned}$$

That is something really beautiful, only with the postulates easily we can get nature of the mechanics for high speeds, and easily see that the nature again follows a group's structure! Having all on count we can get the Lorentz's transform:

$$[\Lambda] = \begin{bmatrix} \gamma & -\gamma v_x & -\gamma v_y & -\gamma v_z \\ -\gamma v_x & 1 + (\gamma - 1)\frac{v_x^2}{v^2} & (\gamma - 1)\frac{v_x v_y}{v^2} & (\gamma - 1)\frac{v_x v_z}{v^2} \\ -\gamma v_y & (\gamma - 1)\frac{v_y v_x}{v^2} & 1 + (\gamma - 1)\frac{v_y^2}{v^2} & (\gamma - 1)\frac{v_y v_z}{v^2} \\ -\gamma v_z & (\gamma - 1)\frac{v_z v_x}{v^2} & (\gamma - 1)\frac{v_z v_y}{v^2} & 1 + (\gamma - 1)\frac{v_z^2}{v^2} \end{bmatrix} \quad (3.132)$$

Lets turn back on the geodesics in manifolds. Let  $t$  the affine parameter for  $\vec{\nabla}_X X = 0$  and suppose another parameter  $u$  such that  $t = t(u)$  and the following relatio is fulfilled

$$\frac{dt}{du} > 0, \quad (3.133)$$

this condition is someway like the convex property on the Legendre's transformation, the reason why the changing rate of the parameter  $t$  respect to  $u$  must be greater than zero is to guarantee that for two different values of  $u$  they wont have the same image, or the tame, the relation  $t = t(u)$  is injective and well behaved.

Lets then find out the geodesics from this other parameterization and what relation must be between these two pictures to they represent the same geodesic see 3.8.

$$Y = hX, \quad h = \frac{dt}{du}, \quad (3.134)$$

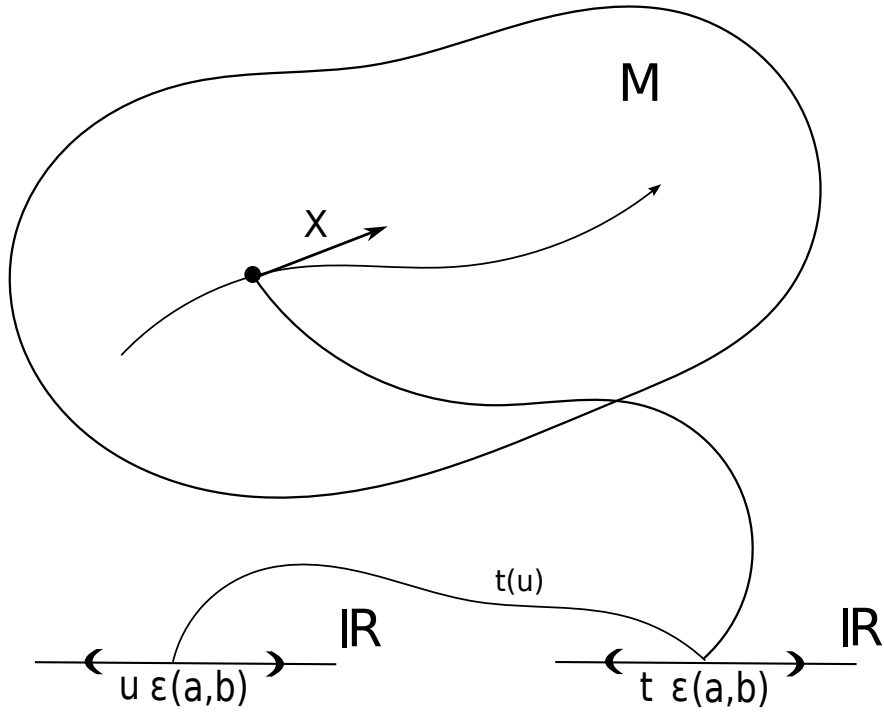


Figure 3.8: Affine parameterization of the geodesics, here we illustrate the path (geodesic) a free particle will follow which is parameterized with a parameter  $t$  but this is some application of the other parameter  $u$ .

$$\vec{\nabla}_Y Y = \vec{\nabla}_{hX}(hX) = h\vec{\nabla}_X(hX) = h^2\vec{\nabla}_X X + hX = h^2\vec{\nabla}_X X + hX\vec{\nabla}_X h, \quad (3.135)$$

$$\vec{\nabla}_Y Y = hX\vec{\nabla}_X h = fY. \quad (3.136)$$

Then we can see it is not parameterized affinely, the affine parameters are linear relations with the other parameters. Both are the same geodesic but the second one has not affine parameterization, but lets see how must be  $h$  to have such a parameterization

$$X(h) = 0 \equiv x^\alpha \partial_\alpha h = 0, \quad (3.137)$$

therefore  $h$  is covariantly constant and  $\frac{dt}{du} = h$  or what is the same thing

$$u = at + b, a > 0. \quad (3.138)$$

Where this is fulfilled  $U$  is then either an affine parameter, thus its geodesics equation is

$$\vec{\nabla}_Y Y = 0. \quad (3.139)$$

In the general relativity the proper time  $\tau$  is an affine parameters and it works to describe the time-like geodesics, *the free particles move on geodesics* this having on count the Levi-Civita connection.

### 3.3 Riemann's tensor

Here we're going to talk about the hearth of the general relativity, the hearth of the differential geometry (from a physical point of view hehe), the Riemann's tensor, inside this beautifull quantity there is the information of the curvature of our space-time the curvature of the geodesics? the curvature of the quantum vacuum? Yes, all this informations can be found inside this tensor, thus lets have fun on this topic.

#### 3.3.1 Curvature of the Levi-Civita connection

Until here we haven't solved the problem of transporting a vector from a tangent space to another vector in another tangent space, here there is the solution of that, what is called the *parallel transport*.

Let  $X$  a tangent vector to a curve, a tensor field  $T$  is transported paralely along the curve if and only if

$$\vec{\nabla}_X T = 0. \quad (3.140)$$

This means the tensor field does not change along the vector, therefore we had already solved the problem of parallel transport implicitly!  $\vec{\nabla}_X X = 0$  this a curve whose tangent vector is paralely transported along the same curve, but but but, we need it to be unique, lets see if this is or not unique. Let  $T$  be a tensor of rank  $\binom{1}{1}$  we're going to define a coordinate chart such that

$$\phi = (x^1, \dots, x^n) \quad X^\mu = \frac{dx^\mu}{d\lambda} \quad (3.141)$$

where  $\lambda$  is the curve parameter;

$$\left(\vec{\nabla}_X T\right)_\nu^\mu = (X^\sigma \nabla_\sigma T_\nu^\mu) = X^\sigma \partial_\sigma T_\nu^\mu + \Gamma_{\rho\sigma}^\mu T_\nu^\rho X^\sigma - \Gamma_{\nu\sigma}^\rho T_{;\rho}^\mu X^\sigma \quad (3.142)$$

$$\left(\vec{\nabla}_X T\right)_\nu^\mu = \frac{dx^\sigma}{d\lambda} \partial_\sigma T_\nu^\mu + \Gamma_{\rho\sigma}^\mu T_\nu^\rho X^\sigma - \Gamma_{\nu\sigma}^\rho T_{rho}^\mu X^\sigma = 0. \quad (3.143)$$

We have then a first order equation of the form

$$\frac{dT^\mu_\nu}{d\lambda} + \dots = 0, \quad (3.144)$$

therefore there is a unique solution once we define the initial condition, then the tensor fields in  $q$  and  $p$  are isomorphic.

**Definition 25** *The tensor of curvature of Riemann of a connection  $\lambda$  is defined by*

$$R^a_{bcd}Z^bX^cY^d = (R(X, Y)Z)^a, \quad (3.145)$$

where  $X, Y, Z$  are vector fields and  $R(X, Y)Z$  is also a vector field defined by

$$R(X, Y)Z = \vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_X Z - \vec{\nabla}_{[X, Y]}Z. \quad (3.146)$$

The Riemann's tensor has the following properties

$$1. \ R(X, Y)Z = -R(Y, X)Z.$$

2.

$$R(fX, Y)Z = \vec{\nabla}_{fX} \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_{fX} Z - \vec{\nabla}_{[fX, Y]}Z \quad (3.147)$$

$$= f\vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y (f\vec{\nabla}_X Z) - \vec{\nabla}_{f[X, Y] - Y(f)X}Z \quad (3.148)$$

$$= f\vec{\nabla}_X \vec{\nabla}_Y Z - f\vec{\nabla}_Y \vec{\nabla}_X Z - Y(f)\vec{\nabla}_X Z - f\vec{\nabla}_{[X, Y]}Z + Y(f)\vec{\nabla}_X Z = fR(X, Y)Z. \quad (3.149)$$

This means the Riemann's curvature tensors is multilinear in his two arguments, the demonstrations for the second argument is trivial therefore.

$$R(X, fY)Z = fR(X, Y)Z. \quad (3.150)$$

3.

$$\begin{aligned} R(X, Y)fZ &= \vec{\nabla}_X \vec{\nabla}_Y (fZ) - \vec{\nabla}_Y \vec{\nabla}_X (fZ) - \vec{\nabla}_{X, Y}fZ \\ &= f\vec{\nabla}_X \vec{\nabla}_Y Z - f\vec{\nabla}_Y \vec{\nabla}_X Z + \vec{\nabla}_X Z \vec{\nabla}_Y f + \vec{\nabla}_X f \vec{\nabla}_Y Z + Z \vec{\nabla}_X \vec{\nabla}_Y f \\ &\quad - \vec{\nabla}_Y Z \vec{\nabla}_X f - \vec{\nabla}_Y f \vec{\nabla}_X Z - Z \vec{\nabla}_Y \vec{\nabla}_X f - \vec{\nabla}_{X, Y}fZ \\ &= f(\vec{\nabla}_X \vec{\nabla}_Y Z - \vec{\nabla}_Y \vec{\nabla}_X Z - f\vec{\nabla}_{X, Y}Z), \end{aligned}$$

$$R(X, Y)fZ = fR(X, Y)Z. \quad (3.151)$$

In this way, the Riemann's tensor fulfills all those properties and correctly it defines a tensor, lets see how looks the Riemann's tensor given a coordinate chart and a basis  $\{\mathbf{e}_\mu = \frac{\partial}{\partial x^\mu}\}$  and assuming a free of torsion theory

$$[\mathbf{e}_\mu, \mathbf{e}_\nu] = 0, \quad (3.152)$$

$$R(\mathbf{e}_\rho, \mathbf{e}_\sigma)\mathbf{e}_\mu = \vec{\nabla}_{\mathbf{e}_\rho} \vec{\nabla}_{\mathbf{e}_\sigma} \mathbf{e}_\mu - \vec{\nabla}_{\mathbf{e}_\sigma} \vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\mu, \quad (3.153)$$

$$\begin{aligned} R(\mathbf{e}_\rho, \mathbf{e}_\sigma) &= \vec{\nabla}_{\mathbf{e}_\rho} (\Gamma^\tau_{\mu\sigma} \mathbf{e}_\tau) - \vec{\nabla}_{\mathbf{e}_\sigma} (\Gamma^\tau_{\rho\mu} \mathbf{e}_\tau) \\ &= \vec{\nabla}_{\mathbf{e}_\rho} \Gamma^\tau_{\mu\sigma} \mathbf{e}_\tau + \Gamma^\tau_{\mu\sigma} \vec{\nabla}_{\mathbf{e}_\rho} \mathbf{e}_\tau - \vec{\nabla}_{\mathbf{e}_\sigma} \Gamma^\tau_{\rho\mu} \mathbf{e}_\tau - \Gamma^\tau_{\rho\mu} \vec{\nabla}_{\mathbf{e}_\sigma} \mathbf{e}_\tau, \end{aligned}$$

$$\begin{aligned} R(\mathbf{e}_\rho, \mathbf{e}_\sigma)\mathbf{e}_\mu &= \partial_\rho \Gamma^\tau_{\mu\sigma} \mathbf{e}_\tau + \Gamma^\tau_{\mu\sigma} \Gamma^\nu_{\tau\rho} \mathbf{e}_\nu - \partial_\sigma \Gamma^\tau_{\mu\rho} \mathbf{e}_\tau - \Gamma^\tau_{\mu\rho} \Gamma^\nu_{\sigma\tau} \mathbf{e}_\nu \\ &= \{\partial_\rho \Gamma^\nu_{\mu\sigma} + \Gamma^\tau_{\mu\sigma} \Gamma^\nu_{\tau\rho} - \partial_\sigma \Gamma^\tau_{\mu\rho} - \Gamma^\tau_{\mu\rho} \Gamma^\nu_{\sigma\tau}\} \mathbf{e}_\nu, \end{aligned}$$

$$R^\nu_{\mu\rho\sigma} = \partial_\rho \Gamma^\nu_{\mu\sigma} + \Gamma^\tau_{\mu\sigma} \Gamma^\nu_{\tau\rho} - \partial_\sigma \Gamma^\tau_{\mu\rho} - \Gamma^\tau_{\mu\rho} \Gamma^\nu_{\sigma\tau}. \quad (3.154)$$

**Definition 26** The Ricci's curvature tensor, is a tensor of rank  $\binom{0}{2}$  is defined by

$$R_{ab} = R^c_{acb}. \quad (3.155)$$

With this tensor we can construct the Einstein's field equations, under the assumption of a free of torsion theory, and we can define the *Ricci's identity* as

$$R^a_{bcd} Z^b = \vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a, \quad (3.156)$$

lets prove it

$$\begin{aligned} [R(X, Y)Z]^a &= X^c \vec{\nabla}_c (Y^d \vec{\nabla}_d Z^a) - Y^d \vec{\nabla}_d (X^c \vec{\nabla}_c Z^a) - [X, Y]^c \vec{\nabla}_c Z^a \\ &= X^c Y^d \vec{\nabla}_c \vec{\nabla}_d Z^a + X^c (\vec{\nabla}_c Y^d) \vec{\nabla}_d Z^a - Y^d X^c \vec{\nabla}_d \vec{\nabla}_c Z^a - Y^d (\vec{\nabla}_d X^c) \vec{\nabla}_c Z^a - [X, Y]^c \vec{\nabla}_c Z^a, \end{aligned}$$

$$[R(X, Y)Z]^a = X^c Y^d [\vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a] + X^c (\vec{\nabla}_c Y^d) \vec{\nabla}_d Z^a - Y^d (\vec{\nabla}_d X^c) \vec{\nabla}_c Z^a - [X, Y]^c \vec{\nabla}_c Z^a. \quad (3.157)$$

and for a free of torsion theory we can develop the commutator

$$[X, Y]^c = \left[ \vec{\nabla}_X Y - \vec{\nabla}_Y X \right]^c = X^e \vec{\nabla}_e Y^c - Y^e \vec{\nabla}_e X^c, \quad (3.158)$$

$$[X, Y] \vec{\nabla}_c Z^a = X^e \left( \vec{\nabla}_e Y^c \right) \left( \vec{\nabla}_c Z^a \right) - Y^e \left( \vec{\nabla}_e Y^c \right) \left( \vec{\nabla}_c Z^a \right), \quad (3.159)$$

$$X^c \left( \vec{\nabla}_c Y^d \right) \left( \vec{\nabla}_d Z^a \right) - Y^d \left( \vec{\nabla}_d X^c \right) \left( \vec{\nabla}_c Z^a \right) - [X, Y]^c \vec{\nabla}_c Z^a = 0, \quad (3.160)$$

$$[R(X, Y)Z]^a = \left( \vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a \right) X^c Y^d = R^a_{bcd} Z^b X^c Y^d. \quad (3.161)$$

By comparison it only rests that

$$\vec{\nabla}_c \vec{\nabla}_d Z^a - \vec{\nabla}_d \vec{\nabla}_c Z^a = R^a_{bcd} Z^b. \quad (3.162)$$

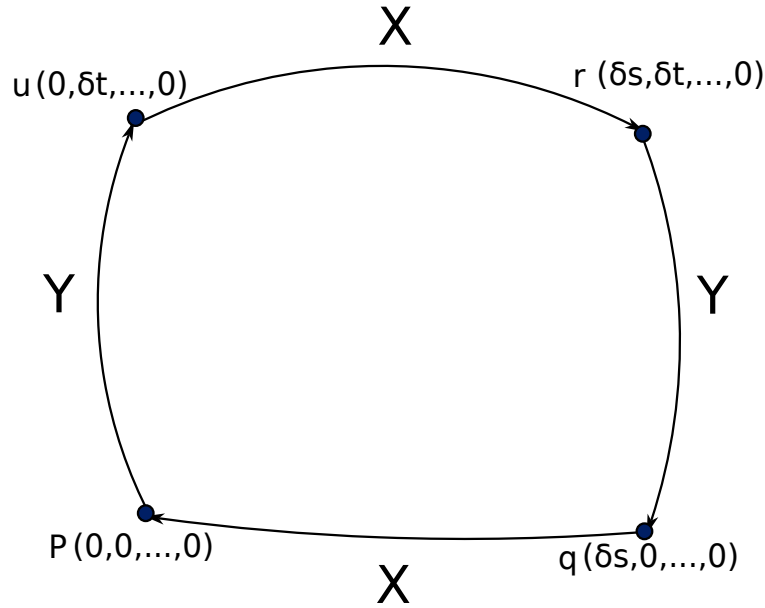


Figure 3.9: Connection between  $q$  and  $p$  with a curve along a variation of  $s$  with  $X$  as tangent vector, in the same way  $q$  and  $r$  can be connected by a curve with  $Y$  as tangent vector,  $p$  and  $u$  can be connected with a curve with  $Y$  as tangent vector and same with  $u$  and  $r$  but with  $X$  as tangent vector.

Now, let's turn back to the relation between the Riemann's tensor and the dependence of the path of the parallel transport. Let  $X$  and  $Y$  vector fields which are linearly independent everywhere such that  $[X, Y] = 0$ . Let  $(s, t, \dots)$  a coordinate chart such that

$$X = \frac{\partial}{\partial s} \text{ \& } Y = \frac{\partial}{\partial t}. \quad (3.163)$$

Let  $p \in M$  and given a coordinate chart  $p$  is represented as  $(0, \dots, 0)$ . Let  $q, r, u$  points with coordinates  $(\delta s, 0, 0, \dots)$ ,  $(\delta s, \delta t, 0, \dots)$ ,  $(0, \delta L, 0, \dots)$ , respectively, where  $\delta s, \delta t$  are little displacement of that coordinate, we can connect the fourth point as is illustrated in the figure 3.9.

Now, let  $Z_p \in T_p(M)$ . Lets transport parallelly  $Z_p$  along  $pqr$  to obtain a vector  $Z_r \in T_r(M)$ , the same thing but to transport  $Z_p$  along  $pur$  to obtain the vector  $Z'_r \in T_r(M)$  and lets calculate the difference  $Z'_r - Z_r$  for a free of torsion connection. it is convenient to introduce a new coordinate chart known as *normal coordinates* in  $p$ ,  $(\Gamma^\mu_{\nu\rho}(p) = 0)$ . The indices  $\mu, \nu$  will refer to this chart.  $S$  and  $t$  will be used as parameters along the curves whose tangent vectors are  $X$  and  $Y$ , respectively.

$pq$  is an integral curve whose tangent vector is  $X$  and parameter  $S$ . Along of  $pq$ ,  $Z$  is transported parallelly:  $\vec{\nabla}_X Z = 0$ , therefore

$$\frac{dZ^\mu}{ds} = -\Gamma^\mu_{\nu\rho} Z^\nu X^\rho, \quad (3.164)$$

and in this way

$$\frac{d^2 Z^\mu}{ds^2} = -(\Gamma^\mu_{\nu\rho} Z^\nu X^\rho)_{,\sigma} X^\sigma. \quad (3.165)$$

Now using Taylor's expansion theorem we get:

$$Z^\mu_q = Z^\mu_q + \left( \frac{dZ^\mu}{ds} \right)_p \delta s + \frac{1}{2} \left( \frac{d^2 Z^\mu}{ds^2} \right)_p \delta s^2 + \mathcal{O}(\delta s^3). \quad (3.166)$$

We must note that:

$$\frac{d^2 Z^\mu}{ds^2} = -\Gamma^\mu_{\nu\rho,\sigma} Z^\nu X^\rho X^\sigma - \Gamma^\mu_{\nu\rho} Z^\nu_{,\sigma} X^\rho X^\sigma - \Gamma^\mu_{\nu\rho} Z^\nu X^\rho_{,\sigma} X^\sigma, \quad (3.167)$$

$$Z^\mu_q = Z^\mu_p - \frac{1}{2} (\Gamma^\mu_{\nu\rho,\sigma} Z^\nu X^\rho X^\sigma)_p \delta s^2 + \mathcal{O}(\delta s^3), \quad (3.168)$$

where we had used  $\Gamma^\mu_{\nu\rho}(p) = 0$  in normal coordinates at  $p$ . Now consider the parallel transport along  $qr$ :

$$\frac{dZ^\mu}{dt} = -\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho, \quad (3.169)$$

$$\frac{d^2 Z^\mu}{dt^2} = -(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} Y^\sigma, \quad (3.170)$$

$$Z_r^\mu = Z_q^\mu - (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_q \delta t - \frac{1}{2} [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} Y^\sigma]_q \delta t^2 + \mathcal{O}(\delta t^3). \quad (3.171)$$

$$(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_Q = (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_p + \left[ \frac{d}{ds} (\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho) \right]_p \delta s + \mathcal{O}(\delta s^2) \quad (3.172)$$

$$= [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma} X^\sigma]_p \delta s + \mathcal{O}(\delta s^2) \quad (3.173)$$

$$= (\Gamma_{\nu\rho}^\mu z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2) \quad (3.174)$$

$$[(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\sigma}]_q = [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho)_{,\Sigma} Y^\sigma]_p + \mathcal{O}(\delta s) = (\Gamma_{\nu\rho,\sigma}^\mu Z^\nu Y^\rho Y^\sigma)_p + \mathcal{O}(\delta s), \quad (3.175)$$

$$Z_r^\mu = Z_q^\mu - [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2)] \delta t - \frac{1}{2} [(\Gamma_{\nu\rho}^\mu Z^\nu Y^\rho X^\sigma)_p \delta s + \mathcal{O}(\delta s^2)] \delta t^2 + \mathcal{O}(\delta t^2), \quad (3.176)$$

$$Z_r^\mu = Z_p^\mu - \frac{1}{2} (\Gamma_{\nu\rho,\sigma}^\mu)_p [Z^\nu (X^\rho X^\sigma \delta s^2 + Y^\rho Y^\sigma \delta t^2 + 2Y^\rho X^\sigma \delta s \delta t)]_p + \mathcal{O}(\delta^3). \quad (3.177)$$

Now, consider the parallel transport along  $pur$ . The result can be obtained from the last expression interchanging  $X$  with  $Y$  and  $s$  with  $t$

$$Z_r'^\mu = Z_p^\mu - \frac{1}{2} (\Gamma_{\nu\rho,\sigma}^\mu)_p [Z^\nu (Y^\rho Y^\sigma \delta t^2 + X^\rho X^\sigma \delta s^2 + 2X^\rho Y^\sigma \delta t \delta s)]_p + \mathcal{O}(\delta^3) \quad (3.178)$$

in this way the difference between the two parallel transport is

$$\Delta Z_r^\mu = Z_r'^\mu - Z_r^\mu \quad (3.179)$$

$$= [\Gamma_{\nu\rho,\sigma}^\mu Z^\nu (Y^\rho X^\sigma - X^\rho Y^\sigma)]_p \delta t \delta s + \mathcal{O}(\delta^3) \quad (3.180)$$

$$= [(\Gamma_{\nu\sigma,\rho}^\mu - \Gamma_{\nu\rho,\sigma}^\mu) Z^\nu X^\rho Y^\sigma]_p \delta s \delta t \mathcal{O}(\delta^3) \quad (3.181)$$

$$= [(R_{\nu\rho\sigma}^\mu - \Gamma_{\nu\sigma}^\tau \Gamma_{\tau\rho}^\mu + \Gamma_{\nu\rho}^\tau \Gamma_{\tau\sigma}^\mu) Z^\nu X^\rho X^\sigma]_p \delta t \delta s + \mathcal{O}(\delta^3), \quad (3.182)$$

$$\Delta Z_r^\mu = (R^\mu_{\nu\rho\sigma} Z^\nu X^\rho X^\sigma)_p \delta s \delta t + \mathcal{O}(\delta^3), \quad (3.183)$$

$$\Delta Z_r^\mu = (R^\mu_{\nu\rho\sigma} Z^\nu X^\rho X^\sigma)_r \delta s \delta t + \mathcal{O}(\delta^3). \quad (3.184)$$

In the last equality is used that the quantities in  $p$  and  $r$  are different by  $\mathcal{O}(\delta)$ . this result is derived in a coordinate basis defined using normal coordinates in  $p$  but now both side take on count tensors in  $r$  therefore our equations is independent of the basis, thus

$$\left( R^a_{bcd} Z^b X^c Y^d \right)_r = \lim_{\delta \rightarrow 0} \frac{\Delta Z_r^\mu}{\delta s \delta t}. \quad (3.185)$$

The Riemann's tensor measures the dependence of the path of the parallel transport. The last result can be reinterpreted describing the effect of the parallel transport of a vector  $Z_r^a$  along a closed curved  $qpur$ , see figure 3.9, to obtain the vector  $Z_r'^a$ . In this way  $\Delta Z_r^a$  measures the change in  $Z_r^a$  when it is transported parallel y around a closed curve.

Lets see the symmetries of the Riemann's tensor

1.  $R^a_{bcd} = -R^a_{bdc}$  it is anti symmetric.
2. If  $\vec{\nabla}$  is a free of torsion connections then

$$R^a_{[bcd]} = \frac{1}{3!} (R^a_{bcd} + R^a_{cdb} + R^a_{dbc} - R^a_{cbd} - R^a_{bdc} - R^a_{dcb}) = 0. \quad (3.186)$$

3. If  $\vec{\nabla}$  is a free of torsion connection then

$$R^a_{b[cd;e]} = 0. \quad (3.187)$$

Thanks to the fact the metric tensor is covariantly constant, then it will allow us to raise and low indices and using these we can see the following properties

1.  $R_{abcd} = R_{cdab}$ .
2.  $R_{ab} = R_{ba}$ .

**Definition 27** The Ricci's scalar is

$$R = g^{ab} R_{ab}. \quad (3.188)$$

**Definition 28** *The Einstein's tensor, is a symmetric tensor of rank  $\binom{0}{2}$  and is defined as*

$$G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R. \quad (3.189)$$

**Proposition:** the covariant derivative of the Einstein's tensor is zero

$$\vec{\nabla}_a G^{ab} = 0. \quad (3.190)$$

**Demonstration:**

$$\begin{aligned} \vec{\nabla}_a G^{ab} &= \vec{\nabla}_a \left[ R^{ab} - \frac{1}{2}g^{ab}R \right] = \vec{\nabla}_a R^{ab} - \frac{1}{2}\vec{\nabla}_a g^{ab}R - \frac{1}{2}g^{ab}\vec{\nabla}_a R \\ &= \vec{\nabla}_a R^{ab} - \frac{1}{2}\vec{\nabla}_a g^{ab} \left( g^{cd}R_{cd} \right) - \frac{1}{2}g^{ab} \left[ \vec{\nabla}_a g^{cd}R_{cd} + g^{cd}\vec{\nabla}_a R_{cd} \right] \\ &= \vec{\nabla}_a R^{ab} - \frac{1}{2}\vec{\nabla}_a g^{ab} \left( g^{cd}R_{cd} \right) - \frac{1}{2}\vec{\nabla}_a g^{cd}g^{ab}R_{cd} - \frac{1}{2}g^{ab}g^{cd}\vec{\nabla}_a R_{cd} \\ &= \vec{\nabla}_a R^{ab} - \frac{1}{2}\vec{\nabla}_a g^{ab} \left( g^{cd}R_{cd} \right) - \frac{1}{2}\vec{\nabla}_a g^{cd}g^{ab}R_{cd} - \frac{1}{2}g^{ab}g^{cd}\vec{\nabla}_a R_{cd}, \end{aligned}$$

and using the Leibniz rule is easy to see that

$$\vec{\nabla}_a \left[ g^{ab}g^{cd}R_{cd} \right] = \vec{\nabla}_a g^{ab} \left[ g^{cd}R_{cd} \right] + g^{ab}R_{cd}\vec{\nabla}_a g^{cd} + g^{ab}g^{cd}\vec{\nabla}_a R_{cd}, \quad (3.191)$$

thus

$$\vec{\nabla}_a G^{ab} = \vec{\nabla}_a R^{ab} + \frac{1}{2}g^{ab}g^{cd}\vec{\nabla}_a R_{cd} - \vec{\nabla}_a R^{ab} - \frac{1}{2}g^{ab}g^{cd}\vec{\nabla}_a R_{cd} = 0. \quad (3.192)$$

### 3.3.1.1 Einstein's equations

These equations holds under some postulates, the known as the general relativity postulates which are

1. The space-time is a pseudo-Riemannian manifold equipped with a Levi-Civita connection  $(M, \mathbb{G}, \vec{\nabla})$ .
2. The free particles follows geodesics.
3. The energy-momentum tensor  $T_{ab}$  is conserved:  $\vec{\nabla}_a T^{ab} = 0$ .

4. The Einstein's tensor satisfies that

$$G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R = \frac{8\pi G}{C^4}T_{ab}. \quad (3.193)$$

We can see that there is no matter thus  $T_{ab} = 0 \rightarrow R = 0 \rightarrow R_{ab} = 0$  lets see this

$$g^{ab} \left( R_{ab} - \frac{1}{2}g_{ab}R \right) = \alpha g^{ab}T_{ab},$$

$$R - 2R = \alpha T \rightarrow T = -\frac{1}{\alpha}R,$$

$$R_{ab} = \alpha T_{ab} + \frac{1}{2}g_{ab}R = \alpha \left( T_{ab} - \frac{1}{2}g_{ab}T \right), \quad (3.194)$$

we can see easily that if there is no matter the Ricci's tensor is zero.

The Einstein's equations are ten second order, non linear and copuled equations of the metric tensor, but this form is not unique. Lets think about a symmetric tensor  $H_{ab}$ , which is covariantly constant, the space-time must be a pseudo-Riemannian manifold of four dimensions, with a levi-civita connections, thus by the **Lovelack theorem**

$$H_{ab} = \alpha G_{ab} + \beta g_{ab}, \quad (3.195)$$

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = \frac{8\pi G}{C^4}T_{\alpha\beta}. \quad (3.196)$$

$\Lambda$  is known as the cosmologic constant and it was first rejected by Einstein but reintroduced by himself to explain the accelerated expansion of the universe, this constant is nowadays studied in relation with the dark energy and dark matter.

### 3.3.2 Properties related to the Riemann's tensor

In this part we're going to study tensor equations related to the Riemann's tensor and some really important identities as the Jacobi and Bianchi's identities, which are really important in the development of general relativity and will guide us to the Einstein's tensor. Let  $\{\mathbf{e}_\alpha\} = \left\{ \frac{\partial}{\partial x^\alpha} \right\}$  a coordinate basis, then in this basis the components of the Riemann's tensor become

$$R^\alpha_{\beta\mu\nu} = \partial_\nu \Gamma^\alpha_{\beta\mu} - \partial_\mu \Gamma^\alpha_{\beta\nu} + \Gamma^\alpha_{\sigma\mu} \Gamma^\sigma_{\beta\nu} - \Gamma^\alpha_{\sigma\nu} \Gamma^\sigma_{\beta\mu}. \quad (3.197)$$

We're going to consider the components of the Riemann's tensor in a locally inertial reference frame known as *normal coordinates*, these are special such that

$$\Gamma_{\sigma\mu}^{\alpha}(p) = 0, \quad \partial_{\mu}\Gamma_{\beta\nu}^{\alpha}(p) \neq 0, \quad (3.198)$$

from these point of view the Riemann's tensor takes the following form:

$$R_{\beta\mu\nu}^{\alpha} = \partial_{\mu}\Gamma_{\beta\nu}^{\alpha} - \partial_{\nu}\Gamma_{\beta\mu}^{\alpha}, \quad (3.199)$$

and we gotta calculate the Cristoffel's symbol, thus

$$\begin{aligned} \Gamma_{\mu\nu}^{\alpha} &= \frac{1}{2}g^{\alpha\beta}(\partial_{\nu}g_{\beta\mu} + \partial_{\mu}g_{\beta\nu} - \partial_{\beta}g_{\mu\nu}). \\ \partial_{\sigma}\Gamma_{\mu\nu}^{\alpha} &= \frac{1}{2}g^{\alpha\beta}(\partial_{\sigma}\partial_{\nu}g_{\beta\mu} + \partial_{\sigma}\partial_{\mu}g_{\beta\nu} - \partial_{\sigma}\partial_{\beta}g_{\mu\nu}) \\ &\quad + \frac{1}{2}(\partial_{\nu}g_{\beta\mu} + \partial_{\mu}g_{\beta\nu} - \partial_{\beta}g_{\mu\nu})\partial_{\sigma}g^{\alpha\beta} \end{aligned}$$

$$\partial_{\sigma}\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta}(\partial_{\sigma}\partial_{\nu}g_{\beta\mu} + \partial_{\sigma}\partial_{\mu}g_{\beta\nu} - \partial_{\sigma}\partial_{\beta}g_{\mu\nu}) + \frac{1}{2}[\beta, \mu\nu]\partial_{\sigma}g^{\alpha\beta}, \quad (3.200)$$

$$\partial_{\sigma}\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta}(\partial_{\sigma}\partial_{\nu}g_{\beta\mu} + \partial_{\sigma}\partial_{\mu}g_{\beta\nu} - \partial_{\sigma}\partial_{\beta}g_{\mu\nu}). \quad (3.201)$$

we're going to use the equation 3.201 on the Riemman's tensor on the point  $p \in M$  in the following way

$$R_{\beta\mu\nu}^{\alpha}|_p = \frac{1}{2}g^{\alpha\sigma}(\partial_{\mu}\partial_{\nu}g_{\sigma\beta} + \partial_{\mu}\partial_{\beta}g_{\sigma\nu} - \partial_{\mu}\partial_{\sigma}g_{\beta\nu})|_p - \frac{1}{2}g^{\alpha\sigma}(\partial_{\nu}\partial_{\mu}g_{\sigma\beta} + \partial_{\nu}\partial_{\beta}g_{\sigma\mu} - \partial_{\mu}\partial_{\sigma}g_{\mu\beta})|_p,$$

$$R_{\beta\mu\nu}^{\alpha}|_p = \frac{1}{2}g^{\alpha\sigma}(\partial_{\beta}\partial_{\mu}g_{\sigma\nu} - \partial_{\nu}\partial_{\beta}g_{\sigma\mu} + \partial_{\nu}\partial_{\sigma}g_{\mu\beta} - \partial_{\mu}\partial_{\sigma}g_{\beta\nu})|_p, \quad (3.202)$$

$$R_{\alpha\beta\mu\nu}|_p = g_{\alpha\lambda}R_{\beta\mu\nu}^{\lambda} = \frac{1}{2}(\partial_{\beta}\partial_{\mu}g_{\alpha\nu} - \partial_{\nu}\partial_{\beta}g_{\alpha\mu} + \partial_{\nu}\partial_{\alpha}g_{\mu\beta} - \partial_{\mu}\partial_{\alpha}g_{\beta\nu})|_p \quad (3.203)$$

From here on we can see the following relations, which are tensor equations and do not depend on the coordinate chart

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \quad (3.204)$$

$$R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu}, \quad (3.205)$$

$$R_{\alpha\beta\mu\nu} R_{\mu\nu\alpha\beta}, \quad (3.206)$$

$$R_{\alpha\beta\mu\nu} + R_{\alpha\nu\beta\mu} + R_{\alpha\mu\nu\beta} = R_{\alpha[\beta\mu\nu]} = 0. \quad (3.207)$$

These relations and symmetries on the Riemann's tensor will allow us to reduce a system of 256 equations to 18 equations. There are more relations beyond theses, the known as the *Bianchi identities* which will be worth to define and arrive the Einstein's tensor. In normal coordinates we have got

$$\vec{\nabla}_\lambda R_{\alpha\beta\mu\nu} = \partial_\lambda R_{\alpha\beta\mu\nu} = \frac{1}{2} (\partial_\lambda \partial_\beta \partial_\mu g_{\alpha\nu} - \partial_\lambda \partial_\nu \partial_\beta g_{\alpha\mu} + \partial_\lambda \partial_\nu \partial_\alpha g_{\mu\beta} - \partial_\lambda \partial_\mu \partial_\alpha g_{\beta\nu})|_p, \quad (3.208)$$

We can find an analogous to the Jacobi's identity with covariant derivatives

$$\vec{\nabla}_\lambda R_{\alpha\beta\mu\nu} + \vec{\nabla}_\nu R_{\alpha\beta\lambda\mu} + \vec{\nabla}_\mu R_{\alpha\beta\nu\lambda} = 0, \quad (3.209)$$

$$R_{\alpha\beta[\mu\nu;\lambda]} = \vec{\nabla}_{[\lambda} R_{\mu\nu]\alpha\beta} = 0. \quad (3.210)$$

All this geometrical developement was done by Riemann with the intention of generalizing the Gauss work on curvature. Lets find out the contracted Bianchi's identities, therefore

$$\left[ \vec{\nabla}_e R_{abbc} + \vec{\nabla}_d R_{abec} + \vec{\nabla}_c R_{abde} = 0 \right] g^{ac}, \quad (3.211)$$

$$\vec{\nabla}_e R_{bd} + \vec{\nabla}_d (g^{ac} R_{abec}) + g^{ac} \vec{\nabla}_c R_{abdc} = 0, \quad (3.212)$$

$$\vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0, \quad (3.213)$$

$$\vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0. \quad (3.214)$$

From the Bianchi's identity for the Riemann's tensor we've found the repective identity for the Ricci's tensor also known as the *contracted Bianchi identity* given in the equations 3.214 but we can contract it one more time and obtaint the Bhianchi identity for the Ricci's scalar wich is related with the Einstein's tensor and also is a clue to the Einstein's equations, so contracting the equation 3.214 we'll then have

$$g^{bd} \left[ \vec{\nabla}_e R_{bd} - \vec{\nabla}_d R_{be} + \vec{\nabla}^a R_{abde} = 0 \right], \quad (3.215)$$

$$\vec{\nabla}_e R - \vec{\nabla}^b R_{be} - \vec{\nabla}^a R_{ae} = 0, \quad (3.216)$$

$$\vec{\nabla}^a R_{ae} = \frac{1}{2} \vec{\nabla}_e R. \quad (3.217)$$

The equation 3.217 is known as the Bianchi's identity twice contracted and playing a bit with this equations we will find something special, lets find out it

$$\begin{aligned} \vec{\nabla}^a R_{ae} - \frac{1}{2} \vec{\nabla}_e R &= 0, \\ \vec{\nabla}^a R_{ae} \frac{1}{2} g_{ae} \vec{\nabla}^a R &= 0, \\ \vec{\nabla}^a \left( R_{ae} - \frac{1}{2} g_{ae} R \right) &= 0, \end{aligned}$$

$$\vec{\nabla}^a G_{ab} = 0 \quad ; \quad G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R, \quad (3.218)$$

There appears naturally the Einstein's tensor! Now, in the curved spaces the geodesics aren't always parallel there are some separation between one or another geodesic but how can we measure or have an idea of how far or deviated is a geodesic from other.

### 3.3.3 Geodesic deviation

**Definition 29** Let  $M$  a manifold with a Levi-Civita connection  $\vec{\nabla}$ . A monoparametric family of geodesics is a map

$$\gamma : I \times I' \rightarrow M, \quad (3.219)$$

where  $I, I'$  are two open intervals on the  $\mathbb{R}$  such that

1. For a  $S$  constant,  $\gamma(s, t) = \gamma_s(t)$  is a geodesic with affine parameters  $t$ , therefore  $S$  would be a parameter that tags the geodesic.
2. The collection of these curves define a 2D smooth surface and is a subset of the manifold  $\Sigma \subset M$  as is represented in the figure 3.10

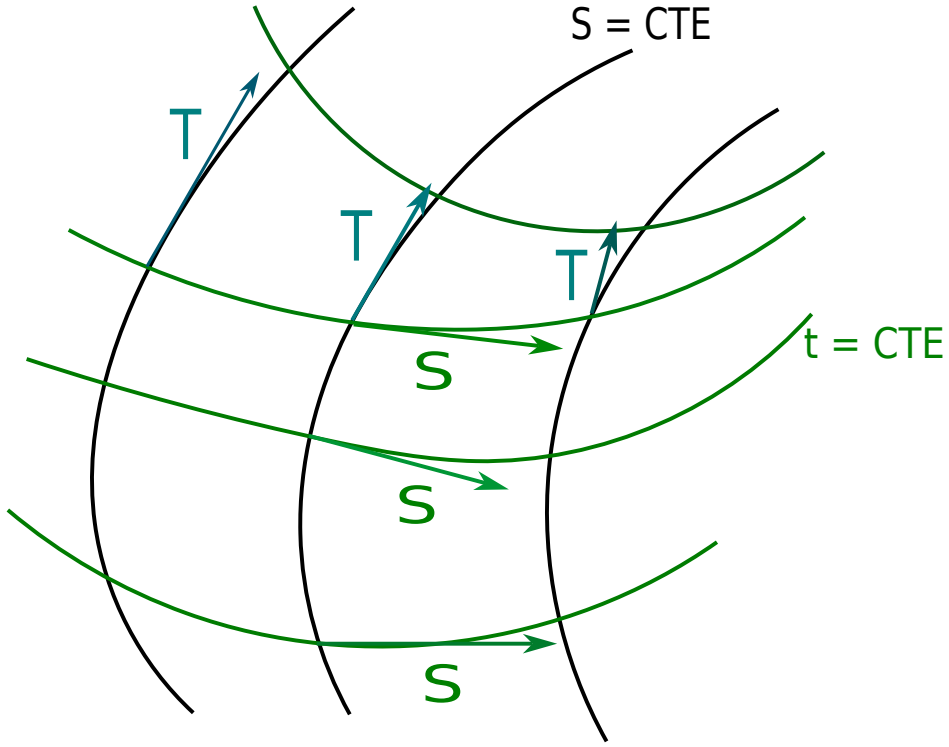


Figure 3.10: Representation of all the possible subsets of  $S$  constant and  $t$  constant, this is an analogous to what is done in the formulation  $3 + 1$  where we separate the space-time and folds as a Cauchy space.

To quantify the deviation between to adjacent geodesics, lets first, suppose a coordinate chart  $x^\mu(s, t) \in M$ , therefore

$$T^\mu = \frac{\partial x^\mu}{\partial t}, \quad S^\mu = \frac{\partial x^\mu}{\partial s}. \quad (3.220)$$

The first vectors  $T^\mu$  are called the tangent vectors to the geodesics, and the other ones  $S^\mu$  are named the deviation vectors. We're going to express the vectors of the chart in terms of the vectors of deviation.

$$x^\mu(s + \delta s, t) = x^\mu(s, t) + \left( \frac{\partial x^\mu}{\partial s} \right) \delta s + \mathcal{O}(\delta s^2). \quad (3.221)$$

We're going to analyze the vector  $S$  and its changing along the geodesic  $T$ . To do it, we can define a relative speed between geodesics

$$V^a = \left( \vec{\nabla}_T S \right)^a = T^b \vec{\nabla}_b S^a, \quad (3.222)$$

if we can talk about the relative speed between geodesics, we can also talk about the acceleration relative between geodesics, then

$$a^a = (\nabla_T V)^a = T^b \vec{\nabla}_b V^a = T^c \vec{\nabla}_c (T^b \vec{\nabla}_b S^a) = a^a = \vec{\nabla}_T \vec{\nabla}_T S^a, \quad (3.223)$$

we can introduce the coordinate basis on  $\Sigma$  as  $S = \partial_s$  and  $T = \partial_t$  and because we're having a theory free of torsion so from the commutator we have

$$[T, S](f) = T[S(f)] - S[T(f)] = \partial_t[S(f)] - \partial_s[T(f)] = 0, \quad (3.224)$$

under these conditions and without torsion then

$$\vec{\nabla}_T S = \vec{\nabla}_S T \rightarrow T^b \nabla_b S^a = S^b \nabla_b T^a, \quad (3.225)$$

$$\vec{\nabla}_T \vec{\nabla}_T S = R(T, S)T. \quad (3.226)$$

Using 3.226 into the acceleration relative into geodesics, then we have

$$a^a = [\vec{\nabla}_T \vec{\nabla}_T S]^a = (R(T, S)T)^a = R^a_{bcd} T^b T^c S^d, \quad (3.227)$$

$$a^a = R^a_{bcd} T^b T^c S^d. \quad (3.228)$$

The equation 3.228 implies something really, really amazing and important ***The acceleration is curvature*** but we gotta read this equations correctly, as cause and effect, the presence of curvature produces and effect we call acceleration, we can denotate the equation 3.228 in other way, as

$$a^a = \frac{D^2}{Dt^2} S^a = R^a_{bcd} T^b T^c S^d. \quad (3.229)$$

### 3.4 Gravitation and the Einstein's field equation

We have found a relation between the curvature tensor and the relative acceleration between geodesics, and knowing a particle test has a geodesics as a path, then different curvatures produces different accelerations, and from the experiments we've talk about in the first part acceleration is indistinguishable from gravity.

Well then, how are going to relate the acceleration to curvature? To do this, we're going to solve the Einsteins field equations in the Newtonian limit to interpret this physically and see if the Newton's theory was plane or curved of if it was, from the beginning a geometrical theory. First we're going to set some properties of the manifolds to set the space-time and use it to develop this theory.

1. The set of all the events in a  $4D$  manifold with pseudo-Riemannian metric and a Levi-Civita connections is a differential manifold called *space-time* denoted by  $(M, \mathbb{G}, \vec{\nabla})$ .
2. The Manifold is Lorentzian.
3. The weak equivalence principle is fulfilled.
4. The Einstein's equivalence principle is fulfilled.

### 3.4.1 Newtonian limit

We gotta then talk about particles whose speed is a lot less than the speed of light  $C$  and the gravitational field is stationary and weak, using geometrical coordinates  $C = 1$  then

$$\frac{d\vec{X}}{d\tau} \ll \frac{dt}{d\tau}, \quad (3.230)$$

$$\frac{d\vec{X}}{d\tau} = \frac{d\vec{X}}{dt} \frac{dt}{d\tau} \ll \frac{dt}{d\tau} \rightarrow \left| \frac{d\vec{X}}{dt} \right| \ll 1. \quad (3.231)$$

Having on count all of this, we're going to see the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (3.232)$$

because we've low speed all the components related to spatial ones when multiplied are really low valued so those terms are going to be ignored, thus

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \frac{dx^0}{d\tau} \frac{dx^0}{d\tau} = 0, \quad (3.233)$$

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu\sigma} (\partial_0 g_{\sigma 0} + \partial_0 g_{0\sigma} - \partial_\sigma g_{00}) \quad (3.234)$$

and because the gravitational field is stationary, the zeroth derivatives are then zero, thus

$$\Gamma^\mu_{00} = -\frac{1}{2} g^{\mu\sigma} \partial_\sigma g_{00}. \quad (3.235)$$

Then the geodesic equation becomes

$$\frac{d^2 x^\mu}{d\tau^2} - \frac{1}{2} g^{\mu\sigma} \partial_\sigma g_{00} \left( \frac{dt}{d\tau} \right)^2 = 0. \quad (3.236)$$

If the gravitational field is then weak, then physically we've got is a plane Minkowskian space-time which in a local region has a perturbation or tensor, then the global metric can be set as

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}; \quad |h_{\alpha\beta}| \ll 1 \quad (3.237)$$

with this we can calculate the Cristoffel symbol:

$$\Gamma_{00}^{\mu} = -\frac{1}{2} (\eta^{\mu\sigma} + h^{\mu\sigma}) \partial_{\sigma} (\eta_{00} + h_{00}) = -\frac{1}{2} (\eta^{\mu\sigma} + h^{\mu\sigma}) \partial_{\sigma} h_{00} \approx \frac{1}{2} \eta^{\mu\sigma} \partial_{\sigma} h_{00}. \quad (3.238)$$

$$\frac{d^2 x^{\mu}}{d\tau^2} = \frac{1}{2} \eta^{\mu\sigma} \partial_{\sigma} h_{00} \left( \frac{dt}{d\tau} \right)^2. \quad (3.239)$$

Because the Minkowski metric is a diagonal metric then we can split it into two equations

1.  $\mu = \sigma = 0$  we find that *in the Newton's theory  $t$  is an affine parameter* and  $t$  and  $\tau$  are related affinely

$$\frac{d^2 t}{d\tau^2} = 0. \quad (3.240)$$

2.  $\mu = \sigma = i$

$$\frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \quad (3.241)$$

In the right hand side of the last equation we see a gradient, this is looking like a Newton's second law, let's find out what is  $h_{00}$ , therefore

$$\frac{d^2 \vec{X}}{dt^2} = -\vec{\nabla} \left( -\frac{GM}{r} \right), \quad (3.242)$$

$$h_{00} = -\frac{2GM}{r} + cte. \quad (3.243)$$

And far from the source it must be asymptotically plane, or Minkowski then the constant we add to  $h_{00}$  is then zero, thus the metric becomes

$$g_{00} = -(1 + 2\phi), \quad (3.244)$$

$$g_{ii} = (1 + 2\phi). \quad (3.245)$$

Thanks to these we can introduce the *covariance principle* which sets that if we develop a theory in a locally plane vicinity we can generalize it changing the usual derivatives by covariant derivatives.

### 3.4.2 Einstein's field equations

Using the covariance principle we can generalize the following electrodynamic equations

$$\partial_\alpha T^{\alpha\beta} = 0 \rightarrow \vec{\nabla}_\alpha T^{\alpha\beta} = 0, \quad (3.246)$$

$$\partial_\beta F^{\alpha\beta} = 4\phi J^\alpha \rightarrow \vec{\nabla}_\beta F^{\alpha\beta} = 4\pi J^\alpha, \quad (3.247)$$

$$\partial_\beta^* F^{\alpha\beta} = 0 \rightarrow \vec{\nabla}_\beta^* F^{\alpha\beta} = 0. \quad (3.248)$$

We see from one side the matter and the other the geometry, the presence of matter implies a change in geometry? We will turn back to this question later. Now, we're going to start from the Einstein's equations in the Newtonian limit

$$\nabla^2 \phi = 4\phi G\rho, \quad \nabla^2 = \delta^{ij} \partial_i \partial_j, \quad (3.249)$$

and introducing the energy-momentum tensor which contains the information of the energy density, momentum flux and strains then the first component of that tensor is the mass density (remembering from the mass-energy equivalence) is  $T^{00} = \rho$ , and in some point in the Newtonian limit then

$$\nabla^2 h_{00} = -8\phi GT_{00}. \quad (3.250)$$

But we've using usual derivatives and we must work with tensor equations, then to solve this problem with the Laplacian we are going to introduce the D'Alembertian  $\square = \nabla^a \nabla_a$  and because we've got a second order equation we gotta use a second order geometry equation, therefore the Riemann's tensor must be introduced but this has a rank  $\binom{1}{3}$  then the Ricci's tensor is a better option. Einstein's proposed the Ricci's tensor is directly proportional to the energy-momentum tensor  $R_{ab} = kT_{ab}$  but from the conservation law we've got

$$\vec{\nabla}^a T_{ab} = 0 \rightarrow \vec{\nabla}^a R_{ab} = 0, \quad (3.251)$$

but

$$\vec{\nabla}^a R_{ab} = \frac{1}{2} \vec{\nabla}_b R, \quad (3.252)$$

$$\vec{\nabla}^a R_{ab} = \frac{1}{2} \vec{\nabla}_b R = \frac{k}{2} \vec{\nabla}_b T = \frac{k}{2} \partial_b T = 0. \quad (3.253)$$

This implies then  $T$  is a constant along the space-time, but this is not always true, thus this assumption between the Ricci's tensor and the energy-momentum tensor is discarded, then we're going to propose the Einstein's tensor (which is covariantly constant), then

$$\vec{\nabla}^a \left( R_{ab} - \frac{1}{2} g_{ab} R \right) = 0 \rightarrow R_{ab} - \frac{1}{2} g_{ab} R = k T_{ab}, \quad (3.254)$$

$$g^{ab} \left( R_{ab} - \frac{1}{2} g_{ab} R \right) = k g^{ab} T_{ab} \rightarrow R = -k T, \quad (3.255)$$

then finally the relation between the curvature and the matter is

$$R_{ab} = k \left( T_{ab} - \frac{1}{2} g_{ab} T \right) \quad (3.256)$$

we're going to see what happens in the Newtonian limit, then the greatest interaction is due to the component  $T_{00}$  or what is the same, due to the mass density.

$$\begin{aligned} R_{00} &= k \left( T_{00} - \frac{1}{2} g_{00} T \right), \\ T &= g^{ab} T_{ab} = g^{00} T_{00} = -(1 + h_{00}) \rho = -T_{00}, \\ R_{00} &= k \left( T_{00} - \frac{1}{2} g_{00} T_{00} \right) = k \left( T_{00} + \frac{1}{2} (h_{00} - 1) T_{00} \right) = \frac{1}{2} k T_{00} = R_{00}. \end{aligned}$$

Now, we are going to calculate the Ricci's tensor

$$\begin{aligned} R_{\mu k} &= \partial_\lambda \Gamma_{\mu k}^\lambda - \partial_k \Gamma_{\mu \lambda}^\lambda + \Gamma_{\mu k}^\rho \Gamma_{\lambda \rho}^\lambda - \Gamma_{\mu \lambda}^\rho \Gamma_{k \rho}^\lambda, \\ R_{00} &= \partial_\lambda \Gamma_{00}^\lambda - \partial_0 \Gamma_{0 \lambda}^\lambda + \Gamma_{00}^\rho \Gamma_{\lambda \rho}^\lambda - \Gamma_{0 \lambda}^\rho \Gamma_{0 \rho}^\lambda, \end{aligned}$$

$$R_{00} = \partial_\lambda \Gamma_{00}^\lambda, \quad (3.257)$$

$$R_{00} = R^i_{0i0} = \partial_i \Gamma^i_{00} = \partial_i \left[ -\frac{1}{2} g^{i\sigma} \partial_\sigma g_{00} \right] = \partial_i \left[ -\frac{1}{2} g^{ij} \partial_j g_{00} \right] \quad (3.258)$$

$$= -\frac{1}{2} \partial_i \left[ (\eta^{ij} + h^{ij}) \partial_j (\eta_{00} + h_{00}) \right] \quad (3.259)$$

$$= -\frac{1}{2} \eta^{ij} \partial_i \partial_j h_{00} = -\frac{1}{2} \nabla^2 h_{00}, \quad (3.260)$$

$$\nabla^2 h_{00} = -k T_{00}. \quad (3.261)$$

## 3.4.2.1 Einstein-Hilbert action

Can we deduce the Einstein's equations from a variational principle? from a Hamiltonian formulation? To solve these Einstein and Hilbert arrived to a object called the Einstein-Hilbert action.

Lets define an action and a Lagrangian density as

$$S_H[g] = \int_M \mathcal{L}_H \sqrt{-g} d^4x, \quad \mathcal{L}_H = \frac{1}{16\pi} R, \quad (3.262)$$

$$S_H[g] = \frac{1}{16\pi} \int_M R \sqrt{-g} d^4x = \int L dt. \quad (3.263)$$

In fact, for a general system, we can split the action into a geometrical part and a material part as

$$S = S_H + S_m. \quad (3.264)$$

Now, lets develop more this

$$S_H[g] = \frac{1}{16\pi} \int_M g^{ab} R_{ab} \sqrt{-g} d^4x, \quad (3.265)$$

now we are going to variate the action thus

$$\delta S_H[g] = \frac{1}{16\pi} \int_M [\sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + R \delta \sqrt{-g}] d^4x \quad (3.266)$$

applying a variation of the gravitational field  $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$  we can expand each term of the integrand, thus first lets solve for  $\sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}$

$$\begin{aligned} \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} &= \sqrt{-g} g^{\mu\nu} \delta [\partial_\alpha \Gamma_{\mu\nu}^\alpha - \partial_\nu \Gamma_{\mu\alpha}^\alpha + \Gamma_{\mu\nu}^\tau \Gamma_{\tau\alpha}^\alpha - \Gamma_{\mu\alpha}^\tau \Gamma_{\nu\tau}^\alpha] \\ &= \sqrt{-g} g^{\mu\nu} [\partial_\alpha \delta \Gamma_{\mu\nu}^\alpha - \partial_\nu \delta \Gamma_{\mu\alpha}^\alpha + \delta \Gamma_{\mu\nu}^\tau \Gamma_{\tau\alpha}^\alpha + \Gamma_{\mu\nu}^\tau \delta \Gamma_{\tau\alpha}^\alpha - \delta \Gamma_{\mu\alpha}^\tau \Gamma_{\nu\tau}^\alpha - \Gamma_{\mu\alpha}^\tau \delta \Gamma_{\nu\tau}^\alpha], \end{aligned}$$

remebering the definition of the Cristoffel symbols

$$\Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}), \quad (3.267)$$

$$\begin{aligned} \delta \Gamma_{\nu\rho}^\mu &= \frac{1}{2} \delta g^{\mu\sigma} (\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho}) + \frac{1}{2} g^{\mu\sigma} (\partial_\rho \delta g_{\sigma\nu} + \partial_\nu \delta g_{\sigma\rho} - \partial_\sigma \delta g_{\nu\rho}) \\ &= \frac{1}{2} \delta g^{\mu\sigma} (g_{\sigma\nu,\rho} + g_{\sigma\rho,\nu} - g_{\nu\rho,\sigma}) + \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{,\rho} + (\delta g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma}]. \end{aligned}$$

We are going to focus on the term  $\delta g^{\mu\sigma}$  then

$$\begin{aligned}\delta g^{\mu\sigma} &= \delta \left( g^{\mu k} g_k^\sigma \right) = \delta \left( g^{\mu k} g^{l\sigma} g_{kl} \right) \\ &= \delta g^{\mu k} g^{l\sigma} g_{kl} + g^{\mu k} \delta g^{l\sigma} g_{kl} + g^{\mu k} g^{l\sigma} \delta g_{kl} \\ &= \delta g^{\mu k} \delta_k^\sigma + \delta g^{l\sigma} \delta_l^\mu + g^{\mu k} g^{l\sigma} \delta g_{kl} = -g^{\mu k} g^{l\sigma} \delta g_{kl}.\end{aligned}$$

Using this in the variation of the Crystoffel symbol, then

$$\begin{aligned}\delta \Gamma_{\nu\rho}^\mu &= -\frac{1}{2} g^{\mu k} g^{l\sigma} \delta g_{kl} (g_{\sigma\nu,\rho} + g_{\sigma\rho,\nu} - g_{\nu\rho,\sigma}) + \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{,\rho} + (g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma}] \\ &= -g^{\mu k} \delta g_{kl} \Gamma_{\nu\rho}^l + \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{,\rho} + (g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma}],\end{aligned}$$

$$\begin{aligned}(\delta g_{\alpha\beta})_{,\gamma} &= (\delta g_{\alpha\beta})_{,\gamma} - \Gamma_{\alpha\gamma}^l \delta g_{l\beta} - \Gamma_{\beta\gamma}^l \delta g_{l\alpha}, \\ &(\delta g_{\sigma\nu})_{,\rho} + (\delta g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma} \\ &= (\delta g_{\sigma\nu})_{;\rho} + (\delta g_{\sigma\rho})_{;\nu} - (\delta g_{\nu\rho})_{;\sigma} + \Gamma_{\sigma\rho}^l \delta g_{l\nu} + \Gamma_{\nu\rho}^l \delta g_{l\sigma} + \Gamma_{\sigma\nu}^l \delta g_{l\rho} + \Gamma_{\rho\nu}^l \delta g_{l\sigma} - \Gamma_{\nu\sigma}^l \delta g_{l\rho} - \Gamma_{\rho\sigma}^l \delta g_{l\nu}.\end{aligned}$$

Because we have got a connection free of torsion is fulfilled the Cristoffel symbols commute in their lower indices  $\Gamma_{\beta\gamma}^\alpha = \Gamma_{\gamma\beta}^\alpha$  then

$$(\delta g_{\sigma\nu})_{,\rho} + (\delta g_{\sigma\rho})_{,\nu} - (\delta g_{\nu\rho})_{,\sigma} = (\delta g_{\sigma\nu})_{;\rho} + (g_{\sigma\rho})_{;\nu} - (\delta g_{\nu\rho})_{;\sigma} + 2\Gamma_{\nu\rho}^l \delta g_{l\sigma}, \quad (3.268)$$

$$\begin{aligned}\delta \Gamma_{\nu\rho}^\mu &= -g^{\mu k} \delta g_{kl} \Gamma_{\nu\rho}^l + \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{;\rho} + (\delta g_{\sigma\rho})_{;\nu} - (\delta g_{\nu\rho})_{;\sigma} + 2\Gamma_{\nu\rho}^l \delta g_{l\sigma}] \\ &= \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{;\rho} + (\delta g_{\sigma\rho})_{;\nu} - (\delta g_{\nu\rho})_{;\sigma}]\end{aligned}$$

$$\delta \Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\sigma} [(\delta g_{\sigma\nu})_{;\rho} + (\delta g_{\sigma\rho})_{;\nu} - (\delta g_{\nu\rho})_{;\sigma}], \quad (3.269)$$

now, lest calculate the following covariant derivatives

$$(\delta \Gamma_{\nu\rho}^\mu)_{;\alpha} = (\delta \Gamma_{\nu\rho}^\mu)_{,\alpha} + \Gamma_{\sigma\alpha}^\mu (\delta \Gamma_{\nu\rho}^\sigma) - \Gamma_{\nu\alpha}^\sigma (\delta \Gamma_{\sigma\rho}^\mu) - \Gamma_{\rho\alpha}^\sigma (\delta \Gamma_{\sigma\nu}^\mu) \quad (3.270)$$

$$(\delta \Gamma_{\nu\alpha}^\mu)_{;\rho} = (\delta \Gamma_{\nu\alpha}^\mu)_{,\rho} + \Gamma_{\sigma\rho}^\mu (\delta \Gamma_{\nu\alpha}^\sigma) - \Gamma_{\nu\rho}^\sigma (\delta \Gamma_{\sigma\alpha}^\mu) - \Gamma_{\alpha\rho}^\sigma (\delta \Gamma_{\sigma\nu}^\mu) \quad (3.271)$$

we are going to, in both equation a contraction of  $\mu$  with  $\alpha$  then

$$(\delta\Gamma_{\nu\rho}^\mu)_{;\mu} = (\delta\Gamma_{\nu\rho}^\mu)_{,\mu} + \Gamma_{\sigma\mu}^\mu(\delta\Gamma_{\nu\rho}^\sigma) - \Gamma_{\nu\mu}^\sigma(\delta\Gamma_{\sigma\rho}^\mu) - \Gamma_{\rho\mu}^\sigma(\delta\Gamma_{\sigma\nu}^\mu), \quad (3.272)$$

$$(\delta\Gamma_{\nu\mu}^\mu)_\rho = (\delta\Gamma_{\nu\mu}^\mu)_\rho + \Gamma_{\sigma\rho}^\mu(\delta\Gamma_{\nu\mu}^\sigma) - \Gamma_{\nu\rho}^\sigma(\delta\Gamma_{\sigma\mu}^\mu) - \Gamma_{\mu\rho}^\sigma(\delta\Gamma_{\sigma\nu}^\mu). \quad (3.273)$$

Subtracting both equations then

$$(\delta\Gamma_{\nu\rho}^\mu)_{;\mu} - (\delta\Gamma_{\nu\mu}^\mu)_{;\rho} = (\delta\Gamma_{\nu\rho}^\mu)_{,\mu} - (\delta\Gamma_{\nu\mu}^\mu)_{,\rho} + \Gamma_{\sigma\mu}^\mu(\delta\Gamma_{\nu\rho}^\sigma) + \Gamma_{\nu\rho}^\sigma(\delta\Gamma_{\sigma\mu}^\mu) - \Gamma_{\nu\mu}^\sigma(\delta\Gamma_{\sigma\rho}^\mu) - \Gamma_{\sigma\rho}^\mu(\delta\Gamma_{\nu\mu}^\sigma) \quad (3.274)$$

If  $\mu = \alpha$ ,  $\nu = \mu$ ,  $\rho = \mu$  then

$$\delta R_{\mu\nu} = (\delta\Gamma_{\mu\nu}^\alpha)_{;\alpha} - (\delta\Gamma_{\mu\alpha}^\alpha)_{;\nu}, \quad (3.275)$$

thus

$$\sqrt{-g}g^{\mu\nu}\delta R_{\mu\nu} = \sqrt{-g}[(\delta\Gamma_{\mu\nu}^\alpha)_{;\alpha} - (\delta\Gamma_{\mu\alpha}^\alpha)_{;\nu}]. \quad (3.276)$$

Now, for the second term in the integrand

$$\begin{aligned} \frac{1}{\sqrt{-g}}\delta\sqrt{-g} &= \delta L n(\sqrt{-g}) = \frac{1}{2}\delta L n(-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu) = \frac{1}{2}\frac{\delta(-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu)}{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} \\ &= -\frac{1}{2}\frac{1}{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}[-\delta g_{\mu\nu}]\dot{x}^\mu\dot{x}^\nu, \end{aligned}$$

$$\delta(g^{\mu\nu}g_{\mu\nu}) = 0 = g^{\mu\nu}\delta g_{\mu\nu} + g_{\mu\nu}\delta g^{\mu\nu}, \quad (3.277)$$

$$\frac{1}{\sqrt{-g}}\delta\sqrt{-g} = \frac{1}{2}g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}g_{\mu\nu}\delta g^{\mu\nu}. \quad (3.278)$$

Lets define the following quantity

$$W^\lambda = g^{\mu k}\delta\Gamma_{\mu k}^\lambda - g^{\mu\lambda}\delta\Gamma_{\mu k}^k \quad (3.279)$$

we're going to interpret this quantity later, from now it will be a useful variable change to work on the algebra, if we contract over  $\lambda$  its covariant derivative we get

$$W^\lambda_{;\lambda} = g^{\mu k}\delta R_{\mu k}, \quad (3.280)$$

using these equation on the variation of the Einstein-Hilbert action we got

$$\delta S_H[g] = \frac{1}{16\pi} \int_M \left[ R_{ab} \delta g^{ab} - \frac{R}{2} g_{ab} \delta g^{ab} + \vec{\nabla}_a W^a \right] \sqrt{-g} d^4x, \quad (3.281)$$

$$\delta S_H[g] = \frac{1}{16\pi} \int \left( R_{ab} - \frac{1}{2} R g_{ab} \right) \sqrt{-g} d^4x + \frac{1}{16\pi} \vec{\nabla}_a W^a \sqrt{-g} d^4x. \quad (3.282)$$

We gotta focus on the second term, to do it, first look at the figure 3.11 where the integrand is being integrated on  $M$  but the second term is in fact a divergence over a  $4D$  volume thus in analogy to the  $3D$  Gauss's theorem of divergence, this becomes the following integral

$$\int_M \vec{\nabla}_a W^a \sqrt{-g} d^4x = \oint_{\Sigma M} W^a dS_a. \quad (3.283)$$

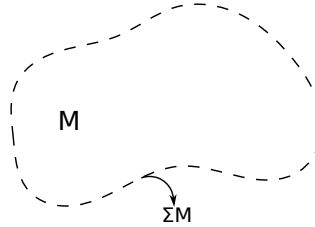


Figure 3.11: Representation of the region (manifold  $M$ ) of integration and its surrounding surface  $\Sigma M$ .

The integral 3.283 is completely analogous to that in electrodynamics where we integrate the divergence of the Poinyng's vector, so we can extend the integration surface  $\Sigma M$  to infinity or what is the same *our manifold is asimptotically plane* and the second integral goes to zero, in analogy to electrodynamics we can interpreted the vector  $W^\lambda$  as a generalized four-momentum or heat or losses vector. The variation of the Einstein-Hilbert action then becomes

$$\delta S_H[g] = \frac{1}{16\pi} \int_M \left( R_{ab} - \frac{1}{2} R g_{ab} \right) \delta g^{ab} d^4x, \quad (3.284)$$

$$\frac{\delta S_H[g]}{\delta g^{ab}} = \frac{G_{ab}}{16\pi}. \quad (3.285)$$

If we extremalize it we find immediately  $G_{ab} = 0$  which are the Einstein's equations in the vacuum.

## 3.4.2.2 Energy-momentum tensor

Here we are going to build up the Energy-momentum tensor the hearth of the Einstein's equations and most of the electrodynamics too, to do it we are going to use the Einstein-Hilbert action and a matter action as follows

$$S = S_H[g] + S_{matter}, \quad (3.286)$$

then lets see how changes the variation of the action

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{ab}} = \frac{1}{\sqrt{-g}} \left[ \frac{\delta S_H}{\delta g^{ab}} + \frac{\delta S_m}{\delta g^{ab}} \right], \quad (3.287)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{ab}} = \frac{1}{\sqrt{-g}} \left( \frac{G_{ab}}{16\pi} + \frac{\delta S_m}{\delta g^{ab}} \right) = 0, \quad (3.288)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{ab}} = -\frac{T_{ab}}{2} \rightarrow T_{ab} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{ab}}, \quad (3.289)$$

$$\delta S_m = -\frac{1}{2} \int_M \sqrt{-g} T_{ab} \delta g^{ab} d^4x, \quad (3.290)$$

applying variational principles we then arrive to

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi T_{ab}. \quad (3.291)$$

Thus we've arrived to the Einstein's field equations from variational principles, now we're going to solve this equations but only under the assumption that *the gravitational field is weak but not stationary*. To do it we have to suppose the metric of our manifold is plane with a certain perturbation in a region, so the global metric will be

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad |h_{\alpha\beta}| \ll 1. \quad (3.292)$$

Before going further we must prove the tensor 3.292 is a metric, lets then prove it. The contravariant components of the metric are given by

$$g^{\alpha\beta} = \eta^{\alpha\beta} - h^{\alpha\beta} \quad (3.293)$$

and given a coordinate chart  $\phi = \{x^i\}$   $i = 0, 1, 2, 3$  and another coordinate chart  $\bar{\phi} = \{\bar{x}^i\}$ ;  $i = 0, 1, 2, 3$  thus the transformation rule for the metric is

$$g_{\bar{\alpha}\bar{\beta}} = \Lambda_{\bar{\alpha}}^{\mu} \Lambda_{\bar{\beta}}^{\nu} g_{\mu\nu} \quad (3.294)$$

where  $\Lambda_{\bar{\beta}}^{\alpha}$  is a Lorentz Boost then what we see is that the transformation rule is a similarity or translation transformation then

$$g_{\bar{\alpha}\bar{\beta}} = \Lambda_{\bar{\alpha}}^{\mu} \Lambda_{\bar{\beta}}^{\nu} \eta_{\mu\nu} + \Lambda_{\bar{\alpha}}^{\mu} \Lambda_{\bar{\beta}}^{\nu} h_{\mu\nu} = \eta_{\bar{\alpha}\bar{\beta}} + h_{\bar{\alpha}\bar{\beta}}. \quad (3.295)$$

Then  $g_{\alpha\beta}$  is effectively a tensor and having a theory invariant under Lorentz transformations implies having at least a special relativistic theory. In the Einstein's equations there is the Ricci's number and tensor, but to calculate them we first have to calculate the Riemann's tensor and the Cristoffel symbols, therefore

$$\begin{aligned} \Gamma_{\nu\rho}^{\mu} &= \frac{1}{2} g^{\mu\sigma} (g_{\sigma\nu,\rho} + g_{\rho\sigma,\nu} - g_{\nu\rho,\sigma}) \\ &= \frac{1}{2} (\eta^{\mu\sigma} - h^{\mu\sigma}) [h_{\sigma\nu,\rho} + h_{\rho\sigma,\nu} - h_{\nu\rho,\sigma}] \\ &= \frac{1}{2} \eta^{\mu\sigma} (h_{\sigma\nu,\rho} + h_{\rho\sigma,\nu} - h_{\nu\rho,\sigma}), \end{aligned}$$

$$R_{\nu\rho\sigma}^{\alpha} = \Gamma_{\nu\sigma,\rho}^{\alpha} - \Gamma_{\nu\rho,\sigma}^{\alpha} + \mathcal{O}(\Gamma^2), \quad (3.296)$$

$$\begin{aligned} \eta_{\mu\alpha} R_{\nu\rho\sigma}^{\alpha} = R_{\mu\nu\rho\sigma} &= \frac{\eta_{\mu\alpha}}{2} \left[ \eta^{\alpha\beta} (h_{\beta\nu,\rho} + h_{\rho\beta,\nu} - h_{\rho\nu,\beta}) \right] - \frac{\eta^{\alpha\beta}}{2} \left[ \eta^{\alpha\beta} (h_{\beta\nu,\sigma} + h_{\sigma\beta,\nu} - h_{\sigma\nu,\beta})_{,\rho} \right. \\ &\quad \left. - \eta^{\alpha\beta} (h_{\beta\nu,\rho} + h_{\rho\beta,\nu} - h_{\rho\nu,\beta})_{,\sigma} \right], \end{aligned}$$

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} \delta_{\mu}^{\beta} (h_{\beta\sigma,\nu\rho} + h_{\rho\nu,\beta\sigma} - h_{\sigma\nu,\beta\rho} - h_{\rho'\beta\sigma,\nu\sigma}), \quad (3.297)$$

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} (h_{\mu\sigma,\nu\rho} + h_{\rho\nu,\mu\sigma} - h_{\sigma\nu,\mu\rho} - h_{\rho\mu,\nu\sigma}) \quad (3.298)$$

There the Ricci's curvature tensor is given by

$$R_{\mu\nu} = \eta_{\alpha}^{\rho} \eta^{\alpha\beta} R_{\alpha\mu\rho\nu} = \frac{1}{2} [\partial_{\mu} \partial_{\rho} h_{\nu}^{\rho} + \partial^{\rho} \partial_{\nu} h_{\rho\mu} - \partial^{\rho} \partial_{\rho} h_{\nu\mu} - \partial_{\mu} \partial_{\nu} h_{\mu}^{\mu}] \quad (3.299)$$

$$R_{\mu\nu} = \frac{1}{2} (\partial_{\rho} \partial_{\mu} h_{\nu}^{\rho} + \partial_{\rho} \partial_{\nu} h_{\mu}^{\rho} - \partial_{\mu} \partial_{\nu} h - \square h_{\mu\nu}), \quad (3.300)$$

where  $\square = \vec{\nabla}_a \vec{\nabla}^a$  is the D'Alembertian and in a Minkowski space it corresponds to  $\square = \partial_\alpha \partial^\alpha$ . Well then the Ricci's number is given by

$$R = \eta^{\mu\nu} R_{\mu\nu} = \frac{1}{2} \eta^{\mu\nu} (\partial_\rho \partial_\mu h^\rho_\nu + \partial_\rho \partial_\nu h^\rho_\mu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}) = \partial_\rho \partial_\mu h^{\rho\mu} - \square h. \quad (3.301)$$

Now, introducing all of this on the Einstein's field equation we will have the following equation

$$\frac{1}{2} (\partial_\rho \partial_\mu h^\rho_\nu + \partial_\rho \partial_\nu h^\rho_\mu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}) - \frac{1}{2} \eta_{\mu\nu} [\partial_\rho \partial_\sigma h^{\rho\sigma} - \square h] = 8\pi T_{\mu\nu}. \quad (3.302)$$

Before going further there is the needing of introducing a special kind of tensor, called the *inverted trace tensor* which are really useful in the general relativity and in fact, the Einstein's tensor is one of this tensor and we are going to prove it in a moment, the inverted trace tensor for the perturbation tensor is given by

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} h_{\mu\nu} h, \quad (3.303)$$

it is easy to see that if we calculated the inverted trace tensor for the Ricci's curvature tensor we get the Einstein's tensor like this:

$$\bar{R}_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R. \quad (3.304)$$

We can then rewrite the perturbation tensor in term of its inverted trace tensor and also its trace as follows

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \bar{h}, \quad (3.305)$$

$$\eta^{\mu\nu} \bar{h}_{\mu\nu} = h - 2h = -h = \bar{h}. \quad (3.306)$$

Then the Einstein's field equations in terms of the inverted trace tensor becomes

$$\begin{aligned} & \partial_\rho \partial_\mu \left( \bar{h}^\rho_\nu - \frac{1}{2} \delta^\rho_\nu \bar{h} \right) + \partial_\rho \partial_\nu \left( \bar{h}^\rho_\mu - \frac{1}{2} \delta^\rho_\mu \bar{h} \right) + \partial_\mu \partial_\nu \bar{h} \\ & - \square \left( \bar{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \bar{h} \right) - \eta_{\mu\nu} \partial_\rho \partial_\sigma \left( \bar{h}^{\rho\sigma} - \frac{1}{2} \eta^{\rho\sigma} \bar{h} \right) - \eta_{\mu\nu} \square \bar{h} = 16\pi T_{\mu\nu} \end{aligned}$$

$$\begin{aligned}
& \partial_\rho \partial_\mu \bar{h}^\rho_\nu - \frac{1}{2} \delta^\rho_\nu \partial_\rho \partial_\mu \bar{h} + \partial_\rho \partial_\nu \bar{h}^\rho_\mu - \frac{1}{2} \delta^\rho_\mu \partial_\rho \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} \\
& \quad - \eta_{\mu\nu} \partial_\rho \partial_\sigma \bar{h}^{\rho\sigma} + \frac{1}{2} \eta_{\mu\nu} \eta^{\rho\sigma} \partial_\rho \partial_\sigma \bar{h} - \eta_{\mu\nu} \square \bar{h} = 16\pi T_{\mu\nu}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \frac{1}{2} \delta^\rho_\nu \partial_\rho \partial_\mu \bar{h} - \frac{1}{2} \delta^\rho_\mu \partial_\rho \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} \\
& \quad - \square \bar{h}_{\mu\nu} + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} + \frac{1}{2} \eta_{\mu\nu} \eta^{\rho\sigma} \partial_\rho \partial_\sigma \bar{h} - \eta_{\mu\nu} \partial_\rho \partial_\sigma \eta^{\rho\alpha} \eta^{\sigma\beta} \bar{h}_{\alpha\beta} - \eta_{\mu\nu} \square \bar{h}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \frac{1}{2} \partial_\nu \partial_\mu \bar{h} - \frac{1}{2} \partial_\mu \partial_\nu \bar{h} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} \\
& \quad + \frac{1}{2} \eta_{\mu\nu} \square \bar{h} + \frac{1}{2} \eta_{\mu\nu} \partial_\rho \partial^\rho \bar{h} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} - \eta_{\mu\nu} \square \bar{h}, \\
& = \partial_\rho \partial_\mu \eta^{\rho\alpha} \bar{h}_{\alpha\nu} + \partial_\rho \partial_\nu \eta^{\rho\alpha} \bar{h}_{\alpha\mu} - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& = \partial_\mu [\partial_\rho (\eta^{\rho\alpha} \bar{h}_{\nu\alpha})] + \partial_\nu [\partial_\rho (\eta^{\rho\alpha} \bar{h}_{\mu\alpha})] - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& \quad = \partial_\mu W_\nu + \partial_\nu W_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \eta^{\sigma\beta} \partial_\rho \partial_\sigma \bar{h}_{\alpha\beta} \\
& \quad = \partial_\mu W_\nu + \partial_\nu W_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \eta^{\rho\alpha} \partial_\rho \left[ \partial_\sigma (\eta^{\sigma\beta} \bar{h}_{\alpha\beta}) \right], \\
& \quad - \left[ \square \bar{h}_{\mu\nu} - \partial_\mu W_\nu - \partial_\nu W_\mu + \eta_{\mu\nu} \eta^{\alpha\beta} \partial_\alpha W_\beta \right] = 16\pi T_{\mu\nu} \tag{3.307}
\end{aligned}$$

Multiplying both sides by  $-1$  and changing  $\mu \rightarrow \alpha$  and  $\nu \rightarrow \beta$  then we have finally

$$\square \bar{h}_{\alpha\beta} - \partial_\beta W_\alpha - \partial_\alpha W_\beta + \eta_{\alpha\beta} \eta^{\mu\nu} \partial_\nu W_\mu = -16\pi T_{\alpha\beta}; \quad W_\alpha = \partial_\mu (\eta^{\mu\nu} \bar{h}_{\alpha\nu}). \tag{3.308}$$

The equation 3.308 is the Einstein's equations and they are really similar to the Maxwell's equations, the terms with  $W_\alpha$  are like heat or losses fluxes on the borders of the manifold. Our intention now is to convert the equation 3.308 into a wave equation, but how do we do it? In the same way we do it in electromagnetism! We work with a Gauge theory, a calibration theory, there we use the Lorenz's Gauge, here we have to find and use this Lorenz's Gauge and later also find out the called *transverse Gauge*.

## *Chapter 4*

### *Gravitational Waves*

*A century after Einstein first wrote them down,  
the ripples of space-time finally reached our  
instruments.*

“Ladies and gentlemen, we have detected gravitational waves. We did it.”

– David Reitze

## 4.1 Lorenz Gauge and gravitational waves

Lets remember how are the conditions of the gravitational field we're working on (there is only one v:) which is a *theory of weak field* this means we have a plane (Minkowski) space-time but there is a certain perturbation on a region we represent it as  $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$   $|h_{\alpha\beta}| \ll 1$ . We can do these perturbations on the coordinate charts as  $x'^\alpha = x^\alpha + \xi^\alpha$  where  $\xi^\alpha = \xi^\alpha(x^\alpha)$  are infinitesimal transformations such that

$$\left| \frac{\partial \xi^\alpha}{\partial x^\beta} \right| \approx |h_{\alpha\beta}| \rightarrow \left| \frac{\partial \xi^\alpha}{\partial x^\beta} \right| \ll 1. \quad (4.1)$$

We are looking for a gauge theory that leaves invariant certain quantity, in our case such quantity is the Riemann's tensor, to do it we gotta see how is the behavior of our space-time

$$g'_{\alpha\beta} = g_{\mu\nu} \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial x^\nu}{\partial x'^\beta}, \quad (4.2)$$

$$\frac{\partial x^\mu}{\partial x'^\alpha} = \delta^\mu_\alpha - \frac{\partial \xi^\mu}{\partial x^\nu} \frac{\partial x^\nu}{\partial x'^\alpha} = \delta^\mu_\alpha - \frac{\partial \xi^\mu}{\partial x^\nu} \left( \delta^\nu_\alpha - \frac{\partial \xi^\nu}{\partial x^\rho} \frac{\partial x^\rho}{\partial x'^\alpha} \right), \quad (4.3)$$

$$\frac{\partial x^\mu}{\partial x'^\alpha} = \delta^\mu_\alpha - \frac{\partial \xi^\mu}{\partial x^\alpha}, \quad (4.4)$$

$$g'_{\alpha\beta} = g_{\mu\nu} \left( \delta^\mu_\alpha - \frac{\partial \xi^\mu}{\partial x^\alpha} \right) \left( \delta^\nu_\beta - \frac{\partial \xi^\nu}{\partial x^\beta} \right) = g_{\alpha\beta} - g_{\alpha\nu} \frac{\partial \xi^\nu}{\partial x^\beta} - g_{\beta\mu} \frac{\partial \xi^\mu}{\partial x^\alpha}, \quad (4.5)$$

$$g'_{\alpha\beta} = (\eta_{\alpha\beta} + h_{\alpha\beta}) - \eta_{\alpha\nu} \frac{\partial \xi^\nu}{\partial x^\beta} - \eta_{\beta\mu} \frac{\partial \xi^\mu}{\partial x^\alpha}, \quad (4.6)$$

$$g'_{\alpha\beta} = \eta_{\alpha\beta} + h'_{\alpha\beta} \rightarrow; h'_{\alpha\beta} = h_{\alpha\beta} - \eta_{\alpha\nu} \frac{\partial \xi^\nu}{\partial x^\beta} - \eta_{\beta\mu} \frac{\partial \xi^\mu}{\partial x^\alpha}. \quad (4.7)$$

**AQUÍ DEBERÍA AGREGAR ALGO SOBRE TEORÍA DE GAUGE EN EL ELECTROMAGNETISMO COMO ANEXO O COMO NO SÉ XD**

In analogy to see how transforms the Riemann's tensor, lets first calculate the following transform of the tensor  $R'^{\alpha\beta}_{\mu\nu}$  this because it will be easier mathematically calculate in this form and in the end using the Minkowski's metric tensor we can lower one index, therefore

$$R'^{\alpha\beta}_{\mu\nu} = \frac{\partial x'^\alpha}{\partial x^\alpha} \frac{\partial x'^\beta}{\partial x^\beta} \frac{\partial x^c}{\partial x'^\mu} \frac{\partial x^d}{\partial x'^\nu} R^{ab}_{cd} \quad (4.8)$$

we gotta calculate the derivative of the prime coordinates respect to the non prime , then

$$\frac{\partial x'^{\alpha}}{\partial x^{\mu}} = \delta_{\mu}^{\alpha} + \frac{\partial \xi^{\alpha}}{\partial x^{\mu}}. \quad (4.9)$$

Using the equation 4.4 and 4.9 into the equation 4.8 we then have

$$\begin{aligned} R'^{\alpha\beta}_{\mu\nu} &= \left[ \delta_a^{\alpha} + \frac{\partial \xi^{\alpha}}{\partial x^a} \right] \left[ \delta_b^{\beta} + \frac{\partial \xi^{\beta}}{\partial x^b} \right] \left[ \delta_{\mu}^c - \frac{\partial \xi^c}{\partial x^{\mu}} \right] \left[ \delta_{\nu}^d - \frac{\partial \xi^d}{\partial x^{\nu}} \right] R^{ab}_{cd} \\ &= \left[ \delta_a^{\alpha} \delta_b^{\beta} + \delta_a^{\alpha} \frac{\partial \xi^{\beta}}{\partial x^b} + \delta_b^{\beta} \frac{\partial \xi^{\alpha}}{\partial x^a} \right] \left[ \delta_{\mu}^c \delta_{\nu}^d - \delta_{\mu}^c \frac{\partial \xi^d}{\partial x^{\nu}} - \delta_{\nu}^d \frac{\partial \xi^c}{\partial x^{\mu}} \right] R^{ab}_{cd} \\ &= \left[ \delta_a^{\alpha} \delta_b^{\beta} \delta_{\mu}^c \delta_{\nu}^d - \delta_a^{\alpha} \delta_b^{\beta} \delta_{\mu}^c \frac{\partial \xi^d}{\partial x^{\nu}} - \delta_a^{\alpha} \delta_b^{\beta} \delta_{\nu}^d \frac{\partial \xi^c}{\partial x^{\mu}} + \delta_a^{\alpha} \delta_{\mu}^c \delta_{\nu}^d \frac{\partial \xi^{\beta}}{\partial x^b} + \delta_b^{\beta} \delta_{\mu}^c \delta_{\nu}^d \frac{\partial \xi^{\alpha}}{\partial x^a} \right] R^{ab}_{cd} \\ &= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^{\beta} \delta_{\mu}^c \left[ \delta_{\nu}^d \frac{\partial \xi^{\alpha}}{\partial x^a} - \delta_a^{\alpha} \frac{\partial \xi^d}{\partial x^{\nu}} \right] + \delta_a^{\alpha} \delta_{\nu}^d \left[ \delta_{\mu}^c \frac{\partial \xi^{\beta}}{\partial x^b} - \delta_b^{\beta} \frac{\partial \xi^c}{\partial x^{\mu}} \right] \right\} R^{ab}_{cd} \\ &= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^{\beta} \delta_{\mu}^c \left[ \delta_{\nu}^d \delta_a^{\alpha} \frac{\partial \xi^{\alpha}}{\partial x^{\nu}} - \delta_a^{\alpha} \delta_{\alpha}^d \frac{\partial \xi^{\alpha}}{\partial x^{\nu}} \right] + \delta_a^{\alpha} \delta_{\nu}^d \left[ \delta_{\mu}^c \delta_b^{\beta} \frac{\partial \xi^{\beta}}{\partial x^{\mu}} - \delta_b^{\beta} \delta_{\beta}^c \frac{\partial \xi^{\beta}}{\partial x^{\mu}} \right] \right\} R^{ab}_{cd} \\ &= R^{\alpha\beta}_{\mu\nu} + \left\{ \delta_b^{\beta} \delta_{\mu}^c \delta_a^{\alpha} \left[ \frac{\partial \xi^{\alpha}}{\partial x^{\nu}} - \frac{\partial \xi^{\alpha}}{\partial x^{\nu}} \right] + \delta_a^{\alpha} \delta_{\nu}^d \delta_b^{\beta} \left[ \frac{\partial \xi^{\beta}}{\partial x^{\mu}} - \frac{\partial \xi^{\beta}}{\partial x^{\mu}} \right] \right\} R^{ab}_{cd}, \end{aligned}$$

$$R'^{\alpha\beta}_{\mu\nu} = R^{\alpha\beta}_{\mu\nu} \quad (4.10)$$

$$\eta_{b\beta} R'^{\alpha\beta}_{\mu\nu} = R'^{\alpha}_{b\mu\nu} = \eta_{b\beta} R^{\alpha\beta}_{\mu\nu} = R^{\alpha}_{b\mu\nu} \quad (4.11)$$

Doing the variable change  $b = \beta$  we then arrive that the Riemann's tensor keep invariant under these infinitesimal transformations:

$$R'^{\alpha}_{\beta\mu\nu} = R^{\alpha}_{\beta\mu\nu}, \quad (4.12)$$

and thus

$$R'_{\alpha\beta} = R_{\alpha\beta} \quad (4.13)$$

and the Lorenz Gauge is

$$\vec{\nabla}_{\alpha} \left( \eta^{\alpha\beta} A_{\beta} \right) = 0 \quad (4.14)$$

and then by analogy with the electromagnetism  $\bar{h}_{\alpha\beta}$  fulfil a Lorenz Gauge as follows

$$\frac{\partial}{\partial x^{\mu}} \left( \eta^{\mu\nu} \bar{h}_{\alpha\nu} \right) = 0. \quad (4.15)$$

Well then, in the Einstein's equation 3.308 there is the trace and the tensors of the trace inverted tensors, so we have to calculate they transformed versions too, in this way lets do it

$$h' = h - 2\partial_\mu \xi^\mu, \quad (4.16)$$

$$\bar{h}'_{\alpha\beta} = h'_{\alpha\beta} - \frac{1}{2}\eta_{\alpha\beta}h' = h_{\alpha\beta} - \frac{\partial\xi_\alpha}{\partial x^\beta} - \frac{\partial\xi_\beta}{\partial x^\alpha} - \frac{1}{2}\eta_{\alpha\beta}\left(h - 2\frac{\partial\xi^\alpha}{\partial x^\mu}\right), \quad (4.17)$$

$$\bar{h}'_{\alpha\beta} = \bar{h}_{\alpha\beta} - \frac{\partial\xi_\alpha}{\partial x^\beta} - \frac{\partial\xi_\beta}{\partial x^\alpha} + \eta_{\alpha\beta}\frac{\partial\xi^\mu}{\partial x^\mu}, \quad (4.18)$$

$$W'_\alpha = \frac{\partial}{\partial x^\mu} \left( \eta^{\mu\nu} \bar{h}'_{\alpha\nu} \right), \quad (4.19)$$

$$W'_\alpha = \partial_\mu \left[ \eta^{\mu\nu} \left( \bar{h}_{\alpha\nu} - \partial_\nu \xi_\alpha - \partial_\alpha \xi_\nu + \eta_{\alpha\nu} \partial_\mu \xi^\mu \right) \right], \quad (4.20)$$

$$W'_\alpha = \partial_\mu \left[ \eta^{\mu\nu} \bar{h}_{\alpha\nu} - \eta^{\mu\nu} \partial_\nu \xi_\alpha - \partial_\alpha \xi^\mu + \delta_\alpha^\mu \partial_\mu \xi^\mu \right] \quad (4.21)$$

$$W'_\alpha = W_\alpha - \eta^{\mu\nu} \frac{\partial^2 \xi}{\partial x^\mu \partial x^\nu} \rightarrow; W'_\alpha = W_\alpha - \square \xi_\alpha, \quad (4.22)$$

$$W'_\alpha = 0 \iff W_\alpha = \square \xi_\alpha. \quad (4.23)$$

We have got a little freedom to choose our  $W_\alpha$  keeping invariant the Riemann's tensor fulfilling then

$$\square \bar{h}_{\alpha\beta} = -16\pi T_{\alpha\beta}, \quad (4.24)$$

and turning back the unities therefore

$$\square \bar{h}_{\alpha\beta} = -\frac{16\pi G}{C^4} T_{\alpha\beta}. \quad (4.25)$$

The equation 4.25 is the wave equation for gravitational waves! Lets see how are their solutions, first we are going to analyze the vacuum case  $T_{\alpha\beta} = 0$  and the simplest solution would be a traveling monochromatic plane wave as

$$\bar{h}_{\alpha\beta}(x^\mu) = A_{\alpha\beta} e^{ik_\mu x^\mu}, \quad (4.26)$$

using this solution in the equation 4.25 we find that  $k_\alpha k^\alpha = 0$  this means the vector  $\mathbb{K}$  is a null-like or light-like vector, this implies that *the gravitational waves travels at the speed of light in vacuum* and from the Lorenz Gauge we find this

$$\frac{\partial}{\partial x^\mu} \left( \eta^{\mu\nu} A_{\alpha\nu} e^{ik_\sigma x^\sigma} \right) = 0, \quad (4.27)$$

$$\eta^{\mu\nu} A_{\alpha\nu} \frac{\partial}{\partial x^\mu} e^{ik_\sigma x^\sigma} = 0, \quad (4.28)$$

$$\eta^{\mu\nu} A_{\alpha\nu} e^{ik_\sigma x^\sigma} i k_\beta \frac{\partial x^\beta}{\partial x^\mu} = 0, \quad (4.29)$$

$$\eta^{\mu\nu} A_{\alpha\nu} i k_\mu e^{ik_\sigma x^\sigma} = 0. \quad (4.30)$$

The equation 4.30 means that the wave vector is orthogonal to the amplitude tensor.

## 4.2 Gravitational waves

Here we are going to keep developing the wave equation for the perturbation  $h_{\alpha\beta}$  see how are the equations for the perturbations of the coordinates components and we will work with a new Gauge, the called *traceless transverse gauge* and we will find out the gravitational waves have got polarization properties.

From our Lorenz gauge we had found that  $W'_\alpha = 0$  but this also means that the perturbations of the coordinates satisfies a wave equation  $W_\alpha = \square \xi_\alpha$  so we can suppose the most basic solution to this equation is a plane wave, therefore

$$\xi^\alpha = B^\alpha e^{ik_\sigma x^\sigma}, \quad (4.31)$$

where  $B^\alpha$  are four infinitesimal constants, and we gotta find them to have a complete solution for the wave equation to do it lets start from the primed and bar perturbation

$$\bar{h}'_{\alpha\beta} = \bar{h}_{\alpha\beta} - \frac{\partial \xi_\alpha}{\partial x^\beta} - \frac{\partial \xi_\beta}{\partial x^\alpha} + \eta_{\alpha\beta} \frac{\partial \xi_\mu}{\partial x^\mu}, \quad (4.32)$$

$$\bar{h}'_{\alpha\beta} = A_{\alpha\beta} e^{ik_\sigma x^\sigma} - i B_\alpha k_\beta e^{ik_\sigma x^\sigma} - i B_\beta k_\alpha e^{ik_\sigma x^\sigma} + i \eta_{\alpha\beta} k_\sigma B^\sigma e^{ik_\sigma x^\sigma} = A'_{\alpha\beta} e^{ik_\sigma x^\sigma}, \quad (4.33)$$

thus

$$A'_{\alpha\beta} = A_{\alpha\beta} - i B_\alpha k_\beta - i B_\beta k_\alpha + i \eta_{\alpha\beta} k_\sigma B^\sigma. \quad (4.34)$$

### 4.2.1 Traceless transverse gauge

Here we are going to choose a new gauge, the *traceless transverse gauge* which will allow us to work as well with  $\bar{h}_{\alpha\beta}$  as  $h_{\alpha\beta}$  because, we will see it in a moment, both satisfies the same wave equations. The traceless transverse Gauge is:

$$\eta^{\mu\nu} A'_{\mu\nu} = 0, \quad (4.35)$$

$$A'_{0i} = 0. \quad (4.36)$$

Calculating the Gauge of the equation 4.34 we have

$$\eta^{\mu\nu} A'_{\mu\nu} = \eta^{\mu\nu} A_{\mu\nu} - i\eta^{\mu\nu} B_\mu k_\nu - i\eta^{\mu\nu} B_\nu k_\mu + i\eta^{\mu\nu} \eta_{\mu\nu} k_\sigma B^\sigma = 0, \quad (4.37)$$

$$A'_{0i} = A_{0i} - iB_0 k_i - iB_i k_0 + \eta_{0i} i k_\sigma B^\sigma = 0, \quad (4.38)$$

$$\eta^{\mu\nu} A'_{\mu\nu} = \eta^{\mu\nu} A_{\mu\nu} + 2ik_\mu B^\mu = 0, \quad (4.39)$$

$$A_{0i} - ik_0 B_i - iB_i k_0 = 0. \quad (4.40)$$

Or we can rewrite them in a more comfortable way as

$$\eta^{\mu\nu} A_{\mu\nu} = -2ik_\mu B^\mu, \quad (4.41)$$

$$k_0 B_i + B_0 k_i = -ia_{0i}. \quad (4.42)$$

Now, from the orthogonality condition 4.30 we can calculate how are the componentes of  $A_{\mu\nu}$  therefore

$$A_{0\beta} k^\beta = A_{00} k^0 + A_{0i} k^i = A_{00} k^0 = 0 \rightarrow A_{00} = 0 \rightarrow A_{0\alpha} = 0, \quad (4.43)$$

$$A_{i\beta} k^\beta = A_{i0} k^0 + A_{ij} k^j = 0 \rightarrow A_{ij} = 0. \quad (4.44)$$

Before going further it will be worth to calculate the trace of the perturbation under these gauge, thus

$$\bar{h}_{0\alpha} = 0, \quad \bar{h} = \eta^{\alpha\beta} h_{\alpha\beta} = \eta^{\alpha\beta} A_{\alpha\beta} = 0 = -h = 0, \quad (4.45)$$

$$\bar{h}_{\alpha\beta} = h_{\alpha\beta}. \quad (4.46)$$

Here we can see that using this gauge the wave equation for  $h_{\alpha\beta}$  is the same that for  $\bar{h}_{\alpha\beta}$  lets write the perturbation in a matrix form under the assumption the wave vector has the following form  $k_\alpha = [\omega/C, 0, 0, -\omega/C]$  thus

$$A_{\alpha\beta}k^\beta = A_{\alpha 0}k^0 + A_{\alpha i}k^i = A_{\alpha 0}\frac{\omega}{C} - A_{\alpha z}\frac{\omega}{C} = 0, \quad (4.47)$$

$$\eta^{\mu\nu}A_{\mu\nu} = 0 \rightarrow A_{xx} + A_{yy} = 0. \quad (4.48)$$

We are going to set that  $A_{xx} = a_+$  called *plus mode* and  $A_{xy} = A_{yx} = a_\times$  called the *crux mode*, there the perturbation written under the traceless transverse gauge is

$$h_{\alpha\beta}^{TT} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (4.49)$$

then we can define now the **gravitational polarization modes** as

$$h_+ = a_+ e^{i\omega(t - \frac{z}{C})}, \quad (4.50)$$

$$h_\times = a_\times e^{i\omega(t - \frac{z}{C})}. \quad (4.51)$$

#### 4.2.1.1 Polarization and Stokes parameters for gravitational waves

We are going to start from the perturbation under the influence of a traceless transverse gauge 4.49 this because it has got a really familiar form so we have to ask ourselves, *what is the difference between light's polarization and gravitational polarization?* Well, lets figure it out.

First, we have to use the fact there is an isomorphism between the vector space  $\mathbb{M}_{n \times n}$  (square  $n \times n$  matrices) and  $\mathbb{R}^n$ , therefore from the perturbation matrix 4.49 we can define our *perturbation state vector*

$$\mathbf{h}^{TT} = \begin{bmatrix} h_+ \\ h_\times \\ h_\times \\ -h_+ \end{bmatrix}. \quad (4.52)$$

This vector will tell us the polarization mode state where our space-time is under the perturbation 4.49 but before going further we must take on count an important fact,  $a_+, a_x$  must have their own spatial phase because not always  $h_+$  and  $h_x$  will be on phase or will represent the same oscillation but with different amplitude, both are different, in general, polarization modes, then if  $A_+, \phi_+, A_x, \phi_x$  are their amplitudes and phases respectively then the polarization mode state vector will be

$$\mathbf{h}^{TT} = \begin{bmatrix} A_+ e^{i\phi_+} \\ A_x e^{i\phi_x} \\ A_x e^{i\phi_x} \\ -A_+ e^{i\phi_+} \end{bmatrix} e^{i(\omega t - kz)}. \quad (4.53)$$

We have got only two degrees of freedom in the state vector 4.53 one for the  $+$  modes and the other for the  $x$  modes, lets see how are they related geometrically. We can see easily the following two equations

$$\frac{a_+}{A_+} e^{-i\phi_x} = e^{i(\omega t - kz)} e^{i\Delta\phi}, \quad \frac{a_x}{A_x} e^{-i\phi_x} = e^{i(\omega t - kz)} e^{-i\Delta\phi} \quad (4.54)$$

where  $\Delta\phi = \phi_+ - \phi_x$  having all of this on count lets calculate the following product

$$\left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right) \left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right)^*, \quad (4.55)$$

$$\begin{aligned} & \left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right) \left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right)^* \\ &= \frac{a_+^2}{A_+^2} - \frac{a_+ a_x^*}{A_+ A_x} e^{i\Delta\phi} - \frac{a_x a_+^*}{A_+ A_x} e^{-i\Delta\phi} + \frac{a_x^2}{A_x^2} \\ &= \frac{a_+^2}{A_+^2} + \frac{a_x^2}{A_x^2} - 2 \frac{\text{Real}\{a_+ a_x^*\}}{A_+ A_x} \cos\Delta\phi, \end{aligned}$$

$$\left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right) \left( \frac{a_+}{A_+} e^{-i\phi_x} - \frac{a_x}{A_x} e^{-i\phi_+} \right)^* = \sin^2 \Delta\phi \quad (4.56)$$

or joining all

$$\frac{a_+^2}{A_+^2} + \frac{a_x^2}{A_x^2} - 2 \frac{\text{Real}\{a_+ a_x^*\}}{A_+ A_x} \cos\Delta\phi = \sin^2 \Delta\phi. \quad (4.57)$$

The polarization modes describe an hyper-ellipse in the complex plane  $\mathbb{C}^2$  in the same way as the light's polarization! Then this implies the polarization mode state vector 4.53 lives, in someway as a vector of this ellipse. Now, we haven't got stationary perturbations, this in the real life usually

oscillates very fast and has some kind of time-dependence, under the same calculation we can arrive to an ellipse in the real space (this time taking its real part and ignoring the imaginary part)

$$\frac{a_+^2(t)}{A_+^2(t)} + \frac{a_x^2(t)}{A_x^2(t)} - 2\frac{a_+(t)a_x(t)}{A_x(t)A_+(t)}\cos\Delta\phi = \sin^2\Delta\phi. \quad (4.58)$$

To solve this problem of the time dependence we can take time averages, why? Because we in fact measure time averages and we, for simplicity, we have suppose monochromatic waves thus

$$\frac{\langle a_+^2(t) \rangle}{A_+^2} + \frac{\langle a_x^2(t) \rangle}{A_x^2} - 2\frac{\langle a_+(t)a_x^*(t) \rangle}{A_x A_+}\cos\Delta\phi = \sin^2\Delta\phi \quad (4.59)$$

where

$$\langle a_i(t)a_j(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T a_i(t)a_j(t)dt \quad i, j = +, x. \quad (4.60)$$

Multiplying the equation 4.59 by  $4A_+^2A_x^2$  we can see that

$$4A_x^2 \langle A_+^2(t) \rangle + 4A_+^2 \langle A_x^2(t) \rangle - 8A_x A_+ \langle a_+(t)a_x(t) \rangle \cos\Delta\phi = (2A_+ A_x \sin\Delta\phi)^2. \quad (4.61)$$

Because in the perturbation 4.49 we have plane and harmonic waves, is easy to see the time averages are

$$\langle a_+(t)^2 \rangle = \frac{1}{2}A_+^2 \quad (4.62)$$

$$\langle a_x(t)^2 \rangle = \frac{1}{2}A_x^2 \quad (4.63)$$

$$\langle a_+(t)a_x(t) \rangle = \frac{1}{2}A_+ A_x \cos\Delta\phi, \quad (4.64)$$

substituting these equation in 4.59 yields to

$$2A_+^2 A_x^2 + 2A_+^2 A_x^2 - (2A_+ A_x \cos\Delta\phi)^2 = (2A_+ A_x \sin\Delta\phi)^2, \quad (4.65)$$

Since we want to express the result in terms of a *gravitational intensity* then we can add to both sides of the equation 4.65 the term  $A_+^4 + A_x^4$  to have then perfect squares thus we arrive to

$$(A_+^2 + A_x^2)^2 - (A_+^2 - A_x^2)^2 - (2A_+ A_x \cos\Delta\phi)^2 = (2A_+ A_x \sin\Delta\phi)^2, \quad (4.66)$$

or defining the following quantities

$$S_0^G = A_+^2 + A_x^2, \quad (4.67)$$

$$S_1^G = A_+^2 - A_x^2, \quad (4.68)$$

$$S_2^G = 2A_+A_x\cos\Delta\phi, \quad (4.69)$$

$$S_3^G = 2A_+A_x\sin\Delta\phi \quad (4.70)$$

letting us to write the equation 4.66 as

$$(S_0^G)^2 = (S_1^G)^2 + (S_2^G)^2 + (S_3^G)^2 \quad (4.71)$$

this is the same equation for the Stokes parameters for a well polarized light beam.

#### 4.2.2 Effects of the gravitational radiation on the matter

To study the effects of the gravitational radiation under the traceless transverse gauge we are going to suppose a particle  $A$  which interacts only gravitationally and follows a geodesic parameterized with its proper time  $\tau$  this particle will stay quite in the origin of coordinates, this means its world line will be a vertical line, therefore given a coordinate chart its coordinates are

$$X_{;A}^0(\tau) = ct, \text{ \& } X_A^i(\tau) = 0, \quad (4.72)$$

lets solve its geodesic equation

$$\frac{d^2 X_A^\alpha}{d\tau^2} + \Gamma_{\mu\nu}^\alpha \dot{x}_A^\mu \dot{x}_A^\nu = 0 \quad (4.73)$$

where

$$\Gamma_{\beta\gamma}^0 = -\frac{1}{2} (h_{0\gamma,\beta} + h_{0\beta,\gamma} - h_{\gamma\beta,0}), \quad (4.74)$$

$$\Gamma_{\beta\gamma}^i = \frac{1}{2} (h_{i\gamma,\beta} + h_{i\beta,\gamma} - h_{\gamma\beta,i}). \quad (4.75)$$

Using the TTG

$$\Gamma^0_{\beta\gamma} = \frac{1}{2C} h^{TT}_{\gamma\beta,t}, \quad (4.76)$$

$$\Gamma^0_{0\alpha} = 0, \quad (4.77)$$

$$\Gamma^0_{ij} = \frac{1}{2C} h^{TT}_{ij,t}, \quad (4.78)$$

$$\Gamma^i_{00} = 0, \quad (4.79)$$

$$\Gamma^i_{0j} = \frac{1}{2C} h^{TT}_{ij,t}, \quad (4.80)$$

$$\Gamma^i_{jk} = \frac{1}{2} (h^{TT}_{ik,j} + h^{TT}_{ji,k} - h^{TT}_{jk,i}). \quad (4.81)$$

When we replace with  $\alpha = 0$  we have got the equations keeps equal, *the gravitational wave does not affect the observer* we will see later this is due that the gravitational waves have a quadru-polar behavior.

### Poincaré Sphere

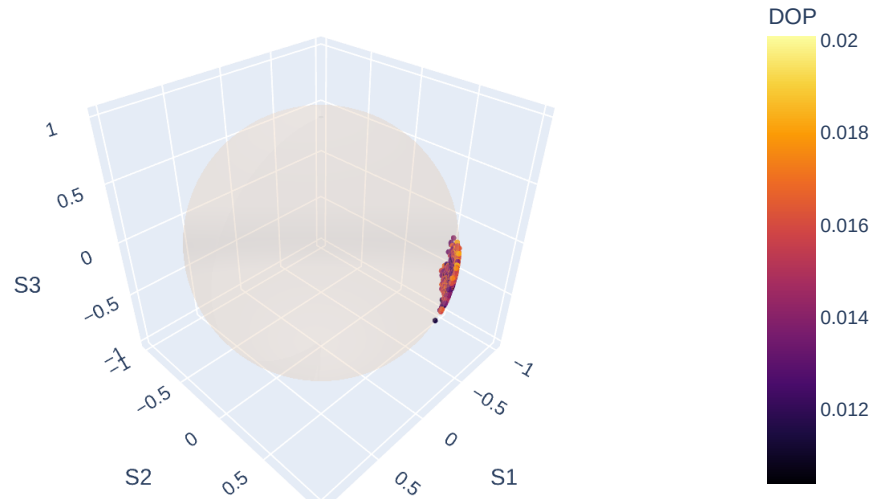


Figure 4.1: Caption



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